# Continuous Curvature Measures on Convex Bodies

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## Abstract

The purpose of this thesis is to introduce the concept of continuous translation-invariant curvature measures which generalizes the notion of smooth translation-invariant curvature measures due to Bernig and Fu. Many of the classical theorems for the space Val of continuous translation-invariant valuations translate to Curv. In particular, it is known that Curv admits a grading analogous to the McMullen decomposition of Val. Moreover, we equip Curv with a Banach space topology and we prove that Curv admits counterparts to the characterization theorems of Hadwiger, McMullen and Klain–Schneider. Since Curv is naturally an infinite dimensional representation of the general linear group $\mathrm{GL}(n,\mathbb{R})$, we introduce the linear subspace of $\mathrm{GL}(n,\mathbb{R})$-smooth vectors and conjecture that this space coincides with the space of smooth translation-invariant curvature measures. This conjecture is verified in certain degrees of homogeneity and in low dimensions. Inspired by Alesker's irreducibility theorem for Val, we also study Jordan–Hölder series in the representation Curv. Our results lead us to conjecture that their length does not exceed two for a given degree and parity.

## Zusammenfassung

Der Zweck dieser Arbeit ist das Konzept stetiger translationsinvarianter Krümmungsmaße einzuführen, was den Begriff der glatten translationsinvarianten Krümmungsmaße nach Bernig und Fu verallgemeinert. Viele klassische Sätze über den Raum Val der stetigen translationsinvarianten Bewertungen lassen sich auf Curv übertragen. Insbesondere ist bekannt, dass Curv analog zur McMullen-Zerlegung von Val graduiert ist. Darüber hinaus statten wir Curv mit einer Banachraum-Topologie aus und wir zeigen, dass Curv Analoga für die Charakterisierungssätze von Hadwiger, McMullen und Klain–Schneider zulässt. Da Curv auf natürliche Weise eine unendlich-dimensionale Darstellung der allgemeinen linearen Gruppe $\mathrm{GL}(n,\mathbb{R})$ ist, führen wir den Untervektorraum der $\mathrm{GL}(n,\mathbb{R})$-glatten Vektoren ein und stellen die Vermutung auf, dass dieser Raum mit dem Raum der glatten translationsinvarianten Krümmungsmaße übereinstimmt. Diese Vermutung wird in gewissen Homogenitätsgraden und in niedrigen Dimensionen bewiesen. Inspiriert durch Aleskers Irreduzibilitätssatz für Val betrachten wir auch Jordan-Hölder-Reihen der Darstellung Curv. Unsere Ergebnisse führen uns zu der Vermutung, dass deren Länge für gegebenen Homogenitätsgrad und gegebene Parität maximal zwei ist.

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# 1 Introduction

## 1.1 History of Curvature Measures

The so called intrinsic volumes are an important and classical notion in convex geometry. For a given convex body $K \subseteq \mathbb{R}^n$, the map

$$(0, \infty) \to \mathbb{C},\ \varepsilon \mapsto \mathrm{vol}_n\left(K + \varepsilon B^n\right)$$

is a polynomial of degree $n$, where $B^n \subseteq \mathbb{R}^n$ denotes the unit ball. Therefore, there exist coefficients $\mu_0(K), \ldots, \mu_n(K) \in \mathbb{C}$ such that the *Steiner formula*

$$\mathrm{vol}_n\left(K + \varepsilon B^n\right) = \sum_{i=0}^{n} \omega_i \mu_{n-i}(K)\, \varepsilon^i$$

holds, where the normalizing constants are given by $\omega_i = \mathrm{vol}_i\left(B^i\right)$. Let $\mathcal{K}$ denote the space of convex bodies in $\mathbb{R}^n$. Then, the map $\mathcal{K} \to \mathbb{C}$, $K \mapsto \mu_i(K)$ is called the $i$-th *intrinsic volume*. The family of intrinsic volumes includes a lot of interesting geometric functionals, such as the Euler characteristic $\mu_0 = \chi = 1$ and the volume $\mu_n = \mathrm{vol}_n$. Moreover, up to constants $\mu_1$ is the mean width and $\mu_{n-1}$ is the surface area.

To understand intrinsic volumes from a bigger picture point of view, the crucial notion is that of a valuation. A map $\varphi\colon \mathcal{K} \to \mathbb{C}$ is called a *valuation* if

$$\varphi(K \cup L) + \varphi(K \cap L) = \varphi(K) + \varphi(L)$$

whenever $K, L, K \cup L \in \mathcal{K}$. Of particular importance is the space Val of continuous and translation-invariant valuations which admits a rich structure theory, see Chapter 2 for details. The general linear group $\mathrm{GL}(n, \mathbb{R})$ acts naturally on the space Val and therefore also any subgroup of $\mathrm{GL}(n, \mathbb{R})$ does. This group action can be used to characterize the intrinsic volumes as was shown by Hadwiger in [Had57].

**Theorem (Hadwiger).** *A valuation $\varphi \in \mathrm{Val}$ is $\mathrm{SO}(n)$-invariant if and only if there exist $\lambda_0, \ldots, \lambda_n \in \mathbb{C}$ such that $\varphi = \lambda_0\mu_0 + \ldots + \lambda_n\mu_n$.*

The notion of curvature measure has its origins in Federer's influential paper [Fed59]. In this paper, he constructs curvature measures as localizations of the intrinsic volumes. For all $K \in \mathcal{K}$ and $i = 0, \ldots, n$, he constructs a complex Borel measure $\Phi_i(K)$ on $\mathbb{R}^n$ with the crucial property that $\Phi_i(K, \mathbb{R}^n) = \mu_i(K)$. In fact, he constructs measures $\Phi_i(A)$ for the much larger class of sets $A \subseteq \mathbb{R}^n$ with positive reach. As we do not study sets of positive reach in this thesis, we omit the definition here. Let $\mathcal{M}$ denote the space of complex Borel measures. In the scope of Federer's work, the assignment

$$\Phi_i\colon \mathcal{K} \to \mathcal{M},\ K \mapsto \Phi_i(K)$$

is called the *$i$-th curvature measure* of $K$. Note that a curvature measure is therefore not a measure itself but a map that assigns a measure to a convex body. Besides introducing the curvature measures, Federer also proves an integral representation for them.

In the paper [Sch78], Schneider proves a localized version of Hadwiger's theorem which characterizes the curvature measures $\Phi_i$ by some simple properties.

**Theorem (Schneider).** *A map* $\Phi\colon \mathcal{K} \to \mathcal{M}$ *satisfies*

$$\Phi = \sum_{i=0}^{n} \lambda_i \Phi_i$$

*for some* $\lambda_0, \ldots, \lambda_n \in \mathbb{C}$ *if and only if* $\Phi$ *satisfies the following list of properties:*

- $\Phi(K + x, \beta + x) = \Phi(K, \beta)$ *for all convex bodies* $K \in \mathcal{K}$*, Borel sets* $\beta \subseteq \mathbb{R}^n$ *and* $x \in \mathbb{R}^n$*.*
- *If* $K, L \subseteq \mathbb{R}^n$ *are convex bodies and* $U \subseteq \mathbb{R}^n$ *is an open set such that* $K \cap U = L \cap U$*, then* $\Phi(K, \beta) = \Phi(L, \beta)$ *for all Borel sets* $\beta \subseteq U$*.*
- $\Phi$ *is weakly continuous in the sense of measures.*
- $\Phi(K \cup L) + \Phi(K \cap L) = \Phi(K) + \Phi(L) \in \mathcal{M}$ *whenever* $K, L, K \cup L \in \mathcal{K}$*.*
- $\Phi(gK, g\beta) = \Phi(K, \beta)$ *for all convex bodies* $K \in \mathcal{K}$*, Borel sets* $\beta \subseteq \mathbb{R}^n$ *and* $g \in \mathrm{SO}(n)$*.*

Of course, the natural next question is for what other valuations $\varphi \in \mathrm{Val}$ a similar localization process is possible. To this end, let us start with some more background on the space Val. In [Ale06a], Alesker introduces a dense linear subspace $\mathrm{Val}^{\mathrm{sm}} \subseteq \mathrm{Val}$ of so called smooth valuations. A valuation $\varphi \in \mathrm{Val}$ is smooth if and only if there exist a constant $c \in \mathbb{C}$ and a translation-invariant differential form $\omega \in \Omega^{n-1}\left(\mathbb{R}^n \times S^{n-1}\right)$ such that

$$\varphi(K) = c \operatorname{vol}_n(K) + \int_{\mathrm{nc}(K)} \omega$$

for all $K \in \mathcal{K}$, where $\mathrm{nc}(K) \subseteq \mathbb{R}^n \times S^{n-1}$ denotes the normal cycle of $K$.

As was first observed by Bernig and Fu in [BF11], smooth valuations do indeed admit a localization. They give the following definition.

**Definition (Bernig–Fu).** A map $\Phi\colon \mathcal{K} \to \mathcal{M}$ is called a *smooth translation-invariant curvature measure* if there exist a constant $c \in \mathbb{C}$ and a translation-invariant differential form $\omega \in \Omega^{n-1}\left(\mathbb{R}^n \times S^{n-1}\right)$ such that

$$\Phi(K, \beta) = c \operatorname{vol}_n(K \cap \beta) + \int_{\mathrm{nc}(K) \cap \pi_1^{-1}(\beta)} \omega$$

for all $K \in \mathcal{K}$ and all Borel sets $\beta \subseteq \mathbb{R}^n$, where $\pi_1\colon \mathbb{R}^n \times S^{n-1} \to \mathbb{R}^n$ denotes the natural projection.

We write $\mathrm{Curv}^{\mathrm{sm}}$ for the space of smooth translation-invariant curvature measures. In [Zäh86], Zähle constructs suitable differential forms $\omega_0, \ldots, \omega_{n-1} \in \Omega^{n-1}\left(\mathbb{R}^n \times S^{n-1}\right)$ such that Federer's curvature measures satisfy

$$\Phi_i(K, \beta) = \int_{\mathrm{nc}(K) \cap \pi_1^{-1}(\beta)} \omega_i$$

for all $K \in \mathcal{K}$, Borel sets $\beta \subseteq \mathbb{R}^n$ and $i = 0, \ldots, n-1$. Since $\Phi_n(K, \beta) = \mathrm{vol}_n(K \cap \beta)$ for all $K$ and $\beta$, it follows that $\Phi_i \in \mathrm{Curv}^{\mathrm{sm}}$ for all $i = 0, \ldots, n$. In other words, the curvature measures introduced by Bernig and Fu generalize the curvature measures introduced by Federer.

From the construction one would expect that the spaces $\mathrm{Curv}^{\mathrm{sm}}$ and $\mathrm{Val}^{\mathrm{sm}}$ are intimately related. This phenomenon is captured by the so called *globalization map*. For any $\Phi \in \mathrm{Curv}^{\mathrm{sm}}$, we let $\mathrm{glob}(\Phi) = \Phi(\cdot, \mathbb{R}^n) \in \mathrm{Val}^{\mathrm{sm}}$. By construction, the induced map $\mathrm{glob}\colon \mathrm{Curv}^{\mathrm{sm}} \to \mathrm{Val}^{\mathrm{sm}}$ is surjective.

Thus, we obtain the following very incomplete diagram which illustrates the relations between the spaces we have discussed.

$$\begin{array}{ccc} \mathrm{Val} & & \\ \uparrow & & \\ \mathrm{Val}^{\mathrm{sm}} & \xleftarrow{\mathrm{glob}} & \mathrm{Curv}^{\mathrm{sm}} \end{array}$$

## 1.2 Contents and Results

The basis of this work is the idea of filling in the top right corner in the diagram above. This means that we want to define a space Curv consisting of continuous translation-invariant curvature measures $\mathcal{K} \to \mathcal{M}$. This space should satisfy $\mathrm{Curv}^{\mathrm{sm}} \subseteq \mathrm{Curv}$ and $\mathrm{glob}(\Phi) = \Phi(\cdot, \mathbb{R}^n) \in \mathrm{Val}$ to ensure that we get a commutative diagram of the following form.

$$\begin{array}{ccc} \mathrm{Val} & \xleftarrow{\mathrm{glob}} & \mathrm{Curv} \\ \uparrow & & \uparrow \\ \mathrm{Val}^{\mathrm{sm}} & \xleftarrow{\mathrm{glob}} & \mathrm{Curv}^{\mathrm{sm}} \end{array}$$

The definition of the space Curv is motivated by the observation that any $\Phi \in \mathrm{Curv}^{\mathrm{sm}}$ satisfies the first four points appearing in the list of properties in Schneider's theorem. Of course there exist many examples of smooth translation-invariant curvature measures which are not invariant under the group $\mathrm{SO}(n)$. Moreover, a result of Weil in [Wei17] implies that we can omit the valuation property as an axiom as it follows from the remaining properties. Therefore, we essentially use the first three points of Schneider's theorem as a definition for continuous translation-invariant curvature measures, see Definition 3.2.

**Definition.** A map $\Phi\colon \mathcal{K} \to \mathcal{M}$ is called a *continuous translation-invariant curvature measure* if it satisfies the following properties.

- $\Phi$ is translation-invariant: $\Phi$ satisfies

$$\Phi(K + x, \beta + x) = \Phi(K, \beta)$$

for all $K \in \mathcal{K}$, $\beta \in \mathcal{B}$ and $x \in \mathbb{R}^n$.

- $\Phi$ is defined locally: Given two convex bodies $K, L \subseteq \mathbb{R}^n$, if there exists an open set $U \subseteq \mathbb{R}^n$ such that $K \cap U = L \cap U$, then

$$\Phi(K, \beta) = \Phi(L, \beta)$$

for all Borel sets $\beta \subseteq U$.

- $\Phi$ is weak* continuous: For a convergent sequence $\left(K_j\right)_{j \in \mathbb{N}} \subseteq \mathcal{K}$ of convex bodies satisfying $K_j \to K$ as $j \to \infty$, we have

$$\lim_{j \to \infty} \Phi\left(K_j, f\right) = \Phi(K, f)$$

for all $f \in C_0(\mathbb{R}^n)$.

Since $\mathcal{M}$ is a complex vector space, one easily verifies that the set Curv of all continuous translation-invariant curvature measures is a complex vector space.

It should be mentioned that we use a seemingly weaker continuity condition, namely weak* continuity. However, it is proved in Proposition 3.6 that weak continuity in the sense of measures in fact follows from weak* continuity for curvature measures.

It is natural to ask which structural results of the space Val have counterparts in the theory of curvature measures. This is the theme of the remainder of this thesis. We will now state the main results of this thesis. For precise definitions and proofs, we refer the reader to the latter part of the text.

Firstly, recall that the space Val admits a natural grading with respect to degree of homogeneity and parity, known as the McMullen decomposition, see [McM77]. It turns out that the same holds true for the space Curv, see Theorem 3.11.

**Theorem (Kiderlen–Weil Decomposition).** *The space* Curv *admits a natural decomposition*

$$\mathrm{Curv} = \bigoplus_{i=1}^{n} \mathrm{Curv}_i^+ \oplus \mathrm{Curv}_i^- .$$

The main part of the proof of this theorem is contained in the paper [KW99] by Kiderlen and Weil.

Surprisingly, Theorem 2.1 of [Wei17] shows that curvature measures automatically satisfy the valuation property. For the convenience of the reader we repeat the argument in this thesis, see Theorem 3.12.

**Theorem (Weil).** *Any* $\Phi \in \mathrm{Curv}$ *is an* $\mathcal{M}$*-valued valuation.*

It should be mentioned that Weil has the additional assumption that the mapping $\mathcal{K} \to \mathbb{C}$, $K \mapsto \Phi(K, \beta)$ is measurable for all Borel sets $\beta \subseteq \mathbb{R}^n$. This is however not needed for the proof of the theorem above.

Moreover, it is well-known that Val is a Banach space. The same turns out to be true

for Curv. In fact, any bounded set $B \subseteq \mathbb{R}^n$ with non-empty interior induces a norm $\|\cdot\|_B$ given by

$$\|\Phi\|_B = \sup_{K \subseteq B} \left|\Phi(K)\right|$$

for all $\Phi \in \mathrm{Curv}$. With respect to these norms the following theorem holds, see Theorem 3.16 and Theorem 3.17.

**Theorem.** *For all bounded sets $B \subseteq \mathbb{R}^n$ with non-empty interior, the space* Curv *equipped with the norm $\|\cdot\|_B$ is a Banach space. Moreover, the norms $\|\cdot\|_B$ and $\|\cdot\|_{B'}$ are equivalent for any two bounded sets $B, B' \subseteq \mathbb{R}^n$ with non-empty interior.*

It should be mentioned that a big difference compared to the setting of valuations is that it is not a priori clear that the norms defined above are in fact finite. As we will see, this follows from a deep theorem in measure theory.

The spaces $\mathrm{Val}_n$ of $n$-homogeneous valuations and $\mathrm{Val}_{n-1}$ of $(n-1)$-homogeneous valuations admit explicit descriptions which are due to Hadwiger, see [Had57], and McMullen, see [McM80], respectively. Moreover, the space of so called simple valuations is characterized by the Klain–Schneider theorem, see [Kla95] and [Sch96]. The same can be done in the setting of curvature measures. The following theorem summarizes the results of Theorems 3.25, 3.27 and 3.29.

**Theorem.** *The following assertions are true:*

- *The space* $\mathrm{Curv}_n$ *is one-dimensional and spanned by the curvature measure* $\mathrm{VOL}_n$ *which is given by* $\mathrm{VOL}_n(K, \beta) = \mathrm{vol}_n(K \cap \beta)$ *for all convex bodies $K \subseteq \mathbb{R}^n$ and all Borel sets $\beta \subseteq \mathbb{R}^n$.*
- *The map $C\left(S^{n-1}\right) \to \mathrm{Curv}_{n-1}$, $f \mapsto \Phi^f$, where*

$$\Phi^f(K, \beta) = \int_{\pi_1^{-1}(\beta)} \pi_2^* f \,\mathrm{d}\,\Theta_{n-1}(K)$$

  *for all $K \in \mathcal{K}$ and all Borel sets $\beta \subseteq \mathbb{R}^n$, is a $\mathbb{C}$-linear isomorphism which preserves parity.*

- *A curvature measure $\Phi \in \mathrm{Curv}$ is simple if and only if*

$$\Phi \in \left(\ker(\mathrm{glob}) \cap \mathrm{Curv}_{n-2}^+\right) \oplus \mathrm{Curv}_{n-1}^- \oplus \mathrm{Curv}_n .$$

It should be mentioned that the first assertion is already contained in [KW99]. However, we give an alternative proof here. Moreover, the third assertion is proved for smooth curvature measures by Solanes and Wannerer in [SW21].

We then switch our point of view a little. Analogous to the space Val, the space Curv also admits an action of the group $\mathrm{GL}(n, \mathbb{R})$. In fact, Curv is a continuous Banach space representation of $\mathrm{GL}(n, \mathbb{R})$ with respect to this action. Therefore, Curv has a natural dense subspace $\mathrm{Curv}^\infty$ of $\mathrm{GL}(n, \mathbb{R})$-smooth vectors. In the setting of valuations, a result due to Alesker states that $\mathrm{Val}^\infty = \mathrm{Val}^{\mathrm{sm}}$. This leads us to the smoothness conjecture, see Conjecture 4.11.

**Conjecture.** $\mathrm{Curv}^{\infty} = \mathrm{Curv}^{\mathrm{sm}}$.

This conjecture in fact decomposes into a family of smaller problems as it is equivalent to $\mathrm{Curv}_i^{\infty} = \mathrm{Curv}_i^{\mathrm{sm}}$ for all $i = 0, \dots, n$. We are able to verify this conjecture in some special cases. The results of Sections 4.2 and 4.3 can be summarized as follows.

**Theorem.** *The following assertions are true:*

- $\mathrm{Curv}_n^{\infty} = \mathrm{Curv}_n^{\mathrm{sm}}$.
- $\mathrm{Curv}_{n-1}^{\infty} = \mathrm{Curv}_{n-1}^{\mathrm{sm}}$.
- *If* $n \leq 2$, *then* $\mathrm{Curv}^{\infty} = \mathrm{Curv}^{\mathrm{sm}}$.

Lastly, Alesker's famous irreducibility theorem states that the subspace $\mathrm{Val}_i^{\varepsilon} \subseteq \mathrm{Val}$ of valuations of degree $i$ and parity $\varepsilon$ is an irreducible $\mathrm{GL}(n, \mathbb{R})$-representation whenever it is non-zero. The same cannot be expected for $\mathrm{Curv}_i^{\varepsilon}$ as we will see in Corollary 4.8. Using the notion of length of Jordan–Hölder series, we complement the irreducibility theorem in the world of curvature measures by the therefore optimal Conjecture 4.27.

**Conjecture.** *For all* $i \in \{0, \dots, n\}$ *and* $\varepsilon \in \{+, -\}$, *the representation* $\mathrm{Curv}_i^{\varepsilon}$ *has length at most two.*

The results of Section 4.4 are concerned with verifying this conjecture in special cases and can be summarized as follows.

**Theorem.** *Let* $i \in \{0, \dots, n\}$ *and* $\varepsilon \in \{+, -\}$. *Moreover, assume that one of the following assumptions is satisfied:* $i = n$, $i = n - 1$ *or* $n \leq 2$. *Then,* $\mathrm{Curv}_i^{\varepsilon}$ *has length at most two.*

## 1.3 Comparison With Related Work

In this section, we want to mention a few more works in connection with curvature measures. They appear both in stochastic and integral geometry.

In the paper [KW99], Kiderlen and Weil prove a decomposition of curvature measures analogous to the McMullen decomposition of valuations. They then use this decomposition to construct *mixed curvature measures* by applying Federer's curvature measures to linear combinations of convex bodies. This construction is a curvature measure version of the classical notion of mixed volumes. Note that they do not use the terminology curvature measure, instead they call maps $\mathcal{K} \to \mathcal{M}$, which satisfy our list of axioms, *translation covariant continuous functionals which are locally defined.*

The work of Kiderlen and Weil has been successfully applied in stochastic geometry. In particular, Weil himself studied Boolean models and Poisson processes using this decomposition theorem, see [Wei15] and [Wei17]. As already mentioned, in the latter paper Weil also proves that the valuation property follows from the remaining axioms. The stochastic applications are also the reason for the additional assumption that $\Phi(\cdot, \beta)$ is measurable.

It should be mentioned that the viewpoint of our work is quite different to that of the three papers [KW99], [Wei15], [Wei17]. Beside the decomposition theorem and the

valuation property, no other structural results are proved in these papers. Following the decomposition theorem, [KW99] concentrates on some explicit constructions, while [Wei15] and [Wei17] are concerned with applications to stochastic geometry. Our goal however is to prove further structural results in analogy to the structure theory of Val.

In a series of papers [HS14], [HS17a], [HS17b], [HS17c], Hug and Schneider study *local tensor valuations*. These can be seen as a generalization of what we call curvature measures as they take values in the space of $\mathbb{T}^p$-valued measures rather than scalar-valued measures. Here $\mathbb{T}^p$ denotes the space of symmetric $p$-tensors of $\mathbb{R}^n$. On polytopes they give a full classification under the assumptions of isometry covariance and local determination. They also address the question which of these admit a weakly continuous continuation to all convex bodies.

Curvature measures also play an important role in the integral geometry of isotropic spaces. At this point it should be mentioned that smooth valuations and curvature measures admit a natural extension to smooth manifolds. An elaborate study of smooth valuations on manifolds can be found in Alesker's series of papers [Ale06a], [Ale06b], [AF08] and [Ale07], whereas smooth curvature measures on manifolds were introduced in [BF11]. A benefit of studying kinematic formulas for curvature measures instead of valuations is that they admit a so called *transfer principal*, see Proposition 2.22 and Theorem 2.23 in [BFS14]. This allows to study kinematic formulas on curved spaces by means of the flat case. For a concise overview on the applications of smooth valuations and curvature measures in integral geometry, the reader may also consult the survey [Ber17] and the references therein.

In his thesis [Sai16], Saienko studies smooth curvature measures from a representation theoretic point of view. One easily sees that $\mathrm{Curv}^{\mathrm{sm}}$ is naturally a representation of $\mathrm{SO}(n)$. Saienko's main result computes the harmonic decomposition of $\mathrm{Curv}^{\mathrm{sm}}$.

Lastly, let us mention the recent work [Knö23] by Knörr in which he studies measure-valued valuations on the space of convex functions $\mathbb{R}^n \to \mathbb{R}$.

## 1.4 Outline of This Work

To conclude this introduction let us provide a brief overview of how this thesis is structured.

In Chapter 2 we recall the relevant background from the theory of valuations. In particular, the space Val is introduced and its structure theory is developed. More precisely, the McMullen decomposition, the topology of Val and the $\mathrm{GL}(n,\mathbb{R})$-action on Val are discussed. We also recall the characterization theorems due to Hadwiger and McMullen on homogeneous valuations of degree $n$ and $n-1$, respectively. Moreover, the characterization of simple valuations due to Klain and Schneider is explained. Lastly, we introduce smooth valuations and smooth curvature measures.

In Chapter 3 we introduce the central notion of this thesis, namely continuous translation-invariant curvature measures. After this we prove analogues for many of the structure results from the setting of valuations. In analogy to the McMullen decomposition we discuss the Kiderlen–Weil decomposition. Moreover, we equip Curv with the aforementioned Banach space topology. We then study the two main technical tools,

namely the globalization and the so called Bernig embedding in Section 3.5. Using these tools, we prove analogues for the aforementioned characterization theorems by Hadwiger, McMullen and Klain–Schneider.

In Chapter 4 we switch our focus towards smooth curvature measures. To do so, we introduce the $\mathrm{GL}(n,\mathbb{R})$-action on Curv and obtain the subspace $\mathrm{Curv}^\infty$ of smooth vectors. We then study Curv from a representation theoretic point of view with the following two goals in mind. We prove the conjecture $\mathrm{Curv}^\infty = \mathrm{Curv}^{\mathrm{sm}}$ in some special cases and we compute the length of the subrepresentations $\mathrm{Curv}_i^\varepsilon$ in some special cases.

# 2 Background on Valuation Theory

In this chapter, we introduce the notion of valuations on convex bodies and discuss some of the most important results in valuation theory. The class of continuous translation-invariant valuations will be of particular importance to us. Moreover, this chapter serves to fix some notation which will be used throughout the thesis. As a general reference we recommend [Sch14], in particular Chapter 6.

## 2.1 Convex Bodies and Valuations

Throughout this thesis, let $n$ denote a positive integer which is reserved for the dimension of the Euclidean space $\mathbb{R}^n$ we are working in. We write $\langle \cdot, \cdot \rangle$ for the standard inner product on $\mathbb{R}^n$, $\|\cdot\|$ for the induced norm and $S^{n-1} = \left\{ x \in \mathbb{R}^n \mid \|x\| = 1 \right\}$ for the unit sphere in $\mathbb{R}^n$. Moreover, we always consider $\mathbb{R}^n$ to be equipped with the standard topology induced by $\|\cdot\|$.

**Definition 2.1.** A set $K \subseteq \mathbb{R}^n$ is called a *convex body* if it is non-empty, compact and convex. The *dimension* $\dim(K)$ of a convex body $K \subseteq \mathbb{R}^n$ is defined as the dimension of its affine hull.

We write $\mathcal{K}$ for the set of convex bodies in $\mathbb{R}^n$. More generally, for a given subset $X \subseteq \mathbb{R}^n$ we write $\mathcal{K}(X)$ for the set of all convex bodies $K \in \mathcal{K}$ which satisfy $K \subseteq X$.

**Examples 2.2.**

- The *unit ball* $B^n = \left\{ x \in \mathbb{R}^n \mid \|x\| \leq 1 \right\}$ is a convex body.
- For any finite set of points $\{x_1, \ldots, x_k\} \subseteq \mathbb{R}^n$, the convex hull

$$P = \operatorname{conv}\{x_1, \ldots, x_k\} = \left\{ \sum_{i=1}^{k} \lambda_i x_i \;\middle|\; \lambda_i \geq 0 \text{ for all } i \text{ and } \sum_{i=1}^{k} \lambda_i = 1 \right\}$$

  is a convex body. Such a convex body is called a *polytope*.
- For two distinct points $x, y \in \mathbb{R}^n$, the convex body $[x, y] = \operatorname{conv}\{x, y\}$ is called a *line segment*.

The set $\mathcal{K}$ of convex bodies comes equipped with two natural operations. These are *Minkowski addition* $\mathcal{K} \times \mathcal{K} \to \mathcal{K}$, which is defined by

$$K + L = \left\{ x + y \mid x \in K,\, y \in L \right\},$$

and *scaling* $\mathbb{R} \times \mathcal{K} \to \mathcal{K}$, which is defined by

$$\lambda K = \left\{ \lambda x \mid x \in K \right\}.$$

If $K \in \mathcal{K}$ and $x \in \mathbb{R}^n$, we write $K + x = K + \{x\}$ and $-K = (-1)K$ for simplicity of notation. Note that Minkowski addition is commutative and associative. Moreover, $K + 0 = K$ for all convex bodies $K \subseteq \mathbb{R}^n$. In other words, $\mathcal{K}$ is a commutative monoid with respect to Minkowski addition.

As an important side remark, note that Minkowski addition and scaling are well-defined for all subsets of $\mathbb{R}^n$. We will later apply these operations to more general sets, too.

Using this, one can define a metric on $\mathcal{K}$ as follows. For any convex body $K \subseteq \mathbb{R}^n$ and any $\varepsilon > 0$, we call the convex body $K_\varepsilon = K + \varepsilon B^n \in \mathcal{K}$ the *$\varepsilon$-neighborhood* of $K$. It is readily verified that

$$d(K, L) = \inf\left\{\varepsilon > 0 \mid K \subseteq L_\varepsilon,\, L \subseteq K_\varepsilon\right\},\ K, L \in \mathcal{K},$$

defines a metric on $\mathcal{K}$ which is called the *Hausdorff metric*. The following useful property of the metric space $(\mathcal{K}, d)$ was proved by Blaschke in [Bla16]. For a modern proof, see Theorem 1.8.7 in [Sch14].

**Proposition 2.3 (Blaschke Selection Theorem).** *Every bounded sequence of convex bodies in $\mathcal{K}$ contains a subsequence which converges in $\mathcal{K}$.*

Let $K \subseteq \mathbb{R}^n$ be a convex body. Then, the function

$$h_K\colon \mathbb{R}^n \to \mathbb{R},\ \xi \mapsto \sup\left\{\langle \xi, x\rangle \mid x \in K\right\}$$

is called the *support function* of $K$. The support function is one of the most important tools for describing convex bodies. This is demonstrated by the following proposition which collects some of its main properties. For proofs we refer the reader to Section 1.7.1 of [Sch14].

**Proposition 2.4.** *Let $K, L \subseteq \mathbb{R}^n$ be convex bodies.*

- *$h_K$ is sublinear which means that*

  $$h_K(x + y) \leq h_K(x) + h_K(y),$$
  $$h_K(\lambda x) = \lambda h_K(y)$$

  *for all $x, y \in \mathbb{R}^n$ and all $\lambda \geq 0$.*

- *$h_K$ is continuous and convex.*

- *$h_{K+L} = h_K + h_L$.*

- *For all $\lambda \geq 0$, $h_{\lambda K} = \lambda h_K$.*

- *Let $\|\cdot\|_\infty$ denote the uniform norm on the unit sphere $S^{n-1} \subseteq \mathbb{R}^n$. Then,*

  $$d(K, L) = \left\|(h_K - h_L)\,|_{S^{n-1}}\right\|_\infty.$$

- *For any sublinear function $h\colon \mathbb{R}^n \to \mathbb{R}$, there exists a convex body $M \subseteq \mathbb{R}^n$ such that $h = h_M$.*

Note that for two convex bodies $K, L \in \mathcal{K}$ which satisfy $K \cup L \in \mathcal{K}$, the intersection $K \cap L$ is necessarily non-empty. Since $K \cap L$ is also convex and compact, it follows that $K \cap L \in \mathcal{K}$. Hence, we can define a valuation as follows.

**Definition 2.5.** Let $S$ be a commutative semigroup. A map $\varphi\colon \mathcal{K} \to S$ is called a *valuation* if it satisfies

$$\varphi(K \cup L) + \varphi(K \cap L) = \varphi(K) + \varphi(L)$$

whenever $K, L, K \cup L \in \mathcal{K}$.

The concept of a valuation dates back at least to the beginning of the twentieth century when Dehn solved Hilbert's third problem. A simplified version of the problem asks whether any two three-dimensional polytopes of equal volume are scissors-congruent. Shortly after Hilbert posed the problem, Dehn was able to answer it negatively, see [Deh01]. In doing so, he constructed a certain valuation on three-dimensional polytopes, now known as the *Dehn invariant*, which is invariant under scissors-congruence. The Dehn invariant of a cube is always equal to zero, while regular tetrahedra have non-zero Dehn invariant. Hence, a cube and a regular tetrahedron of the same volume cannot be scissors-congruent.

Various choices of the codomain $S$ have attracted attention in convex geometry. The most elaborate theory exists for scalar-valued valuations, that is $S = \mathbb{C}$. Many results of this theory remain true if you let $S = \mathbb{C}^m$ be a complex vector space of finite dimension. In this thesis, we will extensively study curvature measures which are valuations with values in the infinite dimensional vector space of complex Borel measures on $\mathbb{R}^n$. Another important class are so called *Minkowski valuations* $\mathcal{K} \to \mathcal{K}$, which were introduced by Ludwig in [Lud05].

Readers looking for a nice overview on valuation theory with a focus on classification results are referred to the recent survey article [Lud21] and the references therein.

For the remainder of this chapter, we want to concentrate on scalar-valued valuations. A valuation $\varphi\colon \mathcal{K} \to \mathbb{C}$ is called *translation-invariant* if it satisfies $\varphi(K + x) = \varphi(K)$ for all $K \in \mathcal{K}$ and $x \in \mathbb{R}^n$. Moreover, we call $\varphi$ *continuous* if it is continuous with respect to the Hausdorff metric on $\mathcal{K}$ and the standard topology on $\mathbb{C}$. We write Val for the set of all continuous translation-invariant valuations $\varphi\colon \mathcal{K} \to \mathbb{C}$. As we will see in the following sections, Val admits a rich structure theory.

**Examples 2.6.** The following are valuations:

- For any commutative semigroup $S$, constant maps $c\colon \mathcal{K} \to S$ are translation-invariant valuations. If $S = \mathbb{C}$, we have $c \in \mathrm{Val}$.
- We define the *volume* $\mathrm{vol}_n$ as the unique Lebesgue measure on $\mathbb{R}^n$ which satisfies $\mathrm{vol}_n\left([0,1]^n\right) = 1$. By abuse of notation, we also write $\mathrm{vol}_n$ for the restriction of the volume to the space $\mathcal{K}$ of convex bodies. Then, we have $\mathrm{vol}_n \in \mathrm{Val}$. For a proof that $\mathrm{vol}_n$ is continuous, see Theorem 1.8.20 in [Sch14].
- The *lattice point enumerator* (for the lattice $\mathbb{Z}^n \subseteq \mathbb{R}^n$)

  $$\mathcal{K} \to \mathbb{C},\ K \mapsto \#(K \cap \mathbb{Z}^n)$$

  is a valuation which is neither translation-invariant nor continuous. It is however invariant with respect to translations by the smaller group $\mathbb{Z}^n \subseteq \mathbb{R}^n$.

- The assignment $K \mapsto h_K$ is a valuation with values in the infinite dimensional real vector space $C(\mathbb{R}^n)$.

## 2.2 McMullen Decomposition

In this section, we start developing the structure theory of Val. One easily verifies that Val forms a complex vector space with respect to pointwise addition and pointwise scalar multiplication.

The first question one might ask is what the dimension of the space Val is. It turns out that

$$\dim(\mathrm{Val}) = \begin{cases} 2, & \text{if } n = 1, \\ \infty, & \text{if } n \geq 2. \end{cases}$$

We will obtain both these dimension formulas at later points of this thesis as byproducts of more general results, see discussion after Theorem 2.11.

For any real number $d$, a valuation $\varphi \in \mathrm{Val}$ is called *$d$-homogeneous* or *of degree $d$* if it satisfies

$$\varphi(\lambda K) = \lambda^d \varphi(K)$$

for all $\lambda > 0$ and $K \in \mathcal{K}$. We write $\mathrm{Val}_d \subseteq \mathrm{Val}$ for the linear subspace of $d$-homogeneous valuations.

As is well-known, the volume is $n$-homogeneous and obviously constant maps $\mathcal{K} \to \mathbb{C}$ are homogeneous of degree zero. However, there are plenty of homogeneous valuations of other degrees as the following example of intrinsic volumes shows. For a thorough discussion, see Section 6.4 of [Gru07].

**Example 2.7 (Intrinsic Volumes).** Let $K \subseteq \mathbb{R}^n$ be a convex body. Then, the map $\varepsilon \mapsto \mathrm{vol}_n(K_\varepsilon)$ is a polynomial of degree $n$ for positive $\varepsilon$. Hence, there exist coefficients $\mu_0(K), \ldots, \mu_n(K) \in \mathbb{C}$ such that

$$\mathrm{vol}_n(K_\varepsilon) = \sum_{i=0}^{n} \omega_i \mu_{n-i}(K)\, \varepsilon^i,$$

where we use the normalizing constant $\omega_i = \mathrm{vol}_i\left(B^i\right)$. This equation is known as the *Steiner formula*. It is readily verified that $\mu_i \colon K \mapsto \mu_i(K)$ is a continuous translation-invariant valuation. Moreover, one easily sees that $\mu_i$ is homogeneous of degree $i$ for all $i = 0, \ldots, n$. In other words, $\mu_i \in \mathrm{Val}_i$. A proof of this can be found in Theorem 6.13 of [Gru07]. Note that plugging in $K = \{0\}$ implies $\mu_0 = 1$ and that taking the limit $\varepsilon \to 0$ yields $\mu_n = \mathrm{vol}_n$.

As a side remark, let us mention that the importance of intrinsic volumes is nicely illustrated by the following theorem, which was proved by Hadwiger in Section 6.1.10 of [Had57].

**Theorem 2.8 (Hadwiger).** *Let $\varphi \in \mathrm{Val}$ be $\mathrm{SO}(n)$-invariant, that is $\varphi(gK) = \varphi(K)$ for all $g \in \mathrm{SO}(n)$. Then, there exist $\lambda_0, \ldots, \lambda_n \in \mathbb{C}$ such that*

$$\varphi = \sum_{i=0}^{n} \lambda_i \mu_i.$$

We have now seen that for all $i \in \{0, \ldots, n\}$, there exist $i$-homogeneous valuations. The following result by McMullen, proved in [McM77], states that there are no homogeneous valuations of other degrees. Moreover, it yields that general valuations can be broken down into simpler homogeneous building blocks.

**Theorem 2.9 (McMullen Decomposition).**

$$\mathrm{Val} = \bigoplus_{i=0}^{n} \mathrm{Val}_i \,.$$

A valuation $\varphi \in \mathrm{Val}$ is called *even* if it satisfies $\varphi(-K) = \varphi(K)$ for all $K \in \mathcal{K}$. Similarly, $\varphi$ is called *odd* if $\varphi(-K) = -\varphi(K)$ for all $K$. We write $\mathrm{Val}^+ \subseteq \mathrm{Val}$ for the subspace of even valuations and $\mathrm{Val}^- \subseteq \mathrm{Val}$ for the subspace of odd valuations.

For any $\varphi \in \mathrm{Val}$, we can define valuations $\varphi^+$ and $\varphi^-$ by

$$\varphi^+(K) = \frac{1}{2}\left(\varphi(K) + \varphi(-K)\right),$$
$$\varphi^-(K) = \frac{1}{2}\left(\varphi(K) - \varphi(-K)\right).$$

One easily sees that $\varphi^+ \in \mathrm{Val}^+$, $\varphi^- \in \mathrm{Val}^-$ and $\varphi = \varphi^+ + \varphi^-$. This construction yields a direct sum decomposition

$$\mathrm{Val} = \mathrm{Val}^+ \oplus \mathrm{Val}^- \,.$$

Moreover, note that for a homogeneous valuation $\varphi$ of degree $i$, the resulting valuations $\varphi^+$ and $\varphi^-$ are homogeneous of degree $i$, too. Hence, we obtain an even finer decomposition, namely

$$\mathrm{Val} = \bigoplus_{i=0}^{n} \mathrm{Val}_i^+ \oplus \mathrm{Val}_i^- ,$$

where we write $\mathrm{Val}_i^\varepsilon = \mathrm{Val}_i \cap \mathrm{Val}^\varepsilon$ for all $i \in \{0, \ldots, n\}$ and $\varepsilon \in \{+, -\}$.

## 2.3 Characterization Theorems for Valuations

The McMullen decomposition morally tells us that it is satisfactory to understand homogeneous valuations. Hence, the main goal of this section is to give explicit descriptions of the spaces $\mathrm{Val}_i^\varepsilon$. However, this is only possible in certain special cases.

The easiest case to settle is the case of 0-homogeneous valuations. Note that $\lambda \to 0$

implies $\lambda K \to \{0\}$ in the Hausdorff metric for all convex bodies $K \subseteq \mathbb{R}^n$. Moreover, a valuation $\varphi \in \mathrm{Val}_0$ satisfies

$$\varphi(K) = \lambda^0 \varphi(K) = \varphi(\lambda K)$$

for all $\lambda > 0$ and $K \in \mathcal{K}$. Using continuity of $\varphi$ and letting $\lambda \to 0$, it follows that $\varphi(K) = \varphi\big(\{0\}\big)$ for all convex bodies $K \subseteq \mathbb{R}^n$. This shows that $\varphi$ is constant. Conversely, any constant map $\mathcal{K} \to \mathbb{C}$ is an even valuation of degree zero. Hence, we have proved the following result.

**Proposition 2.10.** $\mathrm{Val}_0 = \mathrm{Val}_0^+$ *is the one-dimensional space of constant maps* $\mathcal{K} \to \mathbb{C}$ *and* $\mathrm{Val}_0^- = \{0\}$.

The case $\mathrm{Val}_n$ of $n$-homogeneous valuations is more involved, but it is completely solved by a far-reaching result due to Hadwiger. In Satz XIV of [Had57] he shows that all translation-invariant and $n$-homogeneous valuations on polytopes are of the form $c \operatorname{vol}_n$ for some $c \in \mathbb{C}$. Since the set of all polytopes is dense in $\mathcal{K}$, see Theorem 1.8.16 in [Sch14], we obtain the following result.

**Theorem 2.11 (Hadwiger).** $\mathrm{Val}_n = \mathrm{Val}_n^+ = \mathbb{C} \operatorname{vol}_n$ *and* $\mathrm{Val}_n^- = \{0\}$.

In particular, we see that $\mathrm{Val} = \operatorname{span}\{1, \mathrm{vol}_1\}$ if $n = 1$ which implies the first part of the aforementioned dimension formula.

For the following discussion and at later points of this thesis, the following conventions and notations will be useful. For any topological space $X$, we write $C(X)$ for the complex vector space of continuous functions $X \to \mathbb{C}$; this is a Banach space with respect to the uniform norm whenever $X$ is compact. If $X \subseteq \mathbb{C}^m$ for some $m$, we write $L(X) \subseteq C(X)$ for the subspace of restrictions of $\mathbb{C}$-linear functions, that is

$$L(X) = \big\{ f \colon X \to \mathbb{C} \,\big|\, \text{There exists } F \in (\mathbb{C}^m)^* \text{ such that } F|_X = f \big\}.$$

The elements of $L(X)$ will be called *linear functions* on $X$.

Moreover, we always consider $\mathbb{R}^m \subseteq \mathbb{C}^m$ via the natural embedding. In particular, the space $L(Y)$ is well-defined for any subset $Y \subseteq \mathbb{R}^m$.

The last resolved case is the case $\mathrm{Val}_{n-1}$ of homogeneous valuations of degree $n-1$. The abstract result is easily formulated, namely

$$\mathrm{Val}_{n-1} \cong C\left(S^{n-1}\right) \Big/ L\left(S^{n-1}\right).$$

This particularly implies that $\dim(\mathrm{Val}) = \infty$ if $n \geq 2$, since the subspace $\mathrm{Val}_{n-1} \subseteq \mathrm{Val}$ is already infinite dimensional. However, writing down this isomorphism requires some more terminology.

Specifically, we need the so called surface area measure associated with a convex body, for details we refer the reader to Section 4.2 of [Sch14]. Let $P \subseteq \mathbb{R}^n$ be a polytope with non-empty interior. For any $(n-1)$-face of $P$, we denote the unique unit outer normal

vector of $F$ by $N(F) \in S^{n-1}$. Then, we define a Borel measure $S_{n-1}(P, \cdot)$ on the unit sphere by

$$S_{n-1}(P, \omega) = \sum_F \operatorname{vol}_{n-1}(F)\, \delta_{N(F)}(\omega)$$

for all Borel sets $\omega \subseteq S^{n-1}$, where the sum extends over all $(n-1)$-faces $F$ of $P$. This construction can be extended to general convex bodies as follows, see e.g. Theorem 3.2.1 of [Ale18].

**Lemma 2.12.** *Let $K \subseteq \mathbb{R}^n$ be a convex body. Then, there exists a unique non-negative Borel measure $S_{n-1}(K, \cdot)$ on the unit sphere $S^{n-1}$ which is characterized by the following properties:*

- *If $K = P$ is a polytope with non-empty interior, then $S_{n-1}(P, \cdot)$ is defined as above.*
- *Suppose that $(K_m)_{m\in\mathbb{N}} \subseteq \mathcal{K}$ is a convergent sequence satisfying $\lim_{m\to\infty} K_m = K$. Then, the sequence $\left(S_{n-1}(K_m, \cdot)\right)_{m\in\mathbb{N}}$ converges weakly to $S_{n-1}(K, \cdot)$, that is*

$$\lim_{m\to\infty} \int_{S^{n-1}} f(x)\, \mathrm{d}\, S_{n-1}(K_m, x) = \int_{S^{n-1}} f(x)\, \mathrm{d}\, S_{n-1}(K, x)$$

  *for all $f \in C\left(S^{n-1}\right)$.*

For a given convex body $K \subseteq \mathbb{R}^n$, the measure $S_{n-1}(K, \cdot)$ is called the *surface area measure* of $K$. Note that $S_{n-1}(K, \cdot)$ is particularly easy if $\dim(K) \neq n$. If $\dim(K) = n-1$, we have

$$S_{n-1}(K, \omega) = \operatorname{vol}_{n-1}(K) \left(\delta_N(\omega) + \delta_{-N}(\omega)\right),$$

where $N, -N \in S^{n-1}$ denote the two different unit normal vectors to the hyperplane $H = \operatorname{aff}(K) \subseteq \mathbb{R}^n$. For $\dim(K) \leq n-2$, we simply have $S_{n-1}(K, \cdot) = 0$. In particular, we have explicit formulas to compute $S_{n-1}(P, \cdot)$ for all polytopes $P \subseteq \mathbb{R}^n$. Moreover, the last assertion of Theorem 4.2.1 and equation (4.11) in [Sch14] yield that

$$S_{n-1}(K \cup L, \cdot) + S_{n-1}(K \cap L, \cdot) = S_{n-1}(K, \cdot) + S_{n-1}(L, \cdot)$$

whenever $K, L, K \cup L \in \mathcal{K}$.

Fix a function $f \in C\left(S^{n-1}\right)$ for the moment. The equation above implies that $f$ induces a valuation $\varphi_f \colon \mathcal{K} \to \mathbb{C}$ given by

$$\varphi_f(K) = \int_{S^{n-1}} f(x)\, \mathrm{d}\, S_{n-1}(K, x)$$

for all $K \in \mathcal{K}$. From the construction of the surface area measure and the assumption that $f$ is continuous one easily sees that $\varphi_f$ is continuous. Furthermore, one quickly verifies that $\varphi_f$ is translation-invariant. To do so, use the explicit formulas for $S_{n-1}(P, \cdot)$ for polytopes $P$ to show that $\varphi_f$ is translation-invariant on polytopes; the general case then follows by approximation. Lastly, equation (4.17) of [Sch14] yields that $\varphi_f$ is homogeneous of degree $n-1$. Hence, we have shown that $\varphi_f \in \operatorname{Val}_{n-1}$.

The following characterization of $(n-1)$-homogeneous valuations was proved by McMullen in [McM80].

**Theorem 2.13 (McMullen).** *The map*

$$C\left(S^{n-1}\right) \to \mathrm{Val}_{n-1},\ f \mapsto \varphi_f$$

*is $\mathbb{C}$-linear, surjective and has kernel $L\left(S^{n-1}\right)$. Moreover, $\varphi_f$ is odd if and only if $f$ is odd, and $\varphi_f$ is even if and only if*

$$f = f^+ + l$$

*for some $f^+ \in C^+\left(S^{n-1}\right)$ and $l \in L\left(S^{n-1}\right)$.*

We now have an explicit characterization for the valuations of degree $i = 0, n-1, n$. Using McMullen's decomposition, we therefore obtain a formula for all valuations on the plane $\mathbb{R}^2$.

**Corollary 2.14.** *Let $n = 2$. A map $\varphi\colon \mathcal{K} \to \mathbb{C}$ satisfies $\varphi \in \mathrm{Val}$ if and only if there exist constants $c_1, c_2 \in \mathbb{C}$ and a map $f \in C\left(S^1\right)$ such that*

$$\varphi = c_1 + \varphi_f + c_2 \,\mathrm{vol}_2\,.$$

*Moreover, $\varphi$ is even if and only if $\varphi_f$ is even, and $\varphi$ is odd if and only if $c_1 = c_2 = 0$ and $\varphi_f$ is odd.*

Lastly, we want to discuss a classification result of slightly different nature. Instead of studying valuations of a certain degree, we consider so called simple valuations.

**Definition 2.15.** Let $V$ be a complex vector space. A valuation $\varphi\colon \mathcal{K} \to V$ is called *simple* if $\varphi(K) = 0 \in V$ for all convex bodies $K \subseteq \mathbb{R}^n$ satisfying $\dim(K) < n$.

Of course, we have that $\mathrm{vol}_n$ is a simple valuation. Conversely, it follows immediately from Klain's work, namely Theorem 3.4 of [Kla95], that any simple even valuation $\varphi^+ \in \mathrm{Val}^+$ is necessarily of the form $\varphi^+ = c\,\mathrm{vol}_n$ for some $c \in \mathbb{C}$.

Next, let $f\colon S^{n-1} \to \mathbb{C}$ be an odd continuous function and let $\varphi_f \in \mathrm{Val}$ be defined as before. If $\dim(K) \leq n-2$, we have $S_{n-1}(K, \cdot) = 0$ and therefore also $\varphi_f(K) = 0$. If $\dim(K) = n-1$, let $N$ and $-N$ denote the two different unit normal vectors to $H = \mathrm{aff}(K)$. Then, we can compute

$$\varphi_f(K) = \int_{S^{n-1}} f \,\mathrm{d}\,\mathrm{vol}_{n-1}(K)\,(\delta_N + \delta_{-N}) = \mathrm{vol}_{n-1}(K)\left(f(N) + f(-N)\right) = 0,$$

where the last equality follows from the assumption that $f$ is odd. Hence, we have seen that $\varphi_f$ is simple. Conversely, Schneider proved in [Sch96] that any simple odd valuation $\varphi^- \in \mathrm{Val}^-$ is already of this form. Therefore, we obtain the following characterization of simple valuations.

**Theorem 2.16 (Klain–Schneider).** *A valuation $\varphi \in \mathrm{Val}$ is simple if and only if there exist $c \in \mathbb{C}$ and an odd function $f \in C\left(S^{n-1}\right)$ such that*

$$\varphi = c\,\mathrm{vol}_n + \varphi_f.$$

## 2.4 Topology and General Linear Group Action

In this section, we discuss two more useful tools to study the space Val. Firstly, Val can be equipped with a whole family of Banach space norms. While the norms do depend on a certain choice, it turns out that the induced topology does not. Moreover, the real general linear group $\mathrm{GL}(n,\mathbb{R})$ naturally acts on the space Val. The study of this action culminated in Alesker's work, see [Ale00] and [Ale01], which in particular led to a resolution of a conjecture of McMullen.

Let $B \subseteq \mathbb{R}^n$ be a bounded set with non-empty interior. For any valuation $\varphi \in \mathrm{Val}$, we set

$$\|\varphi\|_B = \sup_{K \subseteq B} \left|\varphi(K)\right|.$$

One easily verifies that this defines a norm $\|\cdot\|_B$ on Val. The next theorem follows from the Blaschke selection theorem and the McMullen decomposition. A proof can be found in [Ale18].

**Theorem 2.17.** *For a bounded subset $B \subseteq \mathbb{R}^n$ with non-empty interior, the space* Val *equipped with the norm $\|\cdot\|_B$ is a Banach space. Moreover, the norms $\|\cdot\|_B$ and $\|\cdot\|_{B'}$ are equivalent for any two bounded subsets $B, B' \subseteq \mathbb{R}^n$ with non-empty interior.*

In particular, we have equipped Val with a unique topology, independently of the choice of the set $B$.

As an important remark, note that for any $i \in \{0,\dots,n\}$ and any $\varepsilon \in \{+,-\}$ the subspace $\mathrm{Val}_i^\varepsilon$ is closed in Val, as one easily sees that parity and degree are stable under limits with respect to the norms defined above. In particular, $\mathrm{Val}_i^\varepsilon$ is itself a Banach space.

We let the general linear group $\mathrm{GL}(n,\mathbb{R})$ act on the space Val as follows. For any invertible linear transformation $g \in \mathrm{GL}(n,\mathbb{R})$ and any valuation $\varphi \in \mathrm{Val}$, we define a map $g\varphi\colon \mathcal{K} \to \mathbb{C}$ by

$$(g\varphi)(K) = \varphi\left(g^{-1}K\right)$$

for all convex bodies $K \subseteq \mathbb{R}^n$. One easily verifies that $g\varphi \in \mathrm{Val}$ and that the induced map $\mathrm{GL}(n,\mathbb{R}) \times \mathrm{Val} \to \mathrm{Val}$, $(g,\varphi) \mapsto g\varphi$ is a linear and continuous group action. In other words, Val is a continuous representation of the Lie group $\mathrm{GL}(n,\mathbb{R})$. Since this group action preserves parity and degree of a valuation, it follows that $\mathrm{Val}_i^\varepsilon \subseteq \mathrm{Val}$ is a continuous subrepresentation for all $i$ and $\varepsilon$.

Recall that a continuous representation $V$ of a Lie group $G$ is called *irreducible* if $V \neq \{0\}$ and the only $G$-invariant closed subspaces of $V$ are $\{0\}$ and $V$. In [Ale01], Alesker proved the following remarkable theorem.

**Theorem 2.18 (Irreducibility Theorem).** *For all $i = 1,\dots,n-1$ and $\varepsilon \in \{+,-\}$, the representation $\mathrm{Val}_i^\varepsilon$ of $\mathrm{GL}(n,\mathbb{R})$ is irreducible.*

Note that the same question for the cases $i = 0$ and $i = n$ is answered by Proposition 2.10 and Hadwiger's theorem (Theorem 2.11), respectively. To be more precise, $\mathrm{Val}_0^+$ and

$\mathrm{Val}_n^+$ are irreducible as they are one-dimensional. However, $\mathrm{Val}_0^- = \{0\} = \mathrm{Val}_n^-$ which is not irreducible (by common convention).

It is no exaggeration to say that modern valuation theory is built on the irreducibility theorem. At the time, one of the motivations for Alesker behind the irreducibility theorem was that it implies a conjecture of McMullen. One easily sees that for a fixed convex body $A \subseteq \mathbb{R}^n$ and a fixed valuation $\varphi \in \mathrm{Val}$, the map

$$\varphi_A \colon \mathcal{K} \to \mathbb{C},\ K \mapsto \varphi(K + A)$$

also satisfies $\varphi_A \in \mathrm{Val}$. The following corollary follows from Alesker's irreducibility theorem.

**Corollary 2.19.** *The linear span of*

$$\left\{ K \mapsto \mathrm{vol}_n(K + A) \,\middle|\, A \in \mathcal{K} \right\}$$

*is dense in* Val.

In [McM80], McMullen conjectured that the linear span of so called *mixed volumes* forms a dense subspace of Val. These are functionals of the form

$$\mathcal{K} \to \mathbb{C},\ K \mapsto V\left(K[i], A_1, \ldots, A_{n-i}\right)$$

with fixed reference bodies $A_1, \ldots, A_{n-i} \in \mathcal{K}$, where the notation $K[i]$ means that the argument $K$ is repeated $i$ times. For a construction of mixed volumes, see Theorem 5.1.7 of [Sch14]. Since

$$\mathrm{vol}_n(K + A) = \sum_{i=0}^{n} V\left(K[i], A[n-i]\right)$$

for all $A, K \in \mathcal{K}$, the corollary above implies McMullen's conjecture.

In the proof of the irreducibility theorem, embedding theorems for the spaces $\mathrm{Val}_i^\varepsilon$ play a crucial role. As a similar embedding will be important in our study of curvature measures on the plane later on, we conclude this section with the construction of the so called *Klain embedding*.

For this discussion, fix some $i \in \{0, \ldots, n\}$ and $\varphi \in \mathrm{Val}_i^+$. We write $\mathrm{Gr}_i(\mathbb{R}^n)$ for the Grassmannian of $i$-dimensional linear subspaces of $\mathbb{R}^n$. Hadwiger's characterization of valuations of top degree implies that for all $E \in \mathrm{Gr}_i(\mathbb{R}^n)$ there exists a constant $c = c(\varphi, E) \in \mathbb{C}$, depending on $\varphi$ and $E$, such that

$$\varphi(K) = c\,\mathrm{vol}_i(K)$$

for all $K \in \mathcal{K}(E)$. It easily follows from the continuity of the valuation $\varphi$, that the so called *Klain function*

$$\mathrm{Kl}_\varphi \colon \mathrm{Gr}_i(\mathbb{R}^n) \to \mathbb{C},\ E \mapsto c(\varphi, E)$$

of $\varphi$ is also continuous. Building on his earlier work on simple even valuations, Klain constructed the following useful embedding in Theorem 3.1 of [Kla00].

**Theorem 2.20.** *For all $i = 0, \ldots, n$, the map*

$$\mathrm{Kl} \colon \mathrm{Val}_i^+ \to C\left(\mathrm{Gr}_i(\mathbb{R}^n)\right),\ \varphi \mapsto \mathrm{Kl}_\varphi$$

*is $\mathbb{C}$-linear, continuous and injective.*

## 2.5 Smooth Valuations and Curvature Measures

In this section, we want to discuss one of the most successful applications of the irreducibility theorem, namely the introduction of smooth valuations due to Alesker which first appear in [Ale04] and have been extensively studied since. As was observed by Bernig and Fu in [BF11], smooth valuations can be localized in some precise sense, thus yielding the more general notion of smooth curvature measures.

We first need to recall some general constructions from representation theory. To do so, let $G$ be a Lie group and $\rho\colon G \to \mathrm{GL}(V)$ a continuous representation on a Fréchet space $V$. A vector $v \in V$ is called *smooth* if the map

$$G \to V,\ g \mapsto \rho(g)(v)$$

is smooth. We write $V^{\mathrm{sm}} \subseteq V$ for the subset of smooth vectors. It is well-known that the space $V^{\mathrm{sm}}$ is a $G$-stable and dense linear subspace of $V$. In particular, $V^{\mathrm{sm}} = V$ if $V$ is finite dimensional. However, we are interested in the infinite dimensional case, where in general $V^{\mathrm{sm}} \subsetneq V$. Moreover, the space $V^{\mathrm{sm}}$ is naturally endowed with a Fréchet space structure, for which the topology is finer than the subspace topology induced from $V$. With respect to this topology, the induced representation $\rho\colon G \to \mathrm{GL}(V^{\mathrm{sm}})$ is continuous. The reader not familiar with these constructions is referred to the monograph [Wal88], in particular Section 1.6.

**Definition 2.21.** A valuation $\varphi \in \mathrm{Val}$ is called *smooth* if it is a smooth vector of the representation

$$\mathrm{GL}(n,\mathbb{R}) \to \mathrm{GL}(\mathrm{Val}),\ g \mapsto (\varphi \mapsto g\varphi).$$

As in the general discussion before, the space $\mathrm{Val}^{\mathrm{sm}}$ forms a dense subspace of Val. Moreover, note that the McMullen decomposition is compatible with the space of smooth valuations as the spaces $\mathrm{Val}_i^{\varepsilon}$ are $\mathrm{GL}(n,\mathbb{R})$-stable. More precisely, we have a decomposition

$$\mathrm{Val}^{\mathrm{sm}} = \bigoplus_{i=0}^{n} \mathrm{Val}_i^{+,\mathrm{sm}} \oplus \mathrm{Val}_i^{-,\mathrm{sm}}$$

of the space of smooth translation-invariant valuations.

**Examples 2.22.** Using the characterization results from Section 2.3, we can discuss some examples.

- Let $\varphi \in \mathrm{Val}_0$ be a constant valuation. Then, the map $\mathrm{GL}(n,\mathbb{R}) \to \mathrm{Val}$, $g \mapsto g\varphi$ is constant and therefore smooth, as $g\varphi = \varphi \in \mathrm{Val}$ for all $g \in \mathrm{GL}(n,\mathbb{R})$. Thus, $\varphi$ is smooth.

- Note that the space $\mathrm{Val}_n^{\mathrm{sm}}$ of smooth valuations of degree $n$ forms a dense subspace of $\mathrm{Val}_n$ as it is the subspace of smooth vectors. Since $\mathrm{Val}_n$ has finite dimension, it follows that $\mathrm{Val}_n^{\mathrm{sm}} = \mathrm{Val}_n$. Hence, the valuation $\varphi = c\,\mathrm{vol}_n$ is smooth for all $c \in \mathbb{C}$.

- Let $f\colon S^{n-1} \to \mathbb{C}$ be a continuous function and recall that

$$\varphi_f\colon K \mapsto \int_{S^{n-1}} f(x)\,\mathrm{d}\,S_{n-1}(K,x)$$

  defines a valuation $\varphi_f \in \mathrm{Val}_{n-1}$. Using a more sophisticated $\mathrm{GL}(n,\mathbb{R})$-equivariant description of $\varphi_f$, one can show that $\varphi_f$ is smooth if and only if $f \in C^\infty\left(S^{n-1}\right)$. For details on this, see [Ale18].

Note that the last example in particular yields that, provided $n \geq 2$, $\mathrm{Val}^{\mathrm{sm}}$ is a proper subspace of Val.

Remarkably, smooth valuations admit a fairly explicit geometric description. We write $S\mathbb{R}^n = \mathbb{R}^n \times S^{n-1}$ for the *unit sphere bundle* of $\mathbb{R}^n$. For a given convex body $K \subseteq \mathbb{R}^n$, the set

$$\mathrm{nc}(K) = \left\{(x,u) \in S\mathbb{R}^n \mid x \in K,\, \langle u,y\rangle \leq \langle u,x\rangle \text{ for all } y \in K\right\}$$

is called the *normal cycle* of $K$. One easily sees that $\mathrm{nc}(K)$ consists of precisely those pairs $(x,u) \in S\mathbb{R}^n$ for which $x$ lies in the boundary $\partial K \subseteq K$ and $u$ is a unit outer normal vector to $\partial K$ in $x$.

The study of the normal cycle was initiated by Wintgen in [Win82]. As a side note, let it be mentioned that the normal cycle can be defined for a much larger class of sets than just convex bodies. Much work in this area has been done by Fu, most notably [Fu94] where he proved the existence of a normal cycle for subanalytic sets. In general, the normal cycle provides a suitable framework for studying curvature of singular spaces, see [RZ19]. Its connection to valuation theory first became evident in Zähle's work, see [Zäh86], where she proved that Federer's curvature measures, see [Fed59], could be expressed using integration of certain differential forms over the normal cycle.

Let $M$ be a smooth manifold and $k \in \mathbb{N}_0$. Throughout this text, we write $\Omega^\bullet(M)$ for the complex vector space of smooth complex-valued differential forms on $M$ and $\Omega^k(M) \subseteq \Omega^\bullet(M)$ for the subspace of differential forms of degree $k$.

For a given convex body $K \subseteq \mathbb{R}^n$, the normal cycle $\mathrm{nc}(K)$ is an $(n-1)$-dimensional, compact and naturally oriented Lipschitz submanifold of $S\mathbb{R}^n$. Thus, we obtain a map

$$\mathcal{K} \times \Omega^{n-1}(S\mathbb{R}^n) \to \mathbb{C},\ (K,\omega) \mapsto \int_{\mathrm{nc}(K)} \omega.$$

For the reader familiar with this terminology, this construction satisfies the conditions of a so called current, see [Fu89]. Most importantly for us, it yields a plethora of valuations as the map

$$[\![\omega]\!]:\ \mathcal{K} \to \mathbb{C},\ K \mapsto \int_{\mathrm{nc}(K)} \omega$$

is a continuous valuation for all $\omega \in \Omega^{n-1}(S\mathbb{R}^n)$.

Let $\mathbb{R}^n$ act on the sphere bundle $S\mathbb{R}^n$ by translations in the first component, that is $t \cdot (x,u) = (x+t,u)$ for all $t \in \mathbb{R}^n$ and $(x,u) \in S\mathbb{R}^n$. This naturally lifts to a group

action $\mathbb{R}^n \curvearrowright \Omega^\bullet(S\mathbb{R}^n)$ which allows us to speak of *translation-invariant differential forms* on $S\mathbb{R}^n$. We write

$$\Omega^\bullet(S\mathbb{R}^n)^{\mathrm{tr}} \subseteq \Omega^\bullet(S\mathbb{R}^n)$$

for the subspace of translation-invariant differential forms.

One easily verifies that a valuation of the form $[\![\omega]\!]$ is translation-invariant if the defining differential form $\omega$ is translation-invariant. Moreover, it is not hard to see that $[\![\omega]\!] \in \mathrm{Val}^{\mathrm{sm}}$ in this case. Hence, we have constructed a natural $\mathbb{C}$-linear map

$$\Omega^{n-1}(S\mathbb{R}^n)^{\mathrm{tr}} \to \mathrm{Val}^{\mathrm{sm}},\ \omega \mapsto [\![\omega]\!].$$

Alesker has proved the following explicit description of smooth translation-invariant valuations in Chapter 5 of [Ale06a].

**Theorem 2.23.** *A map $\varphi\colon \mathcal{K} \to \mathbb{C}$ satisfies $\varphi \in \mathrm{Val}^{\mathrm{sm}}$ if and only if there exist a constant $c \in \mathbb{C}$ and a differential form $\omega \in \Omega^{n-1}(S\mathbb{R}^n)^{\mathrm{tr}}$ such that*

$$\varphi = c\,\mathrm{vol}_n + [\![\omega]\!].$$

Based on this characterization of smooth valuations, Bernig and Fu introduced a localized version of smooth valuations, the so called smooth curvature measures, in Section 2.6 of [BF11].

Let $\mathcal{M}$ denote the vector space of complex Borel measures on $\mathbb{R}^n$. For more details on this space, see Section 3.1 of this thesis. Moreover, for any map $\Phi\colon \mathcal{K} \to \mathcal{M}$ we write

$$\Phi(K, \beta) = \Phi(K)(\beta)$$

whenever $K \subseteq \mathbb{R}^n$ is a convex body and $\beta \subseteq \mathbb{R}^n$ is a Borel set. Given a differential form $\omega \in \Omega^{n-1}(S\mathbb{R}^n)^{\mathrm{tr}}$ and a convex body $K \subseteq \mathbb{R}^n$, we define a measure $[\omega](K) \in \mathcal{M}$ by

$$[\omega](K, \beta) = \int_{\mathrm{nc}(K)\cap\pi^{-1}(\beta)} \omega$$

for all Borel sets $\beta \subseteq \mathbb{R}^n$, where $\pi\colon S\mathbb{R}^n \to \mathbb{R}^n$ denotes the natural projection. Hence, we have constructed a map $[\omega]: \mathcal{K} \to \mathcal{M}$.

**Definition 2.24.** A map $\Phi\colon \mathcal{K} \to \mathcal{M}$ is called a *smooth translation-invariant curvature measure* if there exist a constant $c \in \mathbb{C}$ and a differential form $\omega \in \Omega^{n-1}(S\mathbb{R}^n)^{\mathrm{tr}}$, such that

$$\Phi(K, \beta) = c\,\mathrm{vol}_n(K \cap \beta) + [\omega](K, \beta)$$

for all convex bodies $K \subseteq \mathbb{R}^n$ and all Borel sets $\beta \subseteq \mathbb{R}^n$.

We write $\mathrm{Curv}^{\mathrm{sm}}$ for the complex vector space of smooth translation-invariant curvature measures. It follows immediately from the definition that

$$\mathrm{glob}(\Phi) = \Phi(\cdot, \mathbb{R}^n) \in \mathrm{Val}^{\mathrm{sm}}$$

for all $\Phi \in \mathrm{Curv}^{\mathrm{sm}}$. This induces a map $\mathrm{glob}^{\mathrm{sm}} \colon \mathrm{Curv}^{\mathrm{sm}} \to \mathrm{Val}^{\mathrm{sm}}$ which we call the *smooth globalization map*. Note that Theorem 2.23 yields that the smooth globalization map glob is surjective.

Let it be mentioned that the space $\mathrm{Curv}^{\mathrm{sm}}$ does also admit a direct sum decomposition with respect to degree of homogeneity and parity, see Section 2.2 of [BFS14]. We will not go into detail here, as we will discuss such a decomposition for a more general class of curvature measures in the following chapter.

# 3 Curvature Measures

In this chapter, we introduce the space Curv of (continuous, translation-invariant) curvature measures which will be the main focus of this thesis. After the definition we discuss some elementary properties and some examples. In particular, we show that any smooth translation-invariant curvature measure is a curvature measure. It is an easy consequence from the work of Kiderlen and Weil in [KW99] that the space Curv admits a direct sum decomposition induced by degree of homogeneity and parity. This result is an analogue to the McMullen decomposition from the theory of valuations. Weil has furthermore proved that any curvature measure is a valuation. We reproduce the proof of this remarkable result in Theorem 3.12. Moreover, we introduce a Banach space topology on Curv. While the topology is unique, the norm is not. After introducing two of the main technical tools, namely the globalization map and the Bernig embedding, we prove some characterization theorems for curvature measures. We prove that the space $\mathrm{Curv}_n$ of $n$-homogeneous curvature measures is one-dimensional and that the space $\mathrm{Curv}_{n-1}$ is isomorphic to the space $C\left(S^{n-1}\right)$ of continuous functions on the sphere. Finally, we give a characterization of simple curvature measures. These results can be seen as curvature measure counterparts to the theorems of Hadwiger, McMullen and Klain–Schneider, respectively.

## 3.1 Definition

In order to define continuous, translation-invariant curvature measures, we first need to recall some background from the theory of complex measures. As a general reference, we recommend Chapter 6 of [Rud87].

Throughout the remainder of this thesis, we denote the Borel $\sigma$-algebra of $\mathbb{R}^n$ by $\mathcal{B}$. For any complex Borel measure $\mu\colon \mathcal{B} \to \mathbb{C}$, we write $|\mu|\colon \mathcal{B} \to [0,\infty]$ for its *total variation measure*, that is

$$|\mu|(\beta) = \sup \sum_j \left|\mu\left(\beta_j\right)\right|$$

for all $\beta \in \mathcal{B}$, where the supremum is taken over all countable partitions $\beta = \bigsqcup \beta_j$ into pairwise disjoint Borel sets $\beta_j \subseteq \mathbb{R}^n$. One easily shows that $|\mu|$ is a finite (see Theorem 6.4 of [Rud87]) non-negative measure on $\mathbb{R}^n$ which means that $0 \leq |\mu|(\beta) < \infty$ for all Borel sets $\beta \subseteq \mathbb{R}^n$.

Let $\mathcal{M}$ denote the complex vector space of complex Borel measures on $\mathbb{R}^n$. We equip $\mathcal{M}$ with the *total variation norm* which is given by $\|\mu\| = |\mu|(\mathbb{R}^n)$ for all $\mu \in \mathcal{M}$. One easily verifies that $\mathcal{M}$ is a Banach space with respect to this norm.

Using integration the space $\mathcal{M}$ is closely related to a space of functions. For simplicity, let us fix the following notation. Given a measure $\mu \in \mathcal{M}$ and a $\mu$-integrable function $f\colon \mathbb{R}^n \to \mathbb{C}$, we let

$$\mu(f) = \int_{\mathbb{R}^n} f \,\mathrm{d}\,\mu.$$

A function $f\colon \mathbb{R}^n \to \mathbb{C}$ is said to *vanish at infinity* if

$$\lim_{\|x\|\to\infty} f(x) = 0.$$

We write $C_0(\mathbb{R}^n)$ for the space of continuous functions $\mathbb{R}^n \to \mathbb{C}$ which vanish at infinity. One easily sees that any function $f \in C_0(\mathbb{R}^n)$ is necessarily bounded. Thus, we can equip $C_0(\mathbb{R}^n)$ with the uniform norm

$$\|f\|_\infty = \sup_{x\in\mathbb{R}^n} \left|f(x)\right|.$$

Moreover, $C_0(\mathbb{R}^n)$ is closed in the space of all continuous, bounded functions $\mathbb{R}^n \to \mathbb{C}$ (also equipped with the uniform norm). Hence, $C_0(\mathbb{R}^n)$ is a Banach space, in fact it is the completion of the space of all continuous functions with compact support. We write $C_0(\mathbb{R}^n)^*$ for the continuous dual space of $C_0(\mathbb{R}^n)$, equipped with the dual norm

$$\|\nu\| = \sup\left\{\left|\nu(f)\right| \;\middle|\; f \in C_0(\mathbb{R}^n), \|f\| \leq 1\right\}.$$

Let $\mu \in \mathcal{M}$ be an arbitrary complex Borel measure. $\mu$ is called *regular* if its total variation measure $|\mu|$ is regular. This means that $\mu$ is regular if and only if

$$|\mu|(\beta) = \sup\left\{|\mu|(C) \mid C \subseteq \beta,\, C \text{ compact}\right\} = \inf\left\{|\mu|(U) \mid U \supseteq \beta,\, U \text{ is open in } \mathbb{R}^n\right\}$$

for all $\beta \in \mathcal{B}$. Since $|\mu|$ is finite, it follows from Theorem 2.18 of [Rud87] that $|\mu|$ and therefore also $\mu$ are regular.

Using this, we have the following connection between the space $\mathcal{M}$ of (regular) complex Borel measures and the space $C_0(\mathbb{R}^n)^*$, see Theorem 6.19 of [Rud87] and the discussion directly above.

**Lemma 3.1.** *For every $\mu \in \mathcal{M}$, the map $C_0(\mathbb{R}^n) \to \mathbb{C}$, $f \mapsto \mu(f)$ is a bounded linear functional. The induced map*

$$\mathcal{M} \to C_0(\mathbb{R}^n)^*,\ \mu \mapsto \big(f \mapsto \mu(f)\big)$$

*is a $\mathbb{C}$-linear isometric isomorphism.*

The weak* topology is a topology which is naturally defined on the dual space of a Banach space, see for example pp. 62-70 in [Rud91] for a detailed account. To introduce the weak* topology, let $X$ be a Banach space and let

$$T\colon X \to X^{**},\ x \mapsto \big(Tx\colon x^* \mapsto x^*(x)\big)$$

denote the natural inclusion into the double dual space. Then, the weak* topology is the weakest topology on $X^*$ such that $Tx \in X^{**}$ is continuous for all $x \in X$.

Applying this procedure to $X = C_0(\mathbb{R}^n)$ yields a weak* topology on $\mathcal{M}$ by virtue of Lemma 3.1. This topology can be characterized as follows. A sequence $(\mu_k)_{k\in\mathbb{N}} \subseteq \mathcal{M}$ converges to $\mu \in \mathcal{M}$ with respect to the weak* topology if and only if

$$\lim_{k\to\infty} \mu_k(f) = \mu(f)$$

for all $f \in C_0(\mathbb{R}^n)$.

With this in mind, we are finally able to define continuous, translation-invariant curvature measures. This notion will be the central object of study for the remainder of this thesis.

**Definition 3.2.** A map $\Phi\colon \mathcal{K} \to \mathcal{M}$ is called a *continuous translation-invariant curvature measure* if it satisfies the following properties.

- $\Phi$ is translation-invariant: $\Phi$ satisfies

  $$\Phi(K + x, \beta + x) = \Phi(K, \beta)$$

  for all $K \in \mathcal{K}$, $\beta \in \mathcal{B}$ and $x \in \mathbb{R}^n$.

- $\Phi$ is defined locally: Given two convex bodies $K, L \subseteq \mathbb{R}^n$, if there exists an open set $U \subseteq \mathbb{R}^n$ such that $K \cap U = L \cap U$, then

  $$\Phi(K, \beta) = \Phi(L, \beta)$$

  for all Borel sets $\beta \subseteq U$.

- $\Phi$ is weak* continuous: For a convergent sequence $\left(K_j\right)_{j\in\mathbb{N}} \subseteq \mathcal{K}$ of convex bodies satisfying $K_j \to K$ as $j \to \infty$, we have

  $$\lim_{j\to\infty} \Phi\left(K_j, f\right) = \Phi(K, f)$$

  for all $f \in C_0(\mathbb{R}^n)$.

We write Curv for the set of all continuous translation-invariant curvature measures $\mathcal{K} \to \mathcal{M}$. Since $\mathcal{M}$ is a complex vector space, one easily sees that Curv is a complex vector space, too.

In the remainder of this third chapter, we want to develop a structure theory for the space Curv, similar to the structure theory of Val. As we are not interested in more general classes, we will drop the leading adjectives and simply call the elements of Curv *curvature measures*.

Let us again point out that we do not require a valuation property for curvature measures as this is implied by the three given axioms, see Theorem 3.12.

## 3.2 Basic Properties and Examples

As we have defined two different topologies on $\mathcal{M}$, namely the norm topology and the weak* topology, there a priori are two different natural notions of continuity for curvature measures. However, the following example illustrates that asking for norm continuity would be too much.

**Example 3.3.** Let $n = 1$ for this example and define $\Phi\colon \mathcal{K} \to \mathcal{M}$ by

$$\Phi\left([a, b]\right) = \delta_a \in \mathcal{M}$$

for all convex bodies $[a,b] \subseteq \mathbb{R}$. For this construction, one easily sees that $\Phi \in \operatorname{Curv}$. Now assume that $a_k \to a$ is a convergent sequence of real numbers such that $a_k < a$ for all $k \in \mathbb{N}$. Moreover, suppose that $b \geq a$. Then, we obviously have $[a_k, b] \to [a,b]$ in $\mathcal{K}$. However, we also have

$$\Big\|\Phi\big([a_k,b]\big) - \Phi\big([a,b]\big)\Big\| = \big\|\delta_{a_k} - \delta_a\big\| = \sup\Big|\big(\delta_{a_k} - \delta_a\big)(f)\Big| = \sup\big|f(a_k) - f(a)\big|$$

for all $k \in \mathbb{N}$, where the supremum is taken over all functions $f \in C_0(\mathbb{R}^n)$ which satisfy $\|f\| = 1$. Note that for all $k$ there exists such a function $f$ such that $f(a_k) = 1$ and $f(a) = -1$ due to $a_k \neq a$. Therefore, we obtain

$$\Big\|\Phi\big([a_k,b]\big) - \Phi\big([a,b]\big)\Big\| = 2$$

for all $k$ which shows that $\Phi$ is not norm continuous.

Let $\mu\colon \mathcal{B} \to \mathbb{C}$ be a complex Borel measure on $\mathbb{R}^n$ and let $\mathcal{N}(\mu) \subseteq \mathcal{B}$ be the family of all open sets $U \subseteq \mathbb{R}^n$ which satisfy $|\mu|(U) = 0$. Then, the set

$$\operatorname{supp}(\mu) = \mathbb{R}^n \setminus \left( \bigcup_{U \in \mathcal{N}(\mu)} U \right)$$

is called the *support* of $\mu$.

**Proposition 3.4.** *Let $\Phi \in \operatorname{Curv}$ be a curvature measure. Then, $\operatorname{supp}\big(\Phi(K)\big) \subseteq K$ for all $K \in \mathcal{K}$.*

*Proof.* Let $\Phi \in \operatorname{Curv}$ be a curvature measure and let $K \subseteq \mathbb{R}^n$ be a convex body. Consider the map $\mu = \mu_\Phi\colon \mathcal{B} \setminus \{\mathbb{R}^n\} \to \mathbb{C}$ given by $\mu(\beta) = \Phi(\{x\}, \beta)$ for all Borel sets $\beta \subsetneq \mathbb{R}^n$, where $x$ is some point in $\mathbb{R}^n \setminus \beta$.

To see that $\mu$ is well-defined, let $\beta \subseteq \mathbb{R}^n$ be a Borel set for which there exist two distinct points $x \neq y$ in $\mathbb{R}^n \setminus \beta$. Clearly, $V = \mathbb{R}^n \setminus \{x,y\}$ is open in $\mathbb{R}^n$ and satisfies $\beta \subseteq V$. Since $\{x\} \cap V = \emptyset = \{y\} \cap V$, we obtain the required

$$\Phi\big(\{x\}, \beta\big) = \Phi\big(\{y\}, \beta\big)$$

from the assumption that $\Phi$ is defined locally.

To see that this map is $\sigma$-additive, let $\beta = \bigsqcup \beta_j$ be a countable Borel partition of a Borel set $\beta \subsetneq \mathbb{R}^n$. Any choice of a point $x \in \mathbb{R}^n \setminus \beta$ clearly satisfies $x \in \mathbb{R}^n \setminus \beta_j$ for all $j$. Since $\Phi\big(\{x\}, \cdot\big)$ is $\sigma$-additive, this implies that

$$\mu(\beta) = \Phi\big(\{x\}, \beta\big) = \Phi\left(\{x\}, \bigsqcup_j \beta_j\right) = \sum_j \Phi\big(\{x\}, \beta_j\big) = \sum_j \mu\big(\beta_j\big)$$

which indeed shows that $\mu\colon \mathcal{B} \setminus \{\mathbb{R}^n\} \to \mathbb{C}$ is $\sigma$-additive.

Let $\beta \subsetneq \mathbb{R}^n$ be a Borel set and let $t \in \mathbb{R}^n$. For a chosen point $x \in \mathbb{R}^n \setminus \beta$, we obviously have $x + t \notin \beta + t$ which yields

$$\mu(\beta + t) = \Phi\big(\{x+t\}, \beta + t\big) = \Phi\big(\{x\} + t, \beta + t\big) = \Phi\big(\{x\}, \beta\big) = \mu(\beta)$$

by translation-invariance of $\Phi$. Thus, we have proved that $\mu$ is translation-invariant.

We now want to show that in fact $\mu = 0$. To do so, let $\beta \subseteq \mathbb{R}^n$ be a bounded Borel set and let $m \in \mathbb{N}$ be arbitrary. Then, there exist $t_1, \ldots, t_m \in \mathbb{R}^n$ such that the Borel sets

$$\beta + t_1, \ldots, \beta + t_m$$

are pairwise disjoint. Moreover, there exists $x \in \mathbb{R}^n$ such that

$$x \notin \bigcup_{i=1}^{m} (\beta + t_i)$$

because the set on the right is bounded. Since $\mu$ is $\sigma$-additive and translation-invariant, we obtain

$$\Phi\left(\{x\}, \bigcup_{i=1}^{m} (\beta + t_i)\right) = \mu\left(\bigcup_{i=1}^{m} (\beta + t_i)\right) = \sum_{i=1}^{m} \mu(\beta + t_i) = \sum_{i=1}^{m} \mu(\beta) = m\mu(\beta)$$

which yields the estimate

$$m\left|\mu(\beta)\right| = \left|\Phi\left(\{x\}, \bigcup_{i=1}^{m} (\beta + t_i)\right)\right| \leq \left|\Phi\left(\{x\}\right)\right|\left(\bigcup_{i=1}^{m} (\beta + t_i)\right) \leq \left|\Phi\left(\{x\}\right)\right|(\mathbb{R}^n) < \infty.$$

Since $m$ is an arbitrary positive integer, it follows that $\mu(\beta) = 0$. Thus, we have proved that $\mu = 0$ on bounded Borel sets in $\mathbb{R}^n$. Since any arbitrary Borel set $\beta' \subsetneq \mathbb{R}^n$ can be written as a disjoint union

$$\beta' = \bigcup_{j \in \mathbb{N}} \beta_j$$

of bounded sets $\beta_j \in \mathcal{B}$, we see that $\mu\left(\beta'\right) = 0$ for all Borel sets $\beta' \subsetneq \mathbb{R}^n$. Therefore, we indeed have $\mu = 0$.

We can apply this to our situation as follows. Consider the set $W = \mathbb{R}^n \setminus K$ and choose a point $x \in \mathbb{R}^n \setminus W = K$. Firstly, note that $W$ is open in $\mathbb{R}^n$ as $K$ is compact. Moreover, observe that $K \cap W = \emptyset = \{x\} \cap W$. The assumption that $\Phi$ is defined locally therefore implies that

$$\left|\Phi(K)\right|(W) = \sup \sum_j \left|\Phi\left(K, \beta_j\right)\right| = \sup \sum_j \left|\Phi\left(\{x\}, \beta_j\right)\right| = \sup \sum_j \left|\mu\left(\beta_j\right)\right| = 0,$$

where the supremum is taken over all countable Borel partitions $W = \bigsqcup \beta_j$. Thus, we have proved that $W \in \mathcal{N}\left(\Phi(K)\right)$ which implies

$$\operatorname{supp}\left(\Phi(K)\right) = \mathbb{R}^n \setminus \left(\bigcup_{U \in \mathcal{N}\left(\Phi(K)\right)} U\right) \subseteq \mathbb{R}^n \setminus W = K$$

as required. □

The construction of the map $\mu$ can be found in Schneider's paper [Sch78] about rigid motion-invariant curvature measures.

As an immediate consequence, we obtain the following useful property of curvature measures.

**Corollary 3.5.** *Let $\Phi \in \mathrm{Curv}$ be a curvature measure. Then, $\Phi(K, \beta) = \Phi(K, \beta \cap K)$ for all convex bodies $K \in \mathcal{K}$ and Borel sets $\beta \in \mathcal{B}$.*

*Proof.* Obviously, we have the partition

$$\beta = \big(\beta \setminus K\big) \sqcup (\beta \cap K)\,.$$

Moreover, $\beta \setminus K \subseteq \mathbb{R}^n \setminus K \subseteq \mathbb{R}^n \setminus \operatorname{supp}\big(\Phi(K)\big)$, where the last inclusion follows from the previous proposition. This implies $\Phi\big(K, \beta \setminus K\big) = 0$ which combined with the partition of $\beta$ above yields

$$\Phi(K, \beta) = \Phi\big(K, \beta \setminus K\big) + \Phi(K, \beta \cap K) = \Phi(K, \beta \cap K)$$

as required. □

As a last property, let us mention that the continuity property of curvature measures extends to all continuous functions, not just the ones which vanish at infinity.

**Proposition 3.6.** *Let $\Phi \in \mathrm{Curv}$ and let $\big(K_j\big)_{j\in\mathbb{N}} \subseteq \mathcal{K}$ be a convergent sequence of convex bodies satisfying $K_j \to K \in \mathcal{K}$. Then,*

$$\lim_{j\to\infty} \Phi\big(K_j, f\big) = \Phi(K, f)$$

*for all continuous functions $f\colon \mathbb{R}^n \to \mathbb{C}$.*

*Proof.* For all $r > 0$, we write $U(r) \subseteq \mathbb{R}^n$ for the open ball of radius $r$ centered at the origin. Let $\varepsilon > 0$ be arbitrary but fixed. Since $K_j \to K$, there exists $S \in \mathbb{N}$ such that $d(K_s, K) < \varepsilon$ for all $s > S$. By definition of the Hausdorff metric, this implies that $K_s \subseteq K + \varepsilon B^n$ for all $s > S$. Since $K + \varepsilon B^n$ is compact, there exists a radius $r_0 > 0$ such that

$$K_s \subseteq K + \varepsilon B^n \subseteq U(r_0)$$

for all $s > S$. For an arbitrary $t \in \{1, \ldots, S\}$, there exists $r_t > 0$ such that $K_t \subseteq U(r_t)$ by compactness of $K_t$. By construction,

$$r = \max\{r_0, \ldots, r_S\}$$

satisfies $K_j \subseteq U(r)$ for all $j \in \mathbb{N}$ and $K \subseteq U(r)$.

Since $U(r)$ is bounded, there exists a continuous function $g\colon \mathbb{R}^n \to \mathbb{C}$ with compact

support, which satisfies $g(x) = f(x)$ for all $x \in U(r)$. In particular, we have $f|_{K_j} = g|_{K_j}$ for all $j \in \mathbb{N}$. Together with Proposition 3.4, this yields

$$\begin{aligned}\Phi\left(K_j, f\right) &= \int_{\mathbb{R}^n} f \,\mathrm{d}\,\Phi\left(K_j\right) = \int_{K_j} f \,\mathrm{d}\,\Phi\left(K_j\right) \\ &= \int_{K_j} g \,\mathrm{d}\,\Phi\left(K_j\right) = \int_{\mathbb{R}^n} g \,\mathrm{d}\,\Phi\left(K_j\right) = \Phi\left(K_j, g\right)\end{aligned}$$

for all $j$. Analogously, $K \subseteq U(r)$ implies that $\Phi(K, f) = \Phi(K, g)$. Moreover, we have $g \in C_0(\mathbb{R}^n)$ as $g$ is continuous with compact support. Therefore, the axiom of weak* continuity yields the required convergence

$$\Phi\left(K_j, f\right) = \Phi\left(K_j, g\right) \to \Phi(K, g) = \Phi(K, f)$$

as $j \to \infty$. □

The first and easiest example of a curvature measure is obtained from volume. To avoid confusion with the Lebesgue measure $\mathrm{vol}_n$, we use capital letters for this curvature measure. To be precise, we define a map $\mathrm{VOL}_n \colon \mathcal{K} \to \mathcal{M}$ by

$$\mathrm{VOL}_n(K, \beta) = \mathrm{vol}_n(K \cap \beta)$$

for all $K \in \mathcal{K}$ and $\beta \in \mathcal{B}$.

**Proposition 3.7.** $\mathrm{VOL}_n \in \mathrm{Curv}$.

*Proof.* The translation-invariance of $\mathrm{VOL}_n$ follows easily from the translation-invariance of $\mathrm{vol}_n$. Indeed, for $K \in \mathcal{K}$, $\beta \in \mathcal{B}$ and $x \in \mathbb{R}^n$ we have

$$\begin{aligned}&\mathrm{VOL}_n(K + x, \beta + x) = \mathrm{vol}_n\left((K + x) \cap (\beta + x)\right) = \mathrm{vol}_n\left((K \cap \beta) + x\right) \\ &= \mathrm{vol}_n(K \cap \beta) = \mathrm{VOL}_n(K, \beta).\end{aligned}$$

Suppose that $K \cap U = L \cap U$ for some open set $U \subseteq \mathbb{R}^n$ and let $\beta \subseteq U$ be a Borel set. Then, we obviously have $K \cap \beta = K \cap U \cap \beta = L \cap U \cap \beta = L \cap \beta$ which yields

$$\mathrm{VOL}_n(K, \beta) = \mathrm{vol}_n(K \cap \beta) = \mathrm{vol}_n(L \cap \beta) = \mathrm{VOL}_n(L, \beta).$$

Thus, $\mathrm{VOL}_n$ is defined locally.

Lastly, we need to check weak* continuity. To do so, let $f \in C_0(\mathbb{R}^n)$ and let $K_j \to K$ be a convergent sequence of convex bodies. We want to apply the dominated convergence theorem to the sequence $\left(f_j\right)_{j \in \mathbb{N}}$ defined by

$$f_j(x) = f(x)\, \mathbf{1}_{K_j}(x)$$

for all $x \in \mathbb{R}^n$ and $j \in \mathbb{N}$, where $\mathbf{1}_{K_j}$ denotes the indicator function of $K_j$. Clearly, $f_j$ is $\mathrm{vol}_n$-integrable for all $j$ and $f\mathbf{1}_K$ is the pointwise limit of the sequence $\left(f_j\right)$. As we have seen in the previous proof, there exists some $r > 0$ such that $K_j \subseteq U(r)$ for all $j$,

where $U(r)$ again denotes the open ball of radius $r$ centered at the origin. We define $g\colon \mathbb{R}^n \to \mathbb{C}$ by

$$g(x) = \left| f(x)\, \mathbf{1}_{rB^n}(x) \right|$$

for all $x \in \mathbb{R}^n$ which is obviously $\mathrm{vol}_n$-integrable because $rB^n$ is compact and $f$ is continuous. Note that we have $K_j \subseteq U(r) \subseteq rB^n$ for all $j$ which yields $\left|f_j\right| \leq g$. Thus, we have verified that $g$ indeed dominates the sequence $\left(f_j\right)$. Furthermore, note that the definition of $\mathrm{VOL}_n$ immediately implies that

$$\mathrm{VOL}_n(L,h) = \int_{\mathbb{R}^n} h \mathbf{1}_L \,\mathrm{d}\,\mathrm{vol}_n$$

for all convex bodies $L \subseteq \mathbb{R}^n$ and all continuous functions $h\colon \mathbb{R}^n \to \mathbb{C}$. As required, we therefore obtain

$$\mathrm{VOL}_n\left(K_j, f\right) = \int_{\mathbb{R}^n} f \mathbf{1}_{K_j} \,\mathrm{d}\,\mathrm{vol}_n = \int_{\mathbb{R}^n} f_j \,\mathrm{d}\,\mathrm{vol}_n \to \int_{\mathbb{R}^n} f \mathbf{1}_K \,\mathrm{d}\,\mathrm{vol}_n = \mathrm{VOL}_n(K,f)$$

as $j \to \infty$, where the convergence follows from the dominated convergence theorem. □

By the chosen terminology, one of course expects that any smooth translation-invariant curvature measure as defined in Definition 2.24 is in fact a curvature measure. In order to prove this, we first need some properties of the normal cycle.

**Lemma 3.8.** *The normal cycle satisfies the following properties:*

- $\mathrm{nc}(K+t) = t \cdot \mathrm{nc}(K)$ *for all* $K \in \mathcal{K}$ *and all* $t \in \mathbb{R}^n$.
- *Let* $K, L \in \mathcal{K}$ *and let* $U \subseteq \mathbb{R}^n$ *be an open set such that* $K \cap U = L \cap U$. *Then,*

$$\mathrm{nc}(K) \cap \pi^{-1}(U) = \mathrm{nc}(L) \cap \pi^{-1}(U),$$

  *where* $\pi\colon S\mathbb{R}^n \to \mathbb{R}^n$ *denotes the natural projection.*

*Proof.* For the first claim, let $K \in \mathcal{K}$ and $t \in \mathbb{R}^n$. Recall that the action of $\mathbb{R}^n$ on $S\mathbb{R}^n$ is given by $t \cdot (x,u) = (x+t,u)$. Obviously, $x \in K$ if and only if $x + t \in K + t$. Moreover, note that the condition $\langle u, y\rangle \leq \langle u, x\rangle$ for all $y \in K$ is equivalent to $\langle u, z\rangle \leq \langle u, x+t\rangle$ for all $z \in K + t$. Therefore, we obtain

$$\begin{aligned} t \cdot \mathrm{nc}(K) &= \left\{(x+t,u) \in S\mathbb{R}^n \mid (x,u) \in \mathrm{nc}(K)\right\} \\ &= \left\{(x+t,u) \mid x+t \in K+t,\ \langle u, z\rangle \leq \langle u, x+t\rangle \text{ for all } z \in K+t\right\} \\ &= \mathrm{nc}(K+t) \end{aligned}$$

as required.

For the second claim, assume we are given an open set $U \subseteq \mathbb{R}^n$ and two convex bodies $K, L \in \mathcal{K}$ such that $K \cap U = L \cap U$. Obviously, the situation is symmetrical in $K$ and $L$ which implies that it is satisfactory to show that

$$\mathrm{nc}(K) \cap \pi^{-1}(U) \subseteq \mathrm{nc}(L) \cap \pi^{-1}(U).$$

To do so, let $(x,u) \in \operatorname{nc}(K) \cap \pi^{-1}(U)$. Then, $x \in K \cap U \subseteq U$ which of course implies $(x,u) \in \pi^{-1}(U)$. To show that $(x,u) \in \operatorname{nc}(L)$, let $y \in L \setminus \{x\}$ be an arbitrary point. Since $U \subseteq \mathbb{R}^n$ is open and $x \neq y$, there exists a point $y' \in [x,y] \cap L \cap U = [x,y] \cap K \cap U$ such that $y' \neq x$. In particular, there exists $\lambda \in (0,1]$ such that $y' = (1-\lambda)\,x + \lambda y$. Since $y' \in K$ and $(x,u) \in \operatorname{nc}(K)$, we obtain

$$\langle u, x\rangle \geq \langle u, y'\rangle = (1-\lambda)\,\langle u, x\rangle + \lambda\,\langle u, y\rangle\,.$$

This is equivalent to $\langle u,x\rangle \geq \langle u,y\rangle$ because of $\lambda \neq 0$. Since the inequality is trivially satisfied for $y = x$, we obtain $\langle u,x\rangle \geq \langle u,y\rangle$ for all $y \in L$. Since we also have

$$x \in K \cap U = L \cap U \subseteq L,$$

the required $(x,u) \in \operatorname{nc}(L)$ follows. □

With this in mind, we are now able to prove the following theorem.

**Theorem 3.9.** $\mathrm{Curv}^{\mathrm{sm}} \subseteq \mathrm{Curv}$.

*Proof.* Proposition 3.7 yields that $c\,\mathrm{VOL}_n \in \mathrm{Curv}$ for all $c \in \mathbb{C}$. Hence, all that is left to show is that $[\omega] \in \mathrm{Curv}$ for any given $\omega \in \Omega^{n-1}(S\,\mathbb{R}^n)^{\mathrm{tr}}$.

First, we want to verify that $[\omega]$ is translation-invariant. To do so, let $K \in \mathcal{K}$, $\beta \in \mathcal{B}$ and $t \in \mathbb{R}^n$. Obviously, we have $\pi^{-1}(\beta + t) = (\beta + t) \times S^{n-1} = t \cdot \big(\beta \times S^{n-1}\big) = t \cdot \pi^{-1}(\beta)$. Therefore, the first point of the previous lemma yields

$$[\omega]\,(K + t, \beta + t) = \int_{\operatorname{nc}(K+t)\cap\pi^{-1}(\beta+t)} \omega = \int_{t\cdot\left(\operatorname{nc}(K)\cap\pi^{-1}(\beta)\right)} \omega.$$

Let us consider the diffeomorphism $D_t \colon S\,\mathbb{R}^n \to S\,\mathbb{R}^n$ induced by translation by $t$, that is $D_t\,(x,u) = t \cdot (x,u) = (x+t,u)$ for all $(x,u) \in S\,\mathbb{R}^n$. By construction, we have that $\int_{t\cdot A} \omega = \int_A D_t^*\omega$ for any Borel set $A \subseteq S\,\mathbb{R}^n$. Moreover, we have $D_t^*\omega = \omega$ due to translation-invariance. Combining this with the computation above, we obtain

$$[\omega]\,(K + t, \beta + t) = \int_{\operatorname{nc}(K)\cap\pi^{-1}(\beta)} D_t^*\omega = \int_{\operatorname{nc}(K)\cap\pi^{-1}(\beta)} \omega = [\omega]\,(K,\beta)$$

which indeed shows that $[\omega]$ is translation-invariant.

The locality property follows almost at once from the second bullet point in the previous lemma. Let the convex bodies $K, L \subseteq \mathbb{R}^n$ satisfy $K \cap U = L \cap U$ for some open $U \subseteq \mathbb{R}^n$ and $\beta \subseteq U$ be a Borel set. Since $\beta$ is contained in $U$, we of course also have $\pi^{-1}(\beta) \subseteq \pi^{-1}(U)$. The previous lemma therefore yields

$$\begin{aligned}&\operatorname{nc}(K) \cap \pi^{-1}(\beta) = \operatorname{nc}(K) \cap \pi^{-1}(U) \cap \pi^{-1}(\beta)\\ &= \operatorname{nc}(L) \cap \pi^{-1}(U) \cap \pi^{-1}(\beta) = \operatorname{nc}(L) \cap \pi^{-1}(\beta)\,.\end{aligned}$$

By definition of $[\omega]$, this immediately yields the required $[\omega]\,(K,\beta) = [\omega]\,(L,\beta)$.

Lastly, the weak* continuity follows from the fact that the normal cycle is continuous with respect to the so called flat norm of currents. This was originally proved in the more

general setting of sets of positive reach in Theorem 3.1 of [RZ01]. In Proposition 7.1 of [RZ19], Rataj and Zähle also show how flat continuity of the normal cycle implies weak* continuity of curvature measures for certain special differential forms $\varphi_k$. However, in this proof they do not use the special nature of those differential forms. Therefore, the general result follows verbatim. □

## 3.3 Kiderlen–Weil Decomposition and Valuation Property

In this section, we prove that the space Curv admits a direct sum decomposition with respect to degree and parity analogous to the McMullen decomposition of valuations. Since the main part of the proof is contained in the paper [KW99] by Kiderlen and Weil, we will refer to it as the *Kiderlen–Weil decomposition*. Moreover, we discuss and prove Weil's remarkable theorem that any curvature measure is an $\mathcal{M}$-valued valuation.

Let $d \in \mathbb{R}$ be an arbitrary real number. A curvature measure $\Phi \in \mathrm{Curv}$ is called *d-homogeneous* or *of degree d* if

$$\Phi(\lambda K, \lambda\beta) = \lambda^d \Phi(K, \beta)$$

for all $\lambda > 0$, $K \in \mathcal{K}$ and $\beta \in \mathcal{B}$. Moreover, we say that $\Phi$ is *even* if it satisfies $\Phi(-K, -\beta) = \Phi(K, \beta)$ for all $K \in \mathcal{K}$ and $\beta \in \mathcal{B}$, and similarly we say that $\Phi$ is *odd* if $\Phi(-K, -\beta) = -\Phi(K, \beta)$ for all $K$ and $\beta$. For all $d \in \mathbb{R}$ and $\varepsilon \in \{+, -\}$, this defines the following list of linear subspaces of Curv:

$$\begin{aligned}
\mathrm{Curv}_d &= \big\{\Phi \in \mathrm{Curv} \mid \Phi \text{ is of degree } d\big\}, \\
\mathrm{Curv}^+ &= \big\{\Phi \in \mathrm{Curv} \mid \Phi \text{ is even}\big\}, \\
\mathrm{Curv}^- &= \big\{\Phi \in \mathrm{Curv} \mid \Phi \text{ is odd}\big\}, \\
\mathrm{Curv}_d^\varepsilon &= \mathrm{Curv}_d \cap \mathrm{Curv}^\varepsilon .
\end{aligned}$$

**Example 3.10.** It is well-known that the Lebesgue measure $\mathrm{vol}_n$ is $n$-homogeneous and even on the Borel $\sigma$-algebra $\mathcal{B}$. This immediately implies $\mathrm{VOL}_n \in \mathrm{Curv}_n^+$.

**Theorem 3.11.** *The space of curvature measures decomposes as*

$$\mathrm{Curv} = \bigoplus_{i=0}^{n} \mathrm{Curv}_i .$$

*Moreover, we have* $\mathrm{Curv}_i = \mathrm{Curv}_i^+ \oplus \mathrm{Curv}_i^-$ *for all* $i = 0, \dots, n$.

*Proof.* Let $\Phi \in \mathrm{Curv}$ be an arbitrary curvature measure. It follows from Theorem 3.1 in [KW99] that there exist a constant $c \in \mathbb{C}$ and homogeneous curvature measures $\Phi_0 \in \mathrm{Curv}_0, \dots, \Phi_{n-1} \in \mathrm{Curv}_{n-1}$ such that

$$\Phi = \sum_{j=0}^{n-1} \Phi_j + c\,\mathrm{VOL}_n .$$

We know from the preceding example that $c\,\mathrm{VOL}_n \in \mathrm{Curv}_n$. Therefore, we obtain $\mathrm{Curv} = \sum \mathrm{Curv}_i$.

To see that this sum is direct, let $i \in \{0, \dots, n\}$ be arbitrary. For any polytope $P \subseteq \mathbb{R}^n$, it follows from equation (3.2) of [KW99] that $\Phi_i(P) \in \mathcal{M}$ is uniquely determined by $\Phi$, $i$ and $P$. Since the set $\mathcal{P}$ of polytopes is dense in $\mathcal{K}$, it follows from weak* continuity that $\Phi_i \in \mathrm{Curv}_i$ is uniquely determined by $\Phi$ and $i$. Therefore, any curvature measure $\Phi \in \mathrm{Curv}$ can be written uniquely as a sum $\Phi = \Phi_0 + \ldots + \Phi_n$ for homogeneous curvature measures $\Phi_0 \in \mathrm{Curv}_0, \dots, \Phi_n \in \mathrm{Curv}_n$ which means that

$$\mathrm{Curv} = \bigoplus_{i=0}^{n} \mathrm{Curv}_i$$

as required.

For the decomposition with respect to parity, suppose we are given a curvature measure $\Phi \in \mathrm{Curv}_i$ for some $i \in \{0, \dots, n\}$. One easily verifies that the assignment

$$(K, \beta) \mapsto \Phi(-K, -\beta)$$

also defines a curvature measure of degree $i$. We consider the curvature measures $\Phi^+, \Phi^- \in \mathrm{Curv}_i$ given by

$$\begin{aligned}
\Phi^+(K, \beta) &= \frac{1}{2}\left(\Phi(K, \beta) + \Phi(-K, -\beta)\right), \\
\Phi^-(K, \beta) &= \frac{1}{2}\left(\Phi(K, \beta) - \Phi(-K, -\beta)\right)
\end{aligned}$$

for all $K \in \mathcal{K}$ and $\beta \in \mathcal{B}$. By construction, we have $\Phi = \Phi^+ + \Phi^-$. Moreover, $\Phi^+$ is even and $\Phi^-$ is odd as the computations

$$\begin{aligned}
\Phi^+(-K, -\beta) &= \frac{1}{2}\left(\Phi(-K, -\beta) + \Phi(K, \beta)\right) = \Phi^+(K, \beta), \\
\Phi^-(-K, -\beta) &= \frac{1}{2}\left(\Phi(-K, -\beta) - \Phi(K, \beta)\right) = -\frac{1}{2}\left(\Phi(K, \beta) - \Phi(-K, -\beta)\right) \\
&= -\Phi^-(K, \beta)
\end{aligned}$$

show. Thus, we have $\mathrm{Curv}_i = \mathrm{Curv}_i^+ + \mathrm{Curv}_i^-$.

Lastly, assume that $\Phi \in \mathrm{Curv}_i$ is both even and odd. This implies that

$$\Phi(K, \beta) = \Phi(-K, -\beta) = -\Phi(K, \beta)$$

for all convex bodies $K \in \mathcal{K}$ and all Borel sets $\beta \in \mathcal{B}$ which of course yields $\Phi = 0$. This finishes the proof of $\mathrm{Curv}_i = \mathrm{Curv}_i^+ \oplus \mathrm{Curv}_i^-$. □

Next, we recall the following remarkable theorem due to Weil, see Theorem 2.1 of [Wei17]. For the convenience of the reader we also reproduce the proof (with some minor changes).

**Theorem 3.12 (Weil).** *Any $\Phi \in \mathrm{Curv}$ is an $\mathcal{M}$-valued valuation, that is*

$$\Phi(K \cup L) + \Phi(K \cap L) = \Phi(K) + \Phi(L) \in \mathcal{M}$$

*whenever $K, L, K \cup L \in \mathcal{K}$.*

*Proof.* Throughout this proof we write $\mathcal{P} \subseteq \mathcal{K}$ for the set of convex polytopes $P \subseteq \mathbb{R}^n$. For given $m \in \{0, \ldots, n-1\}$, we furthermore write $\mathcal{S}_m^{n-1}$ for the set of spherical polytopes $p \subseteq S^{n-1}$ of dimension at most $m$. Moreover, let $\Phi = \Phi_0 + \ldots + \Phi_n$ be the Kiderlen–Weil decomposition of $\Phi$. Clearly, it is enough to show that the homogeneous part $\Phi_i$ is a valuation for all $i = 0, \ldots, n$.

By formula (3.1) of [KW99], $\Phi_n = c\,\mathrm{VOL}_n$ for some $c \in \mathbb{C}$. Therefore, we have

$$\begin{aligned}
&\Phi_n(K \cup L, \beta) = c\,\mathrm{VOL}_n\big((K \cup L) \cap \beta\big) = c\,\mathrm{VOL}_n\big((K \cap \beta) \cup (L \cap \beta)\big) \\
&= c\,\mathrm{VOL}_n(K \cap \beta) + c\,\mathrm{VOL}_n(L \cap \beta) - c\,\mathrm{VOL}_n(K \cap L \cap \beta) \\
&= \Phi_n(K, \beta) + \Phi_n(L, \beta) + \Phi_n(K \cap L, \beta)
\end{aligned}$$

for all $K, L, K \cup L \in \mathcal{K}$ and all $\beta \in \mathcal{B}$ which shows that $\Phi_n$ is an $\mathcal{M}$-valued valuation.

Now assume that $i \in \{0, \ldots, n-1\}$. By formula (3.2) of [KW99], there exists a continuous, simple function $f_i \colon \mathcal{S}_{n-1-i}^{n-1} \to \mathbb{C}$, uniquely determined by $\Phi_i$, such that

$$\Phi_i(P) = \sum_{F \in \mathcal{F}_i(P)} f_i\big(n(P, F)\big)\,\mathrm{vol}_F \in \mathcal{M}$$

holds for all polytopes $P \in \mathcal{P}$, where $\mathcal{F}_i(P)$ denotes the set of all $i$-faces of $P$ and $n(P, F) \subseteq S^{n-1}$ denotes the set of all outer unit normals of $P$ at the face $F$.

One easily sees that additivity of $f_i$ implies additivity of $\Phi_i$. Moreover, by Lemma 1.3 of [McM93], it is enough to show that $f_i$ is weakly additive. Therefore, let $p \in \mathcal{S}_{n-1-i}^{n-1}$ and let $H \subseteq \mathbb{R}^n$ be a $(n-1)$-dimensional linear subspace which divides $p$ into the pieces $p_1, p_2 \in \mathcal{S}_{n-1-i}^{n-1}$. Since $f_i$ is simple, we can assume that $p$ is $(n-1-i)$-dimensional without loss of generality. We have to show that $f_i(p) = f_i(p_1) + f_i(p_2)$.

To do so, choose a polytope $P \in \mathcal{P}$ and a face $F \in \mathcal{F}_i(P)$ such that $p = n(P, F)$. Moreover, choose a point $x \in \mathrm{relint}(F)$ and let $\varepsilon > 0$ be small enough such that the open ball $U = B_\varepsilon(x)$ satisfies $U \cap F \subseteq \mathrm{relint}(F)$ and

$$\mathrm{cl}_{\mathbb{R}^n}(U) \cap F' = \emptyset$$

for all $F' \in \mathcal{F}_i(P) \setminus \{F\}$. The ball $U$ clearly satisfies $\mathrm{vol}_F(\partial U) = 0$. Since $\mathrm{cl}(U)$ does not meet any other $i$-faces of $P$, we obtain

$$\Phi_i(P, \partial U) = f_i(p)\,\mathrm{vol}_F(\partial U) = 0.$$

Let us choose $u \in S^{n-1}$ such that $H = u^\perp$. For $t > 0$ we consider the polytope $P_t = P + t[0, u]$. For small enough $t$, the polytope $P_t$ has the two $i$-faces $F = F + 0u$ and $F_t = F + tu$. By potentially replacing $u$ by $-u$, we can assume that these two faces satisfy

$$n(P_t, F) = p_1, \quad n(P_t, F_t) = p_2.$$

Since $P_t$ converges to $P$ and $F_t$ converges to $F$ as $t \to 0$, we get the weak convergences

$$\Phi_i(P_t) \xrightarrow{w} \Phi_i(P), \quad \mathrm{vol}_{F_t} \xrightarrow{w} \mathrm{vol}_F$$

in the space $\mathcal{M}$. Moreover, we have already seen that $\Phi_i(P, \partial U) = 0$ and $\mathrm{vol}_F(\partial U) = 0$. Therefore, the Portmanteau theorem yields

$$\Phi_i(P_t, U) \to \Phi_i(P, U), \quad \mathrm{vol}_{F_t}(U) \to \mathrm{vol}_F(U)$$

as $t \to 0$. For sufficiently small $t$, we may assume that $\mathrm{cl}(U)$ meets exactly the two $i$-faces $F$ and $F_t$ of $P$. Using the formula of Kiderlen–Weil, this implies

$$\Phi_i(P_t, U) = f_i(p_1)\,\mathrm{vol}_F(U) + f_i(p_2)\,\mathrm{vol}_{F_t}(U).$$

Applying the same formula to $P$ and $U$, we obtain $\Phi_i(P, U) = f_i(p)\,\mathrm{vol}_F(U)$. Letting $t \to 0$, we therefore obtain

$$f_i(p_1)\,\mathrm{vol}_F(U) + f_i(p_2)\,\mathrm{vol}_F(U) = \Phi_i(P, U) = f_i(p)\,\mathrm{vol}_F(U)$$

which implies the claim, since $U$ satisfies $\mathrm{vol}_F(U) > 0$ by construction. □

## 3.4 Topology

The goal of this section is to introduce a topology on Curv. To do so, we first prove that the total variation norm of curvature measures is bounded on uniformly bounded families of convex bodies. This allows us to introduce a natural family of norms on Curv which turn Curv into a Banach space. As we will see all these norms are equivalent, hence inducing a unique topology on Curv. This construction is in many ways analogous to Theorem 2.17 which is concerned with the topology on Val.

For the following lemma, let us first discuss some background from the theory of measures. Let $X$ be a complete separable metric space and let $\mathcal{A}$ be a family of complex Borel measures on $X$. Prokhorov's theorem states that $\mathcal{A}$ is weak* sequentially compact if and only if $\mathcal{A}$ is uniformly tight and uniformly bounded in the total variation norm. For the relevant definitions and a proof of Prokhorov's theorem we refer the reader to Section 8.6 of [Bog07]. Note that the metric space $\mathbb{R}^n$ is complete and separable. Therefore, we will make use of the following direct consequence of Prokhorov's theorem. If $\mathcal{A} \subseteq \mathcal{M}$ is sequentially compact in the weak* topology, then $\mathcal{A}$ is uniformly bounded in the total variation norm.

**Lemma 3.13.** *Let $\Phi \in \mathrm{Curv}$ and let $B \subseteq \mathbb{R}^n$ be bounded. Then,*

$$\sup_{K \subseteq B} \left\|\Phi(K)\right\| < \infty.$$

*Proof.* Note that $B$ is bounded if and only if its closure $\bar{B}$ is bounded. Moreover, we obviously have

$$\sup_{K \subseteq B} \left\|\Phi(K)\right\| \leq \sup_{K \subseteq \bar{B}} \left\|\Phi(K)\right\|$$

because of $B \subseteq \bar{B}$. Therefore, we can assume that $B$ is closed without loss of generality. This yields that $B \subseteq \mathbb{R}^n$ is compact as it is closed and bounded. In the following,

we consider the family of convex bodies $\mathcal{K}_B = \{K \in \mathcal{K} \mid K \subseteq B\}$ and the family of measures $\mathcal{M}_B = \{\Phi(K) \in \mathcal{M} \mid K \subseteq B\}$. By the Blaschke selection theorem, $\mathcal{K}_B$ is sequentially compact. Since the curvature measure $\Phi$ is weak* continuous, it follows that $\mathcal{M}_B = \Phi(\mathcal{K}_B)$ is sequentially compact with respect to the weak* topology. From the discussion above it follows that $\mathcal{M}_B$ is uniformly bounded in the total variation norm. Thus, we obtain

$$\sup_{K \subseteq B} \left\|\Phi(K)\right\| = \sup_{\mu \in \mathcal{M}_B} \|\mu\| < \infty$$

as required. □

Assume now that $B \subseteq \mathbb{R}^n$ is a bounded set with non-empty interior. For a given curvature measure $\Phi \in \mathrm{Curv}$, we set

$$\|\Phi\|_B = \sup_{K \subseteq B} \left\|\Phi(K)\right\|.$$

**Lemma 3.14.** *For any bounded set $B \subseteq \mathbb{R}^n$ with non-empty interior, $\|\cdot\|_B$ is a norm on* Curv.

*Proof.* Using the previous lemma and the fact that the total variation norm is indeed a norm on $\mathcal{M}$, we obtain

$$0 \leq \|\Phi\|_B < \infty$$

for all $\Phi \in \mathrm{Curv}$.

Next, assume that $\|\Phi\|_B = 0$ for some curvature measure $\Phi \in \mathrm{Curv}$. Moreover, we write $\Phi = \Phi_0 + \ldots + \Phi_n$ for the Kiderlen–Weil decomposition of $\Phi$. The assumption implies that $\left\|\Phi(K)\right\| = 0$ for all convex bodies $K \subseteq B$ and hence $\Phi(K) = 0 \in \mathcal{M}$ for all $K \subseteq B$. Let $L \in \mathcal{K}$ be an arbitrary convex body now. Since $B$ has non-empty interior, there exist $\varepsilon_0 > 0$ and $x \in \mathbb{R}^n$ such that $\varepsilon L + x \subseteq B$ for all $\varepsilon \in (0, \varepsilon_0)$. Therefore, we obtain

$$0 = \Phi(\varepsilon L + x, \varepsilon\beta + x) = \Phi(\varepsilon L, \varepsilon\beta) = \sum_{i=0}^{n} \varepsilon^i \Phi_i(L, \beta)$$

for all Borel sets $\beta \subseteq \mathbb{R}^n$, all indices $i \in \{0, \ldots, n\}$ and all $\varepsilon \in (0, \varepsilon_0)$. Note that the right hand side defines a polynomial

$$(0, \varepsilon_0) \to \mathbb{C},\ \varepsilon \mapsto \sum_{i=0}^{n} \varepsilon^i \Phi_i(L, \beta).$$

Since $\beta \subseteq \mathbb{R}^n$ is an arbitrary Borel set, comparison of coefficients yields that $\Phi_i(L) = 0$ for all $i$. Therefore, we obtain $\Phi(L) = 0$ as required.

For positive homogeneity, let $\lambda \in \mathbb{C}$ and $\Phi \in \mathrm{Curv}$. Then, we have

$$\|\lambda\Phi\|_B = \sup_{K \subseteq B} \left\|\lambda\Phi(K)\right\| = \sup_{K \subseteq B} |\lambda| \left\|\Phi(K)\right\| = |\lambda| \sup_{K \subseteq B} \left\|\Phi(K)\right\| = |\lambda| \|\Phi\|_B$$

as required.

Lastly, any pair of curvature measures $\Phi, \Psi \in \mathrm{Curv}$ satisfies the triangle inequality as

$$\begin{aligned}\|\Phi+\Psi\|_B &= \sup_{K\subseteq B}\big\|\Phi(K)+\Psi(K)\big\| \leq \sup_{K\subseteq B}\big(\big\|\Phi(K)\big\|+\big\|\Psi(K)\big\|\big)\\ &\leq \sup_{K\subseteq B}\big\|\Phi(K)\big\|+\sup_{L\subseteq B}\big\|\Psi(L)\big\| = \|\Phi\|_B+\|\Psi\|_B\,.\end{aligned}$$

Thus $\|\cdot\|_B$ indeed defines a norm on Curv. □

Next, we want to show that all those norms are equivalent. To do so, we first need to know how the norms behave under translation and positive scaling of the set $B$.

**Lemma 3.15.** *Let $B \subseteq \mathbb{R}^n$ be a bounded set with non-empty interior, $x \in \mathbb{R}^n$ and $\lambda > 0$. Then, the equations*

$$\begin{aligned}\|\Phi\|_{B+x} &= \|\Phi\|_B\,,\\ \|\Phi\|_{\lambda B} &= \left\|\sum_{i=0}^{n}\lambda^i\Phi_i\right\|_B\end{aligned}$$

*hold for all curvature measures $\Phi \in \mathrm{Curv}$, where $\Phi = \Phi_0 + \ldots + \Phi_n$ denotes the Kiderlen–Weil decomposition of $\Phi$.*

*Proof.* To show that the norms of the form $\|\cdot\|_B$ are translation-invariant in $B$, note that a family $\{\beta_j \mid j \in \mathbb{N}\}$ of Borel subsets of $\mathbb{R}^n$ forms a partition of $\mathbb{R}^n$ if and only if $\{\beta_j + x \mid j \in \mathbb{N}\}$ does. Due to translation-invariance of $\Phi$, this yields

$$\begin{aligned}\|\Phi\|_{B+x} &= \sup_{L\subseteq B+x}\big\|\Phi(L)\big\| = \sup_{L\subseteq B+x}\sup\sum_{j\in\mathbb{N}}\Big|\Phi\big(L,\beta_j\big)\Big| = \sup_{K\subseteq B}\sup\sum_{j\in\mathbb{N}}\Big|\Phi\big(K+x,\beta_j\big)\Big|\\ &= \sup_{K\subseteq B}\sup\sum_{j\in\mathbb{N}}\Big|\Phi\big(K+x,\beta_j+x\big)\Big| = \sup_{K\subseteq B}\sup\sum_{j\in\mathbb{N}}\Big|\Phi\big(K,\beta_j\big)\Big| = \|\Phi\|_B\,,\end{aligned}$$

where the second supremum is taken over all countable partitions $\{\beta_j \mid j \in \mathbb{N}\}$ of $\mathbb{R}^n$.

Similarly, $\{\beta_j \mid j \in \mathbb{N}\}$ forms a Borel partition of $\mathbb{R}^n$ if and only if $\{\lambda\beta_j \mid j \in \mathbb{N}\}$ does. Hence, we compute

$$\begin{aligned}\|\Phi\|_{\lambda B} &= \sup_{L\subseteq \lambda B}\big\|\Phi(L)\big\| = \sup_{L\subseteq \lambda B}\sup\sum_{j\in\mathbb{N}}\Big|\Phi\big(L,\beta_j\big)\Big| = \sup_{K\subseteq B}\sup\sum_{j\in\mathbb{N}}\Big|\Phi\big(\lambda K,\beta_j\big)\Big|\\ &= \sup_{K\subseteq B}\sup\sum_{j\in\mathbb{N}}\Big|\Phi\big(\lambda K,\lambda\beta_j\big)\Big| = \sup_{K\subseteq B}\sup\sum_{j\in\mathbb{N}}\left|\sum_{i=0}^{n}\lambda^i\Phi_i\big(K,\beta_j\big)\right| = \left\|\sum_{i=0}^{n}\lambda^i\Phi_i\right\|_B\end{aligned}$$

which yields the claimed behaviour under positive scaling. □

**Theorem 3.16.** *For any two bounded subsets $B_1, B_2 \subseteq \mathbb{R}^n$ with non-empty interior, the induced norms $\|\cdot\|_{B_1}$ and $\|\cdot\|_{B_2}$ on* Curv *are equivalent.*

*Proof.* Let $B \subseteq \mathbb{R}^n$ be a bounded set with non-empty interior. Since $\|\cdot\|_B$ is translation-invariant in $B$, we may assume that $0 \in \operatorname{int}(B)$. By the assumptions on $B$, there exist $\lambda \geq \mu > 0$ such that $\mu B^n \subseteq B \subseteq \lambda B^n$. In particular, we have

$$\|\Phi\|_{\mu B^n} \leq \|\Phi\|_B \leq \|\Phi\|_{\lambda B^n}$$

for all $\Phi \in \operatorname{Curv}$. Thus, it is satisfactory to show that the norms $\|\cdot\|_{\lambda B^n}$ and $\|\cdot\|_{\mu B^n}$ are equivalent for all $\lambda \geq \mu > 0$.

To do so, let $\Phi = \Phi_0 + \ldots + \Phi_n \in \operatorname{Curv}$ be an arbitrary curvature measure. Obviously, we have $\|\Phi\|_{\mu B^n} \leq \|\Phi\|_{\lambda B^n}$ because of $\mu \leq \lambda$. By the previous lemma, we also have

$$\begin{aligned}\|\Phi\|_{\lambda B^n} &= \|\Phi\|_{(\lambda\mu^{-1})\mu B^n} = \left\| \sum_{i=0}^{n} \left(\lambda\mu^{-1}\right)^i \Phi_i \right\|_{\mu B^n} \leq \sum_{i=0}^{n} \left\| \left(\lambda\mu^{-1}\right)^i \Phi_i \right\|_{\mu B^n} \\ &= \sum_{i=0}^{n} \left(\lambda\mu^{-1}\right)^i \|\Phi_i\|_{\mu B^n} \leq \left(\lambda\mu^{-1}\right)^n \sum_{i=0}^{n} \|\Phi_i\|_{\mu B^n},\end{aligned}$$

where the last inequality follows from $\lambda \geq \mu > 0$ which implies $\lambda\mu^{-1} \geq 1$. For any $\rho > 0$, we consider $\Phi(\rho) : \mathcal{K} \to \mathcal{M}$ which is defined by $\Phi(\rho)(K, \beta) = \Phi(\rho K, \rho\beta)$ for all $K$ and $\beta$. From the Kiderlen–Weil decomposition of $\Phi$ it follows that

$$\Phi(\rho) = \sum_{i=0}^{n} \rho^i \Phi_i$$

which particularly shows that $\Phi(\rho) \in \operatorname{Curv}$. Moreover, note that we have

$$\begin{aligned}\left\|\Phi(\rho)\right\|_{\mu B^n} &= \sup_{K \subseteq \mu B^n} \sup \sum_{j \in \mathbb{N}} \left|\Phi\left(\rho K, \rho\beta_j\right)\right| = \sup_{K \subseteq \mu B^n} \sup \sum_{j \in \mathbb{N}} \left|\Phi\left(\rho K, \beta_j\right)\right| \\ &= \sup_{K \subseteq \mu B^n} \left\|\Phi(\rho K)\right\| = \sup_{L \subseteq \rho\mu B^n} \left\|\Phi(L)\right\| = \|\Phi\|_{\rho\mu B^n}\end{aligned}$$

for all $\rho > 0$. Fix $n+1$ real numbers satisfying $0 < \rho_0 < \rho_1 < \ldots < \rho_n \leq 1$. From the Kiderlen–Weil decomposition of $\Phi(\rho_k)$, we obtain the following equation:

$$\begin{pmatrix} \Phi(\rho_0) \\ \Phi(\rho_1) \\ \vdots \\ \Phi(\rho_n) \end{pmatrix} = \begin{pmatrix} 1 & \rho_0 & \cdots & \rho_0^n \\ 1 & \rho_1 & \cdots & \rho_1^n \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \rho_n & \cdots & \rho_n^n \end{pmatrix} \begin{pmatrix} \Phi_0 \\ \Phi_1 \\ \vdots \\ \Phi_n \end{pmatrix}.$$

Let $M$ denote the Vandermonde matrix in the equation above. Since the $\rho_k$ are pairwise distinct and the determinant of a Vandermonde matrix satisfies

$$\det M = \prod_{0 \leq k < l \leq n} (\rho_l - \rho_k),$$

we see that $M$ is invertible. By inverting the matrix $M$ we obtain a constant $C > 0$, which solely depends on the dimension $n$ and the fixed numbers $\rho_k$, such that

$$\|\Phi_i\|_{\mu B^n} \leq C \max_{k=0,\dots,n} \left\|\Phi\left(\rho_k\right)\right\|_{\mu B^n} = C \max_{k=0,\dots,n} \|\Phi\|_{\rho_k \mu B^n} \leq C \|\Phi\|_{\mu B^n}$$

holds for all $i = 0, \dots, n$, where the last inequality follows from the assumption that $\rho_k \leq 1$ for all $k$. Combining the computations above yields that

$$\|\Phi\|_{\mu B^n} \leq \|\Phi\|_{\lambda B^n} \leq \left(\lambda \mu^{-1}\right)^n \sum_{i=0}^{n} \|\Phi_i\|_{\mu B^n} \leq \left(\lambda \mu^{-1}\right)^n (n+1)\, C \|\Phi\|_{\mu B^n}$$

which proves the required equivalence. □

Recall that two norms are equivalent if and only if they induce the same topology. This implies that for all bounded sets $B \subseteq \mathbb{R}^n$ with non-empty interior, the topologies induced by $\|\cdot\|_B$ coincide. Hence, we have equipped Curv with a unique norm topology.

**Theorem 3.17.** *Let $B$ be a bounded subset of $\mathbb{R}^n$ with non-empty interior. Then, $\left(\mathrm{Curv}, \|\cdot\|_B\right)$ is a Banach space.*

*Proof.* We have already seen that $\|\cdot\|_B$ is a norm on Curv. Therefore, only completeness is left to show. By equivalence of norms, see Theorem 3.16, it is moreover satisfactory to show that Curv is complete with respect to the norm induced by one specific bounded set $B$ with non-empty interior. We choose the unit ball $B = B^n$.

To do so, recall from the Vandermonde matrix argument in the previous proof (applied to $\mu = 1$) that there exists a constant $C > 0$ such that $\|\Psi_i\|_{B^n} \leq C \|\Psi\|_{B^n}$ for all $\Psi = \Psi_0 + \ldots + \Psi_n \in \mathrm{Curv}$ and all $i = 0, \dots, n$. Let

$$\left(\Phi^k\right)_{k \in \mathbb{N}} \subseteq \mathrm{Curv}$$

be a Cauchy sequence with respect to $\|\cdot\|_{B^n}$. For all $k$, we let $\Phi^k = \Phi_0^k + \ldots + \Phi_n^k$ denote the Kiderlen–Weil decomposition of $\Phi^k$. Moreover, let $K \subseteq \mathbb{R}^n$ be an arbitrary convex body and choose $r = r(K) > 0$ such that $K \subseteq rB^n$. Applying Lemma 3.15 and the estimate above to this situation, we obtain

$$\begin{aligned}
\left\|\Phi^k(K) - \Phi^l(K)\right\| &\leq \left\|\Phi^k - \Phi^l\right\|_{rB^n} = \left\|\sum_{i=0}^{n} r^i \left(\Phi_i^k - \Phi_i^l\right)\right\|_{B^n} \\
&\leq \sum_{i=0}^{n} r^i \left\|\Phi_i^k - \Phi_i^l\right\|_{B^n} \leq C \left\|\Phi^k - \Phi^l\right\|_{B^n} \sum_{i=0}^{n} r^i
\end{aligned}$$

for all $k, l \in \mathbb{N}$. By assumption, the last term converges to 0 as $k, l \to \infty$. Therefore, we have shown that

$$\left(\Phi^k(K)\right)_{k \in \mathbb{N}}$$

is a Cauchy sequence in $\mathcal{M}$. Since $\mathcal{M}$ is a Banach space, it follows that there exists a unique measure $\Phi(K) \in \mathcal{M}$ such that

$$\lim_{k\to\infty} \left\| \Phi^k(K) - \Phi(K) \right\| = 0.$$

Since $K$ is an arbitrary convex body, this defines a map $\Phi\colon \mathcal{K} \to \mathcal{M}$, $K \mapsto \Phi(K)$.

Let us write $(a_k)_{k\in\mathbb{N}}$ for the real sequence given by

$$a_k = \left\| \Phi^k - \Phi \right\|_{B^n} = \sup_{K\subseteq B^n} \left\| \left(\Phi^k - \Phi\right)(K) \right\|.$$

We have to show that $a_k \to 0$ as $k \to \infty$. To do so, fix an arbitrary $\varepsilon > 0$. Since $\Phi^1, \Phi^2, \ldots$ is a Cauchy sequence, there exists $k_0 \in \mathbb{N}$ such that

$$\left\| \Phi^k(K) - \Phi^l(K) \right\| < \varepsilon$$

for all $k, l \geq k_0$ and for all $K \subseteq B^n$. Letting $l \to \infty$, we obtain

$$\left\| \Phi^k(K) - \Phi(K) \right\| < \varepsilon$$

for all $k \geq k_0$ and all $K \subseteq B^n$. Of course, we can now take the supremum over all $K$ which yields

$$\sup_{K\subseteq B^n} \left\| \Phi^k(K) - \Phi(K) \right\| \leq \varepsilon$$

for all $k \geq k_0$ as required. It is therefore only left to show that $\Phi$ indeed is a curvature measure.

For all $K \in \mathcal{K}$, $\beta \in \mathcal{B}$ and $x \in \mathbb{R}^n$, we compute

$$\Phi(K + x, \beta + x) = \lim_{k\to\infty} \Phi^k(K + x, \beta + x) = \lim_{k\to\infty} \Phi^k(K, \beta) = \Phi(K, \beta)$$

which shows that $\Phi$ is translation-invariant.

Assume that $K \cap U = L \cap U$ for two convex bodies $K, L \subseteq \mathbb{R}^n$ and an open set $U \subseteq \mathbb{R}^n$. For any Borel set $\beta \subseteq U$, we see that

$$\Phi(K, \beta) = \lim_{k\to\infty} \Phi^k(K, \beta) = \lim_{k\to\infty} \Phi^k(L, \beta) = \Phi(L, \beta)$$

which shows that $\Phi$ is defined locally.

Lastly, let $K_1, K_2, \ldots \in \mathcal{K}$ be a convergent sequence of convex bodies with $K_j \to K$ as $j \to \infty$ and let $f \in C_0(\mathbb{R}^n)$. From the fact that a convergent sequence of convex bodies is uniformly bounded, one easily sees that the Moore-Osgood theorem on interchanging limits is applicable to our situation. Thus, we obtain

$$\begin{aligned} &\lim_{j\to\infty} \Phi\left(K_j, f\right) = \lim_{j\to\infty} \lim_{k\to\infty} \Phi^k\left(K_j, f\right) = \lim_{k\to\infty} \lim_{j\to\infty} \Phi^k\left(K_j, f\right) \\ &= \lim_{k\to\infty} \Phi^k(K, f) = \Phi(K, f) \end{aligned}$$

as required. □

Let us conclude this section with the following useful observation.

**Corollary 3.18.** *For all $i = 0, \dots, n$ and $\varepsilon \in \{+, -\}$, the linear subspaces $\mathrm{Curv}_i$, $\mathrm{Curv}^\varepsilon$ and $\mathrm{Curv}_i^\varepsilon$ are closed in $\mathrm{Curv}$. In particular, they are Banach spaces themselves.*

*Proof.* Throughout this proof, let $\Phi^1, \Phi^2, \dots \in \mathrm{Curv}$ be a convergent sequence of curvature measures such that $\Phi^k \to \Phi$ as $k \to \infty$.

If $\Phi^k$ is $i$-homogeneous for all $k$, then so is $\Phi$ since

$$\Phi(\lambda K, \lambda\beta) = \lim_{k\to\infty} \Phi^k(\lambda K, \lambda\beta) = \lambda^i \lim_{k\to\infty} \Phi^k(K, \beta) = \lambda^i \Phi(K, \beta)$$

for all $\lambda > 0$, $K \in \mathcal{K}$ and $\beta \in \mathcal{B}$. Hence, $\mathrm{Curv}_i$ is closed in $\mathrm{Curv}$.

Similarly, assume that all the $\Phi^k$ are of parity $\varepsilon$. Then,

$$\Phi(-K, -\beta) = \lim_{k\to\infty} \Phi^k(-K, -\beta) = \varepsilon \lim_{k\to\infty} \Phi^k(K, \beta) = \varepsilon\Phi(K, \beta)$$

for all $K$ and $\beta$ which shows that $\Phi$ is also of parity $\varepsilon$. Therefore, $\mathrm{Curv}^\varepsilon$ is closed in $\mathrm{Curv}$.

Lastly, $\mathrm{Curv}_i^\varepsilon$ is of course closed in $\mathrm{Curv}$ as $\mathrm{Curv}_i^\varepsilon = \mathrm{Curv}_i \cap \mathrm{Curv}^\varepsilon$. □

## 3.5 Globalization and Bernig Embedding

Throughout this section, we write $\mathcal{V}$ for the set of continuous valuations $\mathcal{K} \to \mathbb{C}$. One easily verifies that $\mathcal{V}$ is a linear subspace of $C(\mathcal{K})$. Moreover, we have $\mathrm{Val} \subseteq \mathcal{V}$ by definition.

Let $\Phi \in \mathrm{Curv}$ be a curvature measure and assume that the function $f\colon \mathbb{R}^n \to \mathbb{C}$ is $\Phi(K)$-integrable for all $K \in \mathcal{K}$. Then, we define $\Phi_f\colon \mathcal{K} \to \mathbb{C}$ by $\Phi_f(K) = \Phi(K, f)$ for all $K$. Note that the assumption is particularly satisfied if $f$ is continuous.

**Lemma 3.19.** *Let $\Phi \in \mathrm{Curv}$ be a curvature measure and $f\colon \mathbb{R}^n \to \mathbb{C}$ a continuous function. Then the following assertions hold:*

- $\Phi_f \in \mathcal{V}$.
- *The mapping $\cdot_f\colon \mathrm{Curv} \to \mathcal{V}$, $\Phi \mapsto \Phi_f$ is $\mathbb{C}$-linear.*
- *If $f$ is constant, then $\Phi_f \in \mathrm{Val}$.*

*Proof.* Let $K_1, K_2, \dots \in \mathcal{K}$ be a convergent sequence with $\lim K_j = K \in \mathcal{K}$. By Proposition 3.6, we have

$$\Phi_f(K_j) = \Phi(K_j, f) \to \Phi(K, f) = \Phi_f(K)$$

as $j \to \infty$ which shows that $\Phi_f$ is continuous. Now, assume that $K, L \in \mathcal{K}$ satisfy $K \cup L \in \mathcal{K}$. Using Lemma 3.1 and the valuation property of curvature measures, we can compute

$$\begin{aligned}\Phi_f(K \cup L) + \Phi_f(K \cap L) &= \big(\Phi(K \cup L) + \Phi(K \cap L)\big)(f)\\ &= \big(\Phi(K) + \Phi(L)\big)(f) = \Phi_f(K) + \Phi_f(L)\end{aligned}$$

which shows that $\Phi_f$ is a valuation. Therefore, we have $\Phi_f \in \mathcal{V}$.

For the second assertion, let $\lambda \in \mathbb{C}$ and $\Phi, \Psi \in \mathrm{Curv}$. Then,

$$(\lambda\Phi + \Psi)_f(K) = (\lambda\Phi + \Psi)(K, f) = \lambda\Phi(K, f) + \Psi(K, f) = \lambda\Phi_f(K) + \Psi_f(K)$$

for all convex bodies $K \subseteq \mathbb{R}^n$ which shows that $\cdot_f$ is $\mathbb{C}$-linear.

Lastly, assume that $f$ is constant, say $f(p) = c \in \mathbb{C}$ for all $p \in \mathbb{R}^n$. For all $K \in \mathcal{K}$ and $x \in \mathbb{R}^n$, this implies

$$\begin{aligned}\Phi_f(K + x) &= \int_{\mathbb{R}^n} c \,\mathrm{d}\,\Phi(K + x) = c\int_{\mathbb{R}^n} 1 \,\mathrm{d}\,\Phi(K + x) = c\Phi(K + x, \mathbb{R}^n)\\ &= c\Phi(K + x, \mathbb{R}^n + x) = c\Phi(K, \mathbb{R}^n) = c\int_{\mathbb{R}^n} 1 \,\mathrm{d}\,\Phi(K) = \int_{\mathbb{R}^n} c \,\mathrm{d}\,\Phi(K) = \Phi_f(K)\end{aligned}$$

which combined with the first bullet point shows that $\Phi_f \in \mathrm{Val}$. □

Of particular importance is the case $f = 1$. For any curvature measure $\Phi \in \mathrm{Curv}$, the induced valuation $\mathrm{glob}(\Phi) = \Phi_1 = \Phi(\cdot, \mathbb{R}^n) \in \mathrm{Val}$ is called the *globalization* of $\Phi$. Moreover, the corresponding $\mathbb{C}$-linear map

$$\mathrm{glob}\colon \mathrm{Curv} \to \mathrm{Val},\ \Phi \mapsto \mathrm{glob}(\Phi)$$

is called the *globalization map*.

Note that the globalization map preserves degree and parity. To see this, assume that $\Phi \in \mathrm{Curv}$ is $i$-homogeneous. Then, we have

$$\mathrm{glob}(\Phi)(\lambda K) = \Phi(\lambda K, \mathbb{R}^n) = \Phi(\lambda K, \lambda\,\mathbb{R}^n) = \lambda^i\Phi(K, \mathbb{R}^n) = \lambda^i\,\mathrm{glob}(\Phi)(K)$$

for all $K \in \mathcal{K}$ and $\lambda > 0$. Similarly, if $\Phi$ is of parity $\varepsilon \in \{+, -\}$, we have

$$\mathrm{glob}(\Phi)(-K) = \Phi(-K, \mathbb{R}^n) = \Phi(-K, -\,\mathbb{R}^n) = \varepsilon\,\mathrm{glob}(\Phi)(K)$$

for all convex bodies $K \subseteq \mathbb{R}^n$.

**Lemma 3.20.** *The globalization map* $\mathrm{glob}\colon \mathrm{Curv} \to \mathrm{Val}$ *is continuous and has dense image.*

*Proof.* Let $B \subseteq \mathbb{R}^n$ be a bounded set with non-empty interior. Then, we have

$$\big\|\mathrm{glob}(\Phi)\big\|_B = \sup_{K\subseteq B}\big|\Phi(K, \mathbb{R}^n)\big| \leq \sup_{K\subseteq B}\big|\Phi(K)\big|(\mathbb{R}^n) = \sup_{K\subseteq B}\big\|\Phi(K)\big\| = \|\Phi\|_B$$

for all $\Phi \in \mathrm{Curv}$ which shows that glob is continuous.

For the second assertion, let $\varphi \in \mathrm{Val}^{\mathrm{sm}}$ be an arbitrary smooth valuation. By Theorem 2.23, there exist a constant $c \in \mathbb{C}$ and a differential form $\omega \in \Omega^{n-1}(S\,\mathbb{R}^n)^{\mathrm{tr}}$ such that $\varphi = c\,\mathrm{vol}_n + [\![\omega]\!]$. We define $\Phi \in \mathrm{Curv}^{\mathrm{sm}} \subseteq \mathrm{Curv}$ as $\Phi = c\,\mathrm{VOL}_n + [\omega]$. Clearly, the globalization of $\Phi$ satisfies

$$\mathrm{glob}(\Phi) = c\,\mathrm{VOL}_n(\cdot, \mathbb{R}^n) + \int_{\mathrm{nc}(\cdot)\cap\pi^{-1}(\mathbb{R}^n)} \omega = c\,\mathrm{vol}_n + \int_{\mathrm{nc}(\cdot)} \omega = c\,\mathrm{vol}_n + [\![\omega]\!] = \varphi$$

which implies that $\mathrm{Val}^{\mathrm{sm}} \subseteq \mathrm{im}(\mathrm{glob})$. Since $\mathrm{Val}^{\mathrm{sm}}$ is dense in Val, the claim follows. □

Unfortunately, the question whether the globalization map is surjective remains open at this point.

Let us introduce some useful terminology. Recall that a function $f\colon \mathbb{R}^n \to \mathbb{C}$ is called *linear* if there exists a $\mathbb{C}$-linear function $F\colon \mathbb{C}^n \to \mathbb{C}$ such that $F(x) = f(x)$ for all $x \in \mathbb{R}^n$, that is $f \in L(\mathbb{R}^n)$. Similarly, we call $f$ *affine* if there exists a $\mathbb{C}$-affine continuation $F\colon \mathbb{C}^n \to \mathbb{C}$ of $f$.

Remarkably, we can decide whether a curvature measure is zero by solely considering affine functions. The argument is due to Bernig and can be found in the context of smooth curvature measures in Lemma 4.5 of [SW21].

**Lemma 3.21.** *A curvature measure $\Phi \in \mathrm{Curv}$ satisfies $\Phi = 0$ if and only if $\Phi_f = 0 \in \mathcal{V}$ for all affine functions $f\colon \mathbb{R}^n \to \mathbb{C}$.*

*Proof.* Firstly, note that one implication is trivially satisfied. Clearly, $\Phi = 0$ implies $\Phi_f = 0$ for all (in particular, all affine) functions $f\colon \mathbb{R}^n \to \mathbb{C}$.

For the converse implication, assume that $\Phi \in \mathrm{Curv}$ satisfies $\Phi_f = 0$ for all affine functions $f\colon \mathbb{R}^n \to \mathbb{C}$. Let $g\colon \mathbb{R}^n \to \mathbb{C}$ be a continuous, piecewise affine function which means that there exists a locally finite triangulation $\{T_i \mid i \in I\}$ of $\mathbb{R}^n$ such that $g|_{T_i}$ is affine for all $i \in I$. Moreover, let $K \subseteq \mathbb{R}^n$ be an arbitrary convex body and let $I_K \subseteq I$ be the set of all indices $i \in I$ for which $K \cap T_i$ is non-empty. Since the triangulation $\{T_i\}$ is locally finite and $K$ is compact, $I_K$ is necessarily finite. By the assumption that $f \mapsto \Phi_f$ vanishes on affine functions, we obtain $\Phi_g(K \cap T_i) = 0$ for all $i \in I_K$. As $\Phi_g \in \mathcal{V}$ it satisfies the inclusion-exclusion principle; this was proved in [Gro78]. Applying this to the finite index set $I_K$, we obtain $\Phi_g(K) = 0$. Since $K$ is an arbitrary convex body, it follows that $\Phi_g = 0$. Since any continuous function can be approximated uniformly on compact sets by continuous, piecewise affine functions, it follows that $\Phi_h = 0$ for all $h \in C_0(\mathbb{R}^n)$. By Lemma 3.1, this yields the required $\Phi = 0$. □

For any curvature measure $\Phi \in \mathrm{Curv}$ and any convex body $K \in \mathcal{K}$, we let

$$B\Phi(K) = \int_{\mathbb{R}^n} x \,\mathrm{d}\,\Phi(K, x) \in \mathbb{C}^n .$$

Applying Lemma 3.19 to the projection maps $p_i\colon \mathbb{R}^n \to \mathbb{C}$, $(x_1, \dots, x_n) \mapsto x_i$ yields that $B\Phi\colon \mathcal{K} \to \mathbb{C}^n$ is a continuous $\mathbb{C}^n$-valued valuation and that the map

$$B\colon \mathrm{Curv} \to \mathcal{V} \otimes \mathbb{C}^n,\ \Phi \mapsto B\Phi$$

is $\mathbb{C}$-linear.

**Lemma 3.22.** *Let $\Phi \in \mathrm{Curv}$ satisfy $\mathrm{glob}(\Phi) = 0$. Then the valuation $B\Phi$ is translation-invariant, that is $B\Phi \in \mathrm{Val} \otimes \mathbb{C}^n$.*

*Proof.* Let us start with the following useful observation. Let $K \subseteq \mathbb{R}^n$ be a convex body, $\beta \subseteq \mathbb{R}^n$ a Borel set and $t \in \mathbb{R}^n$. Of course, the indicator function $\mathbf{1}_\beta$ is $\Phi(A)$-integrable for all convex bodies $A \subseteq \mathbb{R}^n$. Moreover, it satisfies

$$\Phi\left(K + t, \mathbf{1}_\beta\right) = \Phi(K + t, \beta) = \Phi(K, \beta - t) = \Phi\left(K, \mathbf{1}_{\beta - t}\right).$$

Obviously, we have $\mathbf{1}_{\beta-t}(x) = 1$ if and only if $x + t \in \beta$. Therefore, $\Phi$ satisfies $\Phi\left(K + t, \mathbf{1}_\beta\right) = \Phi\left(K, \mathbf{1}_\beta(\cdot + t)\right)$. By a standard approximation argument, it follows that any continuous function $f\colon \mathbb{R}^n \to \mathbb{C}$ satisfies

$$\Phi(K + t, f) = \Phi\left(K, f(\cdot + t)\right).$$

For our purposes, we can apply this formula to the standard inclusion $\mathbb{R}^n \hookrightarrow \mathbb{C}^n$. This yields

$$\begin{aligned} B\Phi(K + t) &= \int_{\mathbb{R}^n} (x + t)\,\mathrm{d}\,\Phi(K, x) = \int_{\mathbb{R}^n} x\,\mathrm{d}\,\Phi(K, x) + \int_{\mathbb{R}^n} t\,\mathrm{d}\,\Phi(K, x) \\ &= B\Phi(K) + \left(\int_{\mathbb{R}^n} 1\,\mathrm{d}\,\Phi(K, x)\right) t = B\Phi(K) + \left(\mathrm{glob}(\Phi)(K)\right) t = B\Phi(K) \end{aligned}$$

as required, where the last equality follows from the assumption that $\mathrm{glob}(\Phi) = 0$. □

We have seen that $B$ restricts to a $\mathbb{C}$-linear map $B\colon \ker(\mathrm{glob}) \to \mathrm{Val} \otimes \mathbb{C}^n$ which we call the *Bernig embedding*. This name is justified by the following theorem.

**Theorem 3.23.** *The Bernig embedding $B\colon \ker(\mathrm{glob}) \to \mathrm{Val} \otimes \mathbb{C}^n$ is injective and continuous.*

*Proof.* To prove injectivity, suppose we are given a curvature measure $\Phi \in \mathrm{Curv}$ which satisfies $\mathrm{glob}(\Phi) = 0$ and $B\Phi = 0$. Moreover, let $f\colon \mathbb{R}^n \to \mathbb{C}$ be an arbitrary affine function. Since $f$ is affine, there exist complex numbers $c, \lambda_1, \ldots, \lambda_n \in \mathbb{C}$ such that

$$f = c + \sum_{i=1}^{n} \lambda_i p_i,$$

where the $p_i$ denote the aforementioned projection maps. Moreover, note that $\mathrm{glob}(\Phi) = 0$ implies that

$$\Phi_c = \int_{\mathbb{R}^n} c\,\mathrm{d}\,\Phi(\cdot, x) = c \int_{\mathbb{R}^n} 1\,\mathrm{d}\,\Phi(\cdot, x) = c\,\mathrm{glob}(\Phi) = 0 \in \mathrm{Val}.$$

Similarly, $\Phi_{p_i} = 0$ for all $i = 1, \ldots, n$ because of

$$\begin{pmatrix} \Phi_{p_1} \\ \vdots \\ \Phi_{p_n} \end{pmatrix} = \begin{pmatrix} \int_{\mathbb{R}^n} p_1(x)\,\mathrm{d}\,\Phi(\cdot, x) \\ \vdots \\ \int_{\mathbb{R}^n} p_n(x)\,\mathrm{d}\,\Phi(\cdot, x) \end{pmatrix} = \int_{\mathbb{R}^n} x\,\mathrm{d}\,\Phi(\cdot, x) = B\Phi = 0.$$

Applying this to $f = c + \lambda_1 p_1 + \ldots + \lambda_n p_n$, we obtain $\Phi_f = \Phi_c + \lambda_1 \Phi_{p_1} + \ldots + \lambda_n \Phi_{p_n} = 0$. Since $f$ is an arbitrary affine function, Lemma 3.21 yields $\Phi = 0$ as required.

To prove continuity, we choose the unit ball $B^n \subseteq \mathbb{R}^n$. Moreover, let the curvature measure $\Phi \in \mathrm{Curv}$ satisfy $\mathrm{glob}(\Phi) = 0$. Then, we have

$$\begin{aligned} \|B\Phi\|_{B^n} &= \sup_{K \subseteq B^n} \left\|B\Phi(K)\right\|_{\mathbb{C}^n} = \sup_{K \subseteq B^n} \left\|\left(\Phi(K, p_1), \ldots, \Phi(K, p_n)\right)\right\|_{\mathbb{C}^n} \\ &= \sup_{K \subseteq B^n} \sqrt{\sum_{i=1}^{n} \left|\Phi(K, p_i)\right|^2} \leq \sup_{K \subseteq B^n} \sum_{i=1}^{n} \left|\Phi(K, p_i)\right|. \end{aligned}$$

Recall that all convex bodies $K \subseteq B^n$ satisfy $\operatorname{supp}\big(\Phi(K)\big) \subseteq K \subseteq B^n$. In particular, we have $\big|p_i(x)\big| \leq 1$ for all $i = 1, \ldots, n$ and all $x \in \operatorname{supp}\big(\Phi(K)\big)$. For all $i$, there therefore exists a function $f_i \in C_0(\mathbb{R}^n)$ such that $\|f_i\| \leq 1$ and $\Phi(K, p_i) = \Phi(K, f_i)$ for all $K \subseteq B^n$. Combining this with the computation above, we obtain

$$\begin{aligned}\|B\Phi\|_{B^n} &\leq \sup_{K \subseteq B^n} \sum_{i=1}^{n} \big|\Phi(K, f_i)\big| \leq n \sup_{K \subseteq B^n} \max_i \big|\Phi(K, f_i)\big| \\ &\leq n \sup_{K \subseteq B^n} \sup_{f \in C_0(\mathbb{R}^n), \|f\|=1} \big|\Phi(K, f)\big| = n \sup_{K \subseteq B^n} \big\|\Phi(K)\big\| = n\|\Phi\|_{B^n},\end{aligned}$$

where the second to last equality follows from equation (2) in Theorem 6.19 of [Rud87]. This shows that $B$ is continuous as required. □

Let us conclude this section by computing how degree and parity behave under the Bernig embedding.

**Proposition 3.24.** *Let $\Phi \in \operatorname{Curv}$ be a curvature measure which satisfies $\operatorname{glob}(\Phi) = 0$. Then, the following assertions hold:*

- *If $\Phi$ is of degree $i$ for some $i \in \{0, \ldots, n\}$, then $B\Phi$ is of degree $i + 1$.*
- *If $\Phi$ is even, then $B\Phi$ is odd; if $\Phi$ is odd, then $B\Phi$ is even.*

*Proof.* Assume that $\Phi \in \operatorname{Curv}_i$ for some $i \in \{0, \ldots, n\}$ and consider a Borel set $\beta \subseteq \mathbb{R}^n$ first. Then, we have

$$\Phi_{\mathbf{1}_\beta}(\lambda K) = \Phi(\lambda K, \beta) = \lambda^i \Phi\left(K, \lambda^{-1}\beta\right) = \lambda^i \Phi\left(K, \mathbf{1}_{\lambda^{-1}\beta}\right)$$

for all convex bodies $K \subseteq \mathbb{R}^n$ and all $\lambda > 0$. Obviously, $\mathbf{1}_{\lambda^{-1}\beta}(x) = 1$ if and only if $\lambda x \in \beta$. Therefore, we obtain $\Phi_{\mathbf{1}_\beta}(\lambda K) = \lambda^i \Phi\big(K, \mathbf{1}_\beta(\lambda \cdot)\big)$. A standard approximation argument now yields this equation for any continuous function. In particular, any linear function $l \colon \mathbb{R}^n \to \mathbb{C}$ satisfies

$$\Phi_l(\lambda K) = \lambda^i \Phi\big(K, l(\lambda \cdot)\big) = \lambda^i \Phi(K, \lambda l) = \lambda^{i+1} \Phi(K, l) = \lambda^{i+1} \Phi_l(K).$$

Thus, we have proved that $\Phi_l$ is of degree $i + 1$ for any linear function $l$. Applying this to the projections $p_i$, we obtain that $B\Phi$ is of degree $i + 1$.

A similar argument proves the second claim. Assume that $\Phi$ is of parity $\varepsilon \in \{+, -\}$ and again consider a Borel set $\beta \subseteq \mathbb{R}^n$ first. To be able to prove both cases uniformly, we use the natural convention

$$-\varepsilon = \begin{cases} -, & \text{if } \varepsilon = +, \\ +, & \text{if } \varepsilon = -. \end{cases}$$

For all convex bodies $K \in \mathcal{K}$, we compute

$$\Phi_{\mathbf{1}_\beta}(-K) = \Phi(-K, \beta) = \varepsilon \Phi(K, -\beta) = \varepsilon \Phi(K, -\beta) = \varepsilon \Phi\big(K, \mathbf{1}_\beta(-\cdot)\big).$$

By the same standard approximation argument as before, we obtain that any linear map $l\colon \mathbb{R}^n \to \mathbb{C}$ satisfies

$$\Phi_l(-K) = \varepsilon\Phi\left(K, l(-\cdot)\right) = \varepsilon\Phi(K, -l) = -\varepsilon\Phi(K, l) = -\varepsilon\Phi_l(K)$$

which shows that $\Phi_l$ is of parity $-\varepsilon$. Applying this to the projections $p_i$ yields that $B\Phi$ is of parity $-\varepsilon$. □

## 3.6 Characterization Theorems for Curvature Measures

In this section, we characterize all $n$-homogeneous and all $(n-1)$-homogeneous curvature measures. These theorems are analogues of Hadwiger's theorem and McMullen's theorem, respectively. Moreover, we characterize simple curvature measures which can be seen as an analogue of the Klain–Schneider theorem.

Let us start with the easiest case of $n$-homogeneous curvature measures. Note that this result is already contained in Theorem 3.1 of [KW99]. However, we give a different proof using the Bernig embedding.

**Theorem 3.25.** *A curvature measure $\Phi \in \mathrm{Curv}$ is homogeneous of degree $n$ if and only if there exists a constant $c \in \mathbb{C}$ such that $\Phi = c\,\mathrm{VOL}_n$. In short, $\mathrm{Curv}_n = \mathrm{Curv}_n^+ = \mathbb{C}\,\mathrm{VOL}_n$ and $\mathrm{Curv}_n^- = \{0\}$.*

*Proof.* Example 3.10 yields that $\mathbb{C}\,\mathrm{VOL}_n \subseteq \mathrm{Curv}_n$. For the converse inclusion, let $\Phi \in \mathrm{Curv}_n$ be arbitrary. Since the globalization map preserves degree of homogeneity, it follows from Hadwiger's theorem that there exists $c \in \mathbb{C}$ such that $\mathrm{glob}(\Phi) = c\,\mathrm{vol}_n \in \mathrm{Val}_n$. By linearity of the globalization map, we obtain

$$\mathrm{glob}(\Phi - c\,\mathrm{VOL}_n) = \mathrm{glob}(\Phi) - c\,\mathrm{glob}(\mathrm{VOL}_n) = c\,\mathrm{vol}_n - c\,\mathrm{vol}_n = 0 \in \mathrm{Val}_n\,.$$

Therefore, we can apply the Bernig embedding to the $n$-homogeneous curvature measure $\Phi - c\,\mathrm{VOL}_n$. Proposition 3.24 yields that

$$B(\Phi - c\,\mathrm{VOL}_n) \in \mathrm{Val}_{n+1} \otimes \mathbb{C}^n\,.$$

Because of $\mathrm{Val}_{n+1} = \{0\}$ it follows that $B(\Phi - c\,\mathrm{VOL}_n) = 0$, and therefore $\Phi = c\,\mathrm{VOL}_n$ by injectivity of $B$. □

Let us quickly digress to introducing the so called support measures and explaining how they can be used to construct curvature measures.

Fix an index $i \in \{0, \dots, n-1\}$ and let

$$\Theta_i\colon \mathcal{K}^n \times \mathcal{B}(S\,\mathbb{R}^n) \to \mathbb{C}$$

denote the $i$-th support measure functional, see Section 4.2 of [Sch14] for a construction and important properties of $\Theta_i$. For any continuous function $f\colon S^{n-1} \to \mathbb{C}$ and any convex body $K \in \mathcal{K}$, we define the complex Borel measure $\Phi_i^f(K)\colon \mathcal{B} \to \mathbb{C}$ by

$$\Phi_i^f(K, \beta) = \int_{\pi_1^{-1}(\beta)} \pi_2^* f \,\mathrm{d}\,\Theta_i(K)$$

for all $\beta \in \mathcal{B}$, where $\pi_1$ and $\pi_2$ denote the natural projections from the sphere bundle $S\,\mathbb{R}^n = \mathbb{R}^n \times S^{n-1}$ onto its first and second factor, respectively. The following lemma follows easily from the properties of $\Theta_i$ proved in the aforementioned reference.

**Lemma 3.26.** *For any $i \in \{0, \dots, n-1\}$ and any continuous function $f\colon S^{n-1} \to \mathbb{C}$, $\Phi_i^f$ is a curvature measure of degree $i$. Moreover, $\Phi_i^f$ is of parity $\varepsilon \in \{+,-\}$ if $f$ is of parity $\varepsilon$.*

Let us mention that one can also prove the previous lemma by constructing a translation-invariant differential form $\omega_i \in \Omega^{n-1}(S\,\mathbb{R}^n)$ such that

$$\Phi_i^f(K,\beta) = \int_{\mathrm{nc}(K)\cap\pi_1^{-1}(\beta)} \pi_2^* f \wedge \omega_i$$

for all $K$ and $\beta$. Then, one can apply the proof of Theorem 3.9 to the continuous $(n-1)$-form $\pi_2^* f \wedge \omega_i$ to see that $\Phi_i^f$ is indeed a curvature measure.

It turns out that the class of curvature measures constructed above already subsumes all curvature measures in degree $i = n-1$. In comparison to the case of valuations however, there does not exist a kernel.

For simplicity of notation, we write $\Phi^f = \Phi_{n-1}^f$ throughout the next theorem and its proof.

**Theorem 3.27.** *The map*

$$C\left(S^{n-1}\right) \to \mathrm{Curv}_{n-1},\ f \mapsto \Phi^f$$

*is a continuous $\mathbb{C}$-linear isomorphism. Moreover, $\Phi^f$ is even if and only if $f$ is even, and $\Phi^f$ is odd if and only if $f$ is odd.*

*Proof.* We start with the simple verification of linearity. To do so, let $\lambda \in \mathbb{C}$ and $f, g \in C\left(S^{n-1}\right)$. Since integration and pullback are both linear, we have

$$\begin{aligned}\Phi^{\lambda f+g}(K,\beta) &= \int_{\pi_1^{-1}(\beta)} \pi_2^*(\lambda f+g)\,\mathrm{d}\,\Theta_{n-1}(K)\\ &= \lambda\int_{\pi_1^{-1}(\beta)} \pi_2^* f\,\mathrm{d}\,\Theta_{n-1}(K) + \int_{\pi_1^{-1}(\beta)} \pi_2^* g\,\mathrm{d}\,\Theta_{n-1}(K) = \lambda\Phi^f(K,\beta) + \Phi^g(K,\beta)\end{aligned}$$

for all $K$ and $\beta$ which shows that the map $f \mapsto \Phi^f$ is indeed linear.

For continuity, let $B \subseteq \mathbb{R}^n$ be bounded with non-empty interior. Since $\Theta_{n-1}(K,\cdot)$ is a non-negative finite measure for all $K$, we have

$$\begin{aligned}\left\|\Phi^f\right\|_B &= \sup_{K\subseteq B^n}\left\|\Phi^f(K)\right\| = \sup_{K\subseteq B}\sup\left|\int_{\mathbb{R}^n} g\,\mathrm{d}\,\Phi^f(K,g)\right|\\ &= \sup_{K\subseteq B}\sup\left|\int_{\mathbb{R}^n\times S^{n-1}} g(x)\,f(u)\,\mathrm{d}\,\Theta_{n-1}\big(K,(x,u)\big)\right|\\ &\le \left(\sup_{K\subseteq B}\sup\left|\int_{\mathbb{R}^n\times S^{n-1}} g(x)\,\mathrm{d}\,\Theta_{n-1}\big(K,(x,u)\big)\right|\right)\|f\|\end{aligned}$$

where the second supremum is taken over all functions $g \in C_0(\mathbb{R}^n)$ satisfying $\|g\| = 1$. This shows that $f \mapsto \Phi^f$ is continuous.

Next, we want to show injectivity. To do so, assume that $f \in C\left(S^{n-1}\right)$ satisfies $\Phi^f = 0$. In particular, we have

$$0 = \Phi^f(B^n, \beta) = \int_{\pi_1^{-1}(\beta)} \pi_2^* f \,\mathrm{d}\,\Theta_{n-1}(B^n) = \int_{\pi_1^{-1}(\beta)\cap \mathrm{nc}(B^n)} \pi_2^* f \,\mathrm{d}\,\Theta_{n-1}(B^n)$$

for all $\beta \in \mathcal{B}$, where the last equality follows from the fact that the measure $\Theta_i(K, \cdot)$ is supported on $\mathrm{nc}(K)$. In order to rewrite this integral, note that

$$\pi_1^{-1}(\beta) \cap \mathrm{nc}(B^n) = \left(\beta \times S^{n-1}\right) \cap \left\{(u,u) \,\middle|\, u \in S^{n-1}\right\} = \left\{(u,u) \,\middle|\, u \in \beta \cap S^{n-1}\right\}.$$

Therefore, a localized version of Lemma 4.2.2 of [Sch14] and the equation above yield that

$$\int_{\beta \cap S^{n-1}} f \,\mathrm{d}\, S_{n-1}(B^n) = 0$$

for all Borel sets $\beta \subseteq \mathbb{R}^n$. Note that equation (4.32) of [Sch14] applied to the unit ball $B^n$ yields that $S_{n-1}(B^n, \omega) = \mathcal{H}^{n-1}(\omega)$ for all Borel sets $\omega \subseteq S^{n-1}$. Combined with the equation above, we obtain that

$$\int_\omega f \,\mathrm{d}\,\mathcal{H}^{n-1} = \int_\omega f \,\mathrm{d}\, S_{n-1}(B^n) = 0$$

for all Borel sets $\omega \subseteq S^{n-1}$. This obviously implies that $f = 0$ as required.

As an auxiliary step towards surjectivity, let us compute the kernel of

$$\mathrm{glob}_{n-1} \colon \mathrm{Curv}_{n-1} \to \mathrm{Val}_{n-1}.$$

To do so, let $f \in C\left(S^{n-1}\right)$ be arbitrary and let $\varphi_f \in \mathrm{Val}_{n-1}$ be the associated valuation given by

$$\varphi_f(K) = \int_{S^{n-1}} f \,\mathrm{d}\, S_{n-1}(K)$$

for all $K \in \mathcal{K}$. Note that we have

$$\mathrm{glob}\left(\Phi^f\right)(K) = \int_{S\,\mathbb{R}^n} \pi_2^* f \,\mathrm{d}\,\Theta_{n-1}(K) = \int_{S^{n-1}} f \,\mathrm{d}\, S_{n-1}(K) = \varphi_f(K)$$

for all convex bodies $K \subseteq \mathbb{R}^n$, where the penultimate equality follows from equation (4.11) of [Sch14]. Further, note that McMullen's theorem yields that $\varphi_g = 0$ for all linear functions $g \colon S^{n-1} \to \mathbb{C}$. Therefore, we have

$$\left\{\Phi^g \,\middle|\, g \in L\left(S^{n-1}\right)\right\} \subseteq \ker\left(\mathrm{glob}_{n-1}\right).$$

Note that the space on the left hand side is $n$-dimensional due to injectivity of $f \mapsto \Phi^f$. On the other hand, the Bernig embedding gives an injective linear map

$$\ker\left(\mathrm{glob}_{n-1}\right) \hookrightarrow \mathrm{Val}_n \otimes \mathbb{C}^n$$

which shows that $\ker\left(\mathrm{glob}_{n-1}\right)$ has dimension at most $n$ due to $\mathrm{Val}_n = \mathbb{C}\,\mathrm{vol}_n$. Therefore, we conclude that

$$\ker\left(\mathrm{glob}_{n-1}\right) = \left\{ \Phi^g \;\middle|\; g \in L\left(S^{n-1}\right) \right\}.$$

For surjectivity, let $\Phi \in \mathrm{Curv}$ be arbitrary. By McMullen's theorem, there exists a function $f \in C\left(S^{n-1}\right)$ such that $\mathrm{glob}\left(\Phi\right) = \varphi_f \in \mathrm{Val}_{n-1}$. Since $\mathrm{glob}(\Phi^f) = \varphi_f$, it follows that $\Phi - \Phi^f \in \ker\left(\mathrm{glob}_{n-1}\right)$. By the previous computation of the kernel, we find a linear function $g \in L\left(S^{n-1}\right)$ such that $\Phi^g = \Phi - \Phi^f$ which is equivalent to

$$\Phi = \Phi^f + \Phi^g = \Phi^{f+g}$$

as required.

For the statement on parity, recall from the previous lemma that $\Phi^f$ is of parity $\varepsilon$ if $f$ is. Since $f \mapsto \Phi^f$ is an isomorphism, it therefore follows that the restricted maps

$$C^\varepsilon\left(S^{n-1}\right) \to \mathrm{Curv}^\varepsilon_{n-1},\ f \mapsto \Phi^f$$

have to be isomorphisms, too. □

Since we have classified all curvature measures of degree $n-1$ and $n$, we of course get an explicit form for all curvature measures on the real line $\mathbb{R}$.

**Corollary 3.28.** *Let $n = 1$ and let $\Phi \in \mathrm{Curv}$. Then there exist $\lambda_1, \lambda_2, \lambda_3 \in \mathbb{C}$ such that*

$$\Phi\left([a,b]\right) = \lambda_1 \delta_a + \lambda_2 \delta_b + \lambda_3\,\mathrm{VOL}_1\left([a,b]\right)$$

*for all convex bodies $K = [a,b] \subseteq \mathbb{R}$.*

*Proof.* We write $\Phi = \Phi_0 + \Phi_1$ for the Kiderlen–Weil decomposition of $\Phi$. Theorem 3.25 yields that there exists $\lambda_3 \in \mathbb{C}$ such that $\Phi_1 = \lambda_3\,\mathrm{VOL}_1$. Moreover, Theorem 3.27 implies that there exists a function $f\colon S^0 \to \mathbb{C}$ such that

$$\Phi_0\left([a,b],\beta\right) = \int_{\pi_1^{-1}(\beta)} \pi_2^* f \,\mathrm{d}\,\Theta_0\left([a,b]\right)$$

for all convex bodies $K = [a,b] \subseteq \mathbb{R}$ and all Borel sets $\beta \subseteq \mathbb{R}$. Using the explicit definition of $\Theta_0$ on polytopes, see formula (4.3) of [Sch14], one can easily evaluate the integral above. This yields

$$\Phi_0\left([a,b],\beta\right) = f\left(-1\right)\delta_a\left(\beta\right) + f\left(1\right)\delta_b\left(\beta\right)$$

for all $[a,b] \subseteq \mathbb{R}$ and $\beta \in \mathcal{B}$. Setting $\lambda_1 = f\left(-1\right)$ and $\lambda_2 = f\left(1\right)$ therefore finishes the proof. □

To conclude this chapter, we study simple curvature measures. Recall from Definition 2.15 that a curvature measure $\Phi \in \operatorname{Curv}$ is called simple if $\Phi(K) = 0 \in \mathcal{M}$ for all convex bodies $K \in \mathcal{K}$ with $\dim(K) < n$. The following theorem can be seen as a curvature measure analogue to the Klain–Schneider characterization of simple valuations. Note that an analogous theorem has been proved for smooth curvature measures in Corollary 4.6 of [SW21].

**Theorem 3.29.** *A curvature measure $\Phi \in \operatorname{Curv}$ is simple if and only if*

$$\Phi \in \ker\left(\operatorname{glob}_{n-2}^{+}\right) \oplus \operatorname{Curv}_{n-1}^{-} \oplus \operatorname{Curv}_{n}.$$

*Proof.* For readability, we write

$$S = \ker\left(\operatorname{glob}_{n-2}^{+}\right) \oplus \operatorname{Curv}_{n-1}^{-} \oplus \operatorname{Curv}_{n}$$

throughout this proof. Given a curvature measure $\Phi \in \operatorname{Curv}$ and an affine subspace $A \subseteq \mathbb{R}^n$, we furthermore write

$$\Phi|_A \in \operatorname{Curv}(A)$$

for the curvature measure on $A$ given by restricting $\Phi$ to convex bodies and Borel sets contained in $A$.

Assume first that $\Phi = \Phi_{n-2} + \Phi_{n-1} + \Phi_n \in S$ and let $K \subseteq \mathbb{R}^n$ be an arbitrary convex body satisfying $\dim(K) < n$. Then, there exists an affine hyperplane $H \subseteq \mathbb{R}^n$ such that $K \subseteq H$. Using Corollary 3.5, we obtain

$$\Phi_n(K, \beta) = \Phi_n(K, \beta \cap K) = \Phi_n|_H(K, \beta \cap K) = 0$$

for all $\beta \in \mathcal{B}$, where the last equality follows from $\Phi_n|_H \in \operatorname{Curv}_n(H) = \{0\}$ because of $\dim(H) = n - 1$. This proves that $\Phi_n$ is simple. Similarly, we have

$$\Phi_{n-1}(K, \beta) = \Phi_{n-1}|_H(K, \beta \cap K) = 0$$

for all Borel sets $\beta \subseteq \mathbb{R}^n$ because of $\Phi_{n-1}|_H \in \operatorname{Curv}_{n-1}^{-}(H) = \{0\}$ which shows that $\Phi_{n-1}$ is simple. To show that $\Phi_{n-2}$ is also simple, we write $\operatorname{glob}_H \colon \operatorname{Curv}(H) \to \operatorname{Val}(H)$ for the globalization map on $H$. Note that

$$\Phi_{n-2}|_H \in \ker(\operatorname{glob}_H) \cap \operatorname{Curv}_{n-1}^{+}(H).$$

Since $\dim(H) = n - 1$, we have seen in the proof of Theorem 3.27 that the restricted globalization map

$$(\operatorname{glob}_H)_{n-2}^{+} \colon \operatorname{Curv}_{n-2}^{+}(H) \to \operatorname{Val}_{n-2}^{+}(H)$$

is an isomorphism. Therefore $\Phi_{n-2}|_H = 0$ which yields

$$\Phi_{n-2}(K, \beta) = \Phi_{n-2}|_H(K, \beta \cap K) = 0$$

for all Borel sets $\beta \subseteq \mathbb{R}^n$. Thus, we have seen that $\Phi_{n-2}$ is simple, too. Obviously, the simplicity of $\Phi_{n-2}$, $\Phi_{n-1}$ and $\Phi_n$ implies that $\Phi = \Phi_{n-2} + \Phi_{n-1} + \Phi_n$ is simple.

For the converse direction, assume that $\Phi \in \operatorname{Curv}$ is simple and let $\varphi = \operatorname{glob}(\Phi)$. For all convex bodies $K \in \mathcal{K}$ of dimension less than $n$, we have

$$\varphi(K) = \Phi(K, \mathbb{R}^n) = 0$$

due to simplicity of $\Phi$. This shows that $\varphi$ is simple which implies that $\varphi \in \operatorname{Val}_{n-1}^- \oplus \operatorname{Val}_n$ by the Klain–Schneider theorem. Since globalization preserves degree and parity, it follows that

$$\Phi \in \left( \bigoplus_{i=0}^{n-2} \ker(\operatorname{glob}_i) \right) \oplus \operatorname{Curv}_{n-1}^- \oplus \operatorname{Curv}_n .$$

We write $\Phi = \Psi + \Phi_{n-1} + \Phi_n$ for the decomposition of $\Phi$ with respect to the three direct summands above. In order to verify the required $\Phi \in S$, it is left to show that $\Psi$ is even and of degree $n-2$. To do so, note that $\Phi_{n-1} \in \operatorname{Curv}_{n-1}^- \subseteq S$ and $\Phi_n \in \operatorname{Curv}_n \subseteq S$. In particular, $\Phi_{n-1}$ and $\Phi_n$ are simple due to the already proved direction. Since $\Phi$ is simple by assumption, it follows that $\Psi = \Phi - \Phi_{n-1} - \Phi_n$ is also simple. Obviously, $\Psi(K) = 0$ implies

$$B\Psi(K) = \int_{\mathbb{R}^n} x \,\mathrm{d}\, \Psi(K, x) = 0 \in \mathbb{C}^n .$$

Hence, $B\Psi$ is a simple $\mathbb{C}^n$-valued valuation. Using the Klain–Schneider classification of simple valuations again, it follows that

$$B\Psi \in \left( \operatorname{Val}_{n-1}^- \oplus \operatorname{Val}_n \right) \otimes \mathbb{C}^n .$$

By applying Proposition 3.24 we obtain

$$\Psi \in \ker\left(\operatorname{glob}_{n-2}^+\right) \oplus \ker\left(\operatorname{glob}_{n-1}\right),$$

where the injectivity of $B$ implies that no other subspaces can appear. Combined with the fact that $\Psi$ is an element of the first direct summand of the decomposition above, we obtain

$$\Psi \in \left( \bigoplus_{i=0}^{n-2} \ker(\operatorname{glob}_i) \right) \cap \left( \ker\left(\operatorname{glob}_{n-2}^+\right) \oplus \ker\left(\operatorname{glob}_{n-1}\right) \right) = \ker\left(\operatorname{glob}_{n-2}^+\right)$$

which indeed shows that $\Psi$ is even and homogeneous of degree $n-2$. □

# 4 Smooth Curvature Measures

In this chapter, we show that Curv admits a natural group action by the general linear group $\mathrm{GL}(n,\mathbb{R})$. This is a curvature measure analogue to the group action discussed in Section 2.4. With respect to this action, Curv is a continuous representation of $\mathrm{GL}(n,\mathbb{R})$. This yields that Curv has a dense subspace $\mathrm{Curv}^\infty$ of $\mathrm{GL}(n,\mathbb{R})$-smooth curvature measures. Our first main conjecture of this section is that $\mathrm{Curv}^\infty = \mathrm{Curv}^{\mathrm{sm}}$, where $\mathrm{Curv}^{\mathrm{sm}}$ denotes the space of smooth curvature measures due to Bernig and Fu. The analogous statement in the theory of valuations is true as was proved by Alesker, see Theorem 2.23. This conjecture will be verified in some special cases, namely for degrees $i = n-1, n$ in general dimension $n$ and for degree 0 in dimension $n = 2$. In view of Alesker's irreducibility theorem, we also study Jordan–Hölder series in the subrepresentations $\mathrm{Curv}_i^\varepsilon \subseteq \mathrm{Curv}$. While one easily sees that $\mathrm{Curv}_i^\varepsilon$ is not irreducible in general, we conjecture that the Jordan–Hölder series of $\mathrm{Curv}_i^\varepsilon$ is of length at most two. This conjecture is verified in the same cases as the smoothness conjecture.

## 4.1 General Linear Group Action

For any curvature measure $\Phi \in \mathrm{Curv}$ and any $g \in \mathrm{GL}(n,\mathbb{R})$, we define $g\Phi\colon \mathcal{K} \to \mathcal{M}$ by

$$g\Phi(K,\beta) = \Phi\left(g^{-1}K,\, g^{-1}\beta\right)$$

for all $K \in \mathcal{K}$ and all $\beta \in \mathcal{B}$.

**Lemma 4.1.** *If $\Phi \in \mathrm{Curv}$ and $g \in \mathrm{GL}(n,\mathbb{R})$, then $g\Phi \in \mathrm{Curv}$. The induced map $\mathrm{GL}(n,\mathbb{R}) \times \mathrm{Curv} \to \mathrm{Curv}$, $(g,\Phi) \mapsto g\Phi$ is a left group action.*

*Proof.* Throughout this proof, fix a curvature measure $\Phi \in \mathrm{Curv}$ and a group element $g \in \mathrm{GL}(n,\mathbb{R})$.

Firstly, we verify translation-invariance. Since $g^{-1}\colon \mathbb{R}^n \to \mathbb{R}^n$ is linear and $\Phi$ is translation-invariant, we see that

$$\begin{aligned} g\Phi(K+x,\beta+x) &= \Phi\left(g^{-1}K + g^{-1}x, g^{-1}\beta + g^{-1}x\right) \\ &= \Phi\left(g^{-1}K, g^{-1}\beta\right) = g\Phi(K,\beta) \end{aligned}$$

for all $K \in \mathcal{K}$, $\beta \in \mathcal{B}$ and $x \in \mathbb{R}^n$.

Let $K, L \in \mathcal{K}$ and let $U \subseteq \mathbb{R}^n$ be an open set such that $K \cap U = L \cap U$. Since $g^{-1}\colon \mathbb{R}^n \to \mathbb{R}^n$ is a homeomorphism, $g^{-1}U$ is open in $\mathbb{R}^n$ and $g^{-1}K \cap g^{-1}U = g^{-1}L \cap g^{-1}U$. Moreover, any Borel set $\beta \subseteq U$ obviously satisfies $g^{-1}\beta \subseteq g^{-1}U$. Therefore, we have

$$g\Phi(K,\beta) = \Phi\left(g^{-1}K, g^{-1}\beta\right) = \Phi\left(g^{-1}L, g^{-1}\beta\right) = g\Phi(L,\beta)$$

because $\Phi$ is defined locally.

For weak* continuity, we first note that $\mathbf{1}_\beta \circ g = \mathbf{1}_{g^{-1}\beta}$ for all Borel sets $\beta \subseteq \mathbb{R}^n$. This yields that

$$g\Phi\left(K,\mathbf{1}_\beta\right) = g\Phi(K,\beta) = \Phi\left(g^{-1}K,\, g^{-1}\beta\right) = \Phi\left(g^{-1}K,\, \mathbf{1}_{g^{-1}\beta}\right) = \Phi\left(g^{-1}K, \mathbf{1}_\beta \circ g\right)$$

for all convex bodies $K \subseteq \mathbb{R}^n$ and Borel sets $\beta \subseteq \mathbb{R}^n$. Now let $f \in C_0(\mathbb{R}^n)$. By approximation, we obtain

$$g\Phi(K, f) = \Phi\left(g^{-1}K, f \circ g\right)$$

from the previous computation. Assume now that $K_j \to K$ is a convergent sequence of convex bodies. Then, we obviously also have $g^{-1}K_j \to g^{-1}K$ and we obtain

$$\lim_{j\to\infty} g\Phi\left(K_j, f\right) = \lim_{j\to\infty} \Phi\left(g^{-1}K_j, f \circ g\right) = \Phi\left(g^{-1}K, f \circ g\right) = g\Phi(K, f).$$

This completes the proof of $g\Phi \in \mathrm{Curv}$.

For the group action claim, let $I \in \mathrm{GL}(n, \mathbb{R})$ denote the unit matrix. Then, we observe that

$$I\Phi(K, \beta) = \Phi\left(I^{-1}K, I^{-1}\beta\right) = \Phi(K, \beta)$$

for all $K \in \mathcal{K}$ and $\beta \in \mathcal{B}$. For another $h \in \mathrm{GL}(n, \mathbb{R})$, we furthermore have

$$\begin{aligned}(gh)\,\Phi(K, \beta) &= \Phi\left((gh)^{-1}K, (gh)^{-1}\beta\right) = \Phi\left(h^{-1}g^{-1}K, h^{-1}g^{-1}\beta\right)\\ &= h\Phi\left(g^{-1}K, g^{-1}\beta\right) = g\left(h\Phi\right)(K, \beta)\end{aligned}$$

for all $K$ and $\beta$ which shows that $\mathrm{GL}(n, \mathbb{R})$ indeed acts on Curv. □

**Corollary 4.2.** Curv *is a complex representation of* $\mathrm{GL}(n, \mathbb{R})$.

*Proof.* Let $g \in \mathrm{GL}(n, \mathbb{R})$. For all $\lambda \in \mathbb{C}$, $\Phi, \Psi \in \mathrm{Curv}$, $K \in \mathcal{K}$ and $\beta \in \mathcal{B}$, we have

$$\begin{aligned}g\left(\lambda\Phi + \Psi\right)(K, \beta) &= \left(\lambda\Phi + \Psi\right)\left(g^{-1}K, g^{-1}\beta\right)\\ &= \lambda\Phi\left(g^{-1}K, g^{-1}\beta\right) + \Psi\left(g^{-1}K, g^{-1}\beta\right) = \lambda\left(g\Phi\right)(K, \beta) + g\Psi(K, \beta).\end{aligned}$$

This shows that the map $\rho(g) : \mathrm{Curv} \to \mathrm{Curv}$, $\Phi \mapsto g\Phi$ is $\mathbb{C}$-linear. In the previous lemma, it was proved that the induced map

$$\rho\colon\ \mathrm{GL}(n, \mathbb{R}) \to \mathrm{GL}(\mathrm{Curv}),\ g \mapsto \rho(g)$$

is a group homomorphism. Therefore, Curv is a representation of $\mathrm{GL}(n, \mathbb{R})$. □

**Proposition 4.3.** *For all* $i = 0, \ldots, n$ *and* $\varepsilon \in \{+, -\}$*, the space* $\mathrm{Curv}_i^{\varepsilon}$ *is invariant under the action of* $\mathrm{GL}(n, \mathbb{R})$.

*Proof.* Throughout this proof, let $g \in \mathrm{GL}(n, \mathbb{R})$ and $\Phi \in \mathrm{Curv}_i^{\varepsilon}$. Due to linearity of $g^{-1}\colon\ \mathbb{R}^n \to \mathbb{R}^n$, we have

$$g\Phi(-K, -\beta) = \Phi\left(-g^{-1}K, -g^{-1}\beta\right) = \varepsilon\Phi\left(g^{-1}K, g^{-1}\beta\right) = \varepsilon\left(g\Phi\right)(K, \beta)$$

for all $K \in \mathcal{K}$ and $\beta \in \mathcal{B}$. This shows that $g\Phi$ is of parity $\varepsilon$. Analogously, we obtain

$$g\Phi(\lambda K, \lambda\beta) = \Phi\left(\lambda g^{-1}K, \lambda g^{-1}\beta\right) = \lambda^i\Phi\left(g^{-1}K, g^{-1}\beta\right) = \lambda^i\left(g\Phi\right)(K, \beta)$$

for all $\lambda > 0$. Thus, we indeed have $g\Phi \in \mathrm{Curv}_i^{\varepsilon}$. □

Next, we want to prove that the representation $(\mathrm{Curv}, \rho)$ of the Lie group $\mathrm{GL}(n, \mathbb{R})$ is continuous. Recall that this representation is continuous if the action map

$$\mathrm{GL}(n, \mathbb{R}) \times \mathrm{Curv} \to \mathrm{Curv},\ (g, \Phi) \mapsto g\Phi$$

is continuous. We start with the following weaker statement.

**Lemma 4.4.** *For all $g \in \mathrm{GL}(n, \mathbb{R})$, the map $\rho(g) : \mathrm{Curv} \to \mathrm{Curv}$ is continuous.*

*Proof.* Fix an arbitrary $g \in \mathrm{GL}(n, \mathbb{R})$. Since $g^{-1} \colon \mathbb{R}^n \to \mathbb{R}^n$ is an isomorphism, we observe two things. Firstly, a countable family $\{\beta_j\}$ of sets $\beta_j \subseteq \mathbb{R}^n$ forms a Borel partition of $\mathbb{R}^n$ if and only if $\{g^{-1}\beta_j\}$ does. Secondly, $g^{-1}B^n$ is bounded with non-empty interior. For all $\Phi \in \mathrm{Curv}$, we therefore obtain

$$\begin{aligned}\|g\Phi\|_{B^n} &= \sup_{K \subseteq B^n} \sup \sum_j \left|g\Phi\left(K, \beta_j\right)\right| = \sup_{K \subseteq B^n} \sup \sum_j \left|\Phi\left(g^{-1}K, g^{-1}\beta_j\right)\right| \\ &= \sup_{K \subseteq B^n} \sup \sum_j \left|\Phi\left(g^{-1}K,\, \beta_j\right)\right| = \sup_{K \subseteq B^n} \left\|\Phi\left(g^{-1}K\right)\right\| = \sup_{L \subseteq g^{-1}B^n} \left\|\Phi(L)\right\| = \|\Phi\|_{g^{-1}B^n},\end{aligned}$$

where the second supremum is taken over all countable Borel partitions $\mathbb{R}^n = \bigsqcup \beta_j$. Hence, we have seen that

$$\rho(g) :\ \left(\mathrm{Curv}, \|\cdot\|_{B^n}\right) \to \left(\mathrm{Curv}, \|\cdot\|_{g^{-1}B^n}\right)$$

is continuous. By equivalence of norms, see Theorem 3.16, the claim follows. □

The following general lemma from the theory of representations on Banach spaces follows from the principle of uniform boundedness, see for example p. 219 of [War72].

**Lemma 4.5.** *Let $G$ be a locally compact Hausdorff group and let $X$ be a Banach space. Moreover, let $\pi \colon G \to \mathrm{GL}(X)$ be a linear representation of $G$ for which the linear map $X \to X$, $x \mapsto \pi(g)\, x$ is continuous for all $g \in G$. Then, $\pi$ is continuous if and only if the map*

$$G \to X,\ g \mapsto \pi(g)\, x$$

*is continuous for all $x \in X$.*

**Theorem 4.6.** *The representation $(\mathrm{Curv}, \rho)$ of $\mathrm{GL}(n, \mathbb{R})$ is continuous.*

*Proof.* By the previous two lemmas, it is only left to show that $\mathrm{GL}(n, \mathbb{R}) \to \mathrm{Curv}$, $g \mapsto g\Phi$ is continuous for all $\Phi \in \mathrm{Curv}$.

To do so, we start with some auxiliary considerations. Let $\Phi \in \mathrm{Curv}$ and $f \in C_0(\mathbb{R}^n)$ be arbitrary. Then, the map

$$\mathcal{K} \to \mathbb{R},\ K \mapsto \left|\Phi(K, f)\right|$$

is continuous due to weak* continuity of $\Phi$. Since the set $\left\{K \in \mathcal{K} \mid K \subseteq B^n\right\}$ is compact due to the Blaschke selection theorem, we have

$$s = \sup_{K \subseteq B^n} \left|\Phi\left(K, f\right)\right| \in \mathbb{R}\,.$$

By continuity of the map $K \mapsto \left|\Phi\left(K, f\right)\right|$ and the fact that the set of all polytopes lies dense in $\mathcal{K}$, it follows that

$$s = \sup_{P \subseteq B^n} \left|\Phi\left(P, f\right)\right|,$$

where the supremum is taken over all polytopes $P \subseteq B^n$. Since $f$ was an arbitrary function, we obtain

$$\|\Phi\|_{B^n} = \sup \sup_{K \subseteq B^n} \left|\Phi\left(K, f\right)\right| = \sup \sup_{P \subseteq B^n} \left|\Phi\left(P, f\right)\right| = \sup_{P \subseteq B^n} \left\|\Phi\left(P\right)\right\|,$$

where the first supremum is taken over all functions $f \in C_0\left(\mathbb{R}^n\right)$ which satisfy $\|f\| = 1$.

Let us now get to the proof of continuity which works by contradiction. To do so, fix an arbitrary $\Phi \in \mathrm{Curv}$ and assume that $g \mapsto g\Phi$ is not continuous. Then, there exists a sequence $\left(g_k\right)_{k \in \mathbb{N}} \subseteq \mathrm{GL}\left(n, \mathbb{R}\right)$ which converges to some $g \in \mathrm{GL}\left(n, \mathbb{R}\right)$ and which satisfies the following property. There exists $\varepsilon > 0$ such that for all $k_0 \in \mathbb{N}$ there exists $k \geq k_0$ such that

$$\|g_k\Phi - g\Phi\| > \varepsilon.$$

In other words, we find an infinite index set $J \subseteq \mathbb{N}$ such that the corresponding subsequence $\left(g_j\right)_{j \in J}$ satisfies

$$\left\|g_j\Phi - g\Phi\right\| > \varepsilon$$

for all $j \in J$. By the previous considerations, it follows that for all $j$ there exists a polytope $P_j \subseteq B^n$ such that

$$\left\|g_j\Phi\left(P_j\right) - g\Phi\left(P_j\right)\right\| \geq \varepsilon.$$

This inequality will later be used to reach a contradiction.

Let $\Phi = \Phi_0 + \ldots + \Phi_n$ denote the Kiderlen–Weil decomposition of $\Phi$. One easily sees that $\mathrm{GL}\left(n, \mathbb{R}\right) \to \mathrm{Curv}$, $g \mapsto g\,\mathrm{VOL}_n$ is continuous. Therefore, we may assume that $\Phi_n = 0$ without loss of generality. Moreover, recall that for any polytope $P \subseteq \mathbb{R}^n$ and any $i$-face $F \subseteq P$, we write $n\left(K, F\right) \subseteq S^{n-1}$ for the set of all unit outer normal vectors to $P$ in $F$. By Theorem 3.1 of [KW99], for all $i \in \{0, \ldots, n-1\}$ there exists a continuous

function $\alpha_i$ on the space of spherical polyhedra of dimension at most $n-1-i$ such that

$$\begin{aligned}
&g_j\Phi_i\left(P_j\right) - g\Phi_i\left(P_j\right) = \Phi_i\left(g_j^{-1}P_j, g_j^{-1}\cdot\right) - \Phi_i\left(g^{-1}P_j, g^{-1}\cdot\right)\\
&= \sum_{F\in\mathcal{F}_i\left(g_j^{-1}P_j\right)} \alpha_i\left(n\left(g_j^{-1}P_j, F\right)\right)\operatorname{vol}_F\left(g_j^{-1}\cdot\right)\\
&\quad - \sum_{F\in\mathcal{F}_i\left(g^{-1}P_j\right)} \alpha_i\left(n\left(g^{-1}P_j, F\right)\right)\operatorname{vol}_F\left(g^{-1}\cdot\right)\\
&= \sum_{F\in\mathcal{F}_i\left(P_j\right)} \alpha_i\left(n\left(g_j^{-1}P_j, g_j^{-1}F\right)\right)\operatorname{vol}_{g_j^{-1}F}\left(g_j^{-1}\cdot\right)\\
&\quad - \sum_{F\in\mathcal{F}_i\left(P_j\right)} \alpha_i\left(n\left(g^{-1}P_j, g^{-1}F\right)\right)\operatorname{vol}_{g^{-1}F}\left(g^{-1}\cdot\right).
\end{aligned}$$

Fix an arbitrary $i\in\{0,\dots,n-1\}$. Due to $g_j\to g$ and continuity of $\alpha_i$, we find $j_0 = j_0(i)\in J$ such that

$$\begin{aligned}
&\left\|\alpha_i\left(n\left(g_{j_0}^{-1}P_{j_0}, g_{j_0}^{-1}F\right)\right)\operatorname{vol}_{g_{j_0}^{-1}F}\left(g_{j_0}^{-1}\cdot\right) - \alpha_i\left(n\left(g^{-1}P_{j_0}, g^{-1}F\right)\right)\operatorname{vol}_{g^{-1}F}\left(g^{-1}\cdot\right)\right\|\\
&< \frac{\varepsilon}{n\operatorname{card}\left(\mathcal{F}_i\left(P_{j_0}\right)\right)}
\end{aligned}$$

for all $i=0,\dots,n-1$ and all $F\in\mathcal{F}_i\left(P_{j_0}\right)$. Applying the triangle inequality to the computation above, we obtain

$$\left\|g_{j_0}\Phi_i\left(P_{j_0}\right) - g\Phi_i\left(P_{j_0}\right)\right\| < \sum_{F\in\mathcal{F}_i\left(P_{j_0}\right)} \frac{\varepsilon}{n\operatorname{card}\left(\mathcal{F}_i\left(P_{j_0}\right)\right)} = \frac{\varepsilon}{n}.$$

Since $\mathrm{GL}(n,\mathbb{R})$ acts linearly on Curv, we therefore obtain

$$\begin{aligned}
\left\|g_{j_0}\Phi\left(P_{j_0}\right) - g\Phi\left(P_{j_0}\right)\right\| &= \left\|\sum_{i=0}^{n-1}\left(g_{j_0}\Phi_i\left(P_{j_0}\right) - g\Phi_i\left(P_{j_0}\right)\right)\right\|\\
&\le \sum_{i=0}^{n-1}\left\|g_{j_0}\Phi_i\left(P_{j_0}\right) - g\Phi_i\left(P_{j_0}\right)\right\| < \varepsilon.
\end{aligned}$$

Since we have previously shown that

$$\left\|g_j\Phi\left(P_j\right) - g\Phi\left(P_j\right)\right\| \ge \varepsilon$$

for all $j\in J$, we have reached our required contradiction. □

To conclude this section, we study how globalization and the Bernig embedding behave with respect to the general linear group action.

**Proposition 4.7.** *The map* glob: $\mathrm{Curv} \to \mathrm{Val}$ *is* $\mathrm{GL}(n,\mathbb{R})$*-equivariant. In particular,* $\ker(\mathrm{glob})$ *is a closed subrepresentation of* Curv.

*Proof.* For all $\Phi \in \mathrm{Curv}$, $g \in \mathrm{GL}(n,\mathbb{R})$ and $K \in \mathcal{K}$, we have

$$\begin{aligned}&\mathrm{glob}(g\Phi)(K) = g\Phi(K,\mathbb{R}^n) = \Phi\left(g^{-1}K, g^{-1}\mathbb{R}^n\right) = \Phi\left(g^{-1}K,\mathbb{R}^n\right)\\ &= \mathrm{glob}(\Phi)\left(g^{-1}K\right) = g\left(\mathrm{glob}(\Phi)\right)(K).\end{aligned}$$

This shows that glob is equivariant.

If $\Phi \in \ker(\mathrm{glob})$, we therefore obtain

$$\mathrm{glob}(g\Phi) = g\left(\mathrm{glob}(\Phi)\right) = 0 \in \mathrm{Val}$$

for all $g \in \mathrm{GL}(n,\mathbb{R})$ which implies $g\Phi \in \ker(\mathrm{glob})$. Thus, $\ker(\mathrm{glob})$ is indeed a subrepresentation of Curv. Since

$$\ker(\mathrm{glob}) = \mathrm{glob}^{-1}\left(\{0\}\right)$$

it follows immediately from the continuity of glob that $\ker(\mathrm{glob})$ is closed in Curv. □

The equivariance of the globalization map particularly implies that a direct analogue of Alesker's irreducibility theorem fails to be true in the setting of curvature measures. More precisely, the representations $\mathrm{Curv}_i^{\varepsilon}$ are not either irreducible or zero for all $i$ and $\varepsilon$ as the next corollary shows.

**Corollary 4.8.** *Let* $n \geq 2$*. Then, the representation* $\mathrm{Curv}_{n-1}^{-}$ *of* $\mathrm{GL}(n,\mathbb{R})$ *is non-zero and not irreducible.*

*Proof.* From Theorem 3.27 we know that

$$\mathrm{Curv}_{n-1}^{-} \cong C^{-}\left(S^{n-1}\right)$$

as complex vector spaces. Therefore, $\mathrm{Curv}_{n-1}^{-}$ is a representation of infinite dimension. However, we have also seen in the proof of Theorem 3.27 that

$$\ker\left(\mathrm{glob}_{n-1}^{-}\right) \cong L\left(S^{n-1}\right)$$

as complex vector spaces. Since the latter space is clearly $n$-dimensional, it follows that $\ker(\mathrm{glob}_{n-1}^{-})$ is a non-trivial subrepresentation of $\mathrm{Curv}_{n-1}^{-}$. Last, note that

$$\ker\left(\mathrm{glob}_{n-1}^{-}\right) = \ker(\mathrm{glob}) \cap \mathrm{Curv}_{n-1}^{-}$$

which shows that $\ker(\mathrm{glob}_{n-1}^{-})$ is closed in $\mathrm{Curv}_{n-1}^{-}$. □

As a side note, let us mention that in dimension $n = 1$ the analogue of Alesker's irreducibility theorem is in fact true as $\mathrm{Curv}_i^\varepsilon$ has dimension one and is therefore irreducible for $(i, \varepsilon) \neq (1, -)$ and $\mathrm{Curv}_1^- = \{0\}$.

Recall that the action of a group $G$ on a tensor product $V \otimes W$ of $G$-representations is given by $g(v \otimes w) = gv \otimes gw$. In the case of the Bernig embedding, this means that the action $\mathrm{GL}(n, \mathbb{R}) \curvearrowright \mathrm{Val} \otimes \mathbb{C}^n$ is given by

$$(g\varphi)(K) = g\left(\varphi\left(g^{-1}K\right)\right)$$

for all $g \in \mathrm{GL}(n, \mathbb{R})$, $\varphi \in \mathrm{Val} \otimes \mathbb{C}^n$ and $K \in \mathcal{K}$.

**Proposition 4.9.** *The Bernig embedding* $B\colon \ker(\mathrm{glob}) \to \mathrm{Val} \otimes \mathbb{C}^n$ *is equivariant with respect to the* $\mathrm{GL}(n, \mathbb{R})$*-action.*

*Proof.* Let $g = (g_{ij}) \in \mathrm{GL}(n, \mathbb{R})$ and let $\mu \in \mathcal{M}$ be an arbitrary measure. Then, linearity of integration yields

$$g\left(\int_{\mathbb{R}^n} x \,\mathrm{d}\,\mu(x)\right) = \begin{pmatrix} g_{11}\int_{\mathbb{R}^n} x_1 \,\mathrm{d}\,\mu(x) + \ldots + g_{1n}\int_{\mathbb{R}^n} x_n \,\mathrm{d}\,\mu(x) \\ \vdots \\ g_{n1}\int_{\mathbb{R}^n} x_1 \,\mathrm{d}\,\mu(x) + \ldots + g_{nn}\int_{\mathbb{R}^n} x_n \,\mathrm{d}\,\mu(x) \end{pmatrix}$$

$$= \int_{\mathbb{R}^n} \begin{pmatrix} g_{11}x_1 + \ldots + g_{1n}x_n \\ \vdots \\ g_{n1}x_1 + \ldots + g_{nn}x_n \end{pmatrix} \mathrm{d}\,\mu(x) = \int_{\mathbb{R}^n} gx \,\mathrm{d}\,\mu(x).$$

Let $\iota\colon \mathbb{R}^n \hookrightarrow \mathbb{C}^n$, $x \mapsto x$ denote the natural inclusion. For all curvature measures $\Phi \in \ker(\mathrm{glob})$ and all convex bodies $K \subseteq \mathbb{R}^n$, we then obtain

$$B(g\Phi)(K) = g\Phi(K, \iota) = \Phi\left(g^{-1}K, \iota \circ g\right) = \int_{\mathbb{R}^n} gx \,\mathrm{d}\,\Phi\left(g^{-1}K, x\right)$$

$$= g\left(\int_{\mathbb{R}^n} x \,\mathrm{d}\,\Phi\left(g^{-1}K, x\right)\right) = g\left(B\Phi\left(g^{-1}K\right)\right) = g(B\Phi)(K)$$

which shows that $B$ is equivariant. □

## 4.2 The Smoothness Conjecture

A curvature measure $\Phi \in \mathrm{Curv}$ is called $\mathrm{GL}(n, \mathbb{R})$*-smooth* if it is a smooth vector of the representation $\mathrm{GL}(n, \mathbb{R}) \to \mathrm{GL}(\mathrm{Curv})$ defined in the previous section. We write $\mathrm{Curv}^\infty \subseteq \mathrm{Curv}$ for the dense linear subspace of $\mathrm{GL}(n, \mathbb{R})$-smooth curvature measures.

**Proposition 4.10.** *Any smooth curvature measure* $\Phi \in \mathrm{Curv}$ *is* $\mathrm{GL}(n, \mathbb{R})$*-smooth. In short,* $\mathrm{Curv}^{\mathrm{sm}} \subseteq \mathrm{Curv}^\infty$.

*Proof.* $\mathrm{VOL}_n \in \mathrm{Curv}^\infty$ is proved in Proposition 4.12 below independently of this proposition. For curvature measures of the form $[\omega]$ for some differential form $\omega \in \Omega^{n-1}(S\mathbb{R}^n)^{\mathrm{tr}}$,

note that there exists a well-known $\mathrm{GL}(n,\mathbb{R})$-equivariant construction of the normal cycle, see e.g. Section 5 of [Ale06a]. Using this construction, the proof of $[\omega] \in \mathrm{Curv}^\infty$ follows verbatim to the analogous proof in the case of valuations which can be found in Proposition 5.1.9(i) of [Ale06a]. □

Note that in the setting of valuations, Theorem 2.23 tells us exactly that the two notions of smoothness coincide. Hence, it is natural to pose the following conjecture in the setting of curvature measures.

**Conjecture 4.11 (Smoothness Conjecture).** *A curvature measure $\Phi \in \mathrm{Curv}$ is smooth if and only if it is $\mathrm{GL}(n,\mathbb{R})$-smooth. In short, $\mathrm{Curv}^\infty = \mathrm{Curv}^{\mathrm{sm}}$.*

Of course, Proposition 4.10 resolves the only if part in full generality. For the converse, note that the Kiderlen–Weil decomposition yields that the smoothness conjecture is equivalent to

$$\mathrm{Curv}_i^{\varepsilon,\infty} = \mathrm{Curv}_i^{\varepsilon,\mathrm{sm}}$$

for all $i \in \{0,\dots,n\}$ and $\varepsilon \in \{+,-\}$.

We will now use the characterization theorems from the previous chapter to verify this in special cases. The easiest case $i = n$, which can be treated independently of parity, is an immediate consequence of the characterization of $n$-homogeneous curvature measures.

**Proposition 4.12.** $\mathrm{Curv}_n^\infty = \mathrm{Curv}_n^{\mathrm{sm}}$.

*Proof.* Recall from Theorem 3.25 that $\mathrm{Curv}_n = \mathbb{C}\,\mathrm{VOL}_n$. On the one hand, $c\,\mathrm{VOL}_n$ is smooth for all $c \in \mathbb{C}$ by definition which implies that $\mathrm{Curv}_n^{\mathrm{sm}} = \mathrm{Curv}_n$. On the other hand, $\mathrm{Curv}_n^\infty$ is a dense subspace of the finite dimensional space $\mathrm{Curv}_n$ which implies that $\mathrm{Curv}_n^\infty = \mathrm{Curv}_n$. Hence, we have $\mathrm{Curv}_n^\infty = \mathrm{Curv}_n^{\mathrm{sm}}$ as required. □

The case of curvature measures of degree $n-1$ is not much harder and follows from the characterization given in Theorem 3.27.

Let us however first include a lemma which will be of use at various points in the following.

**Lemma 4.13.** *For all $i = 0,\dots,n$, the map*

$$C\left(S^{n-1}\right) \to \mathrm{Curv}_i,\ f \mapsto \Phi_i^f$$

*is $\mathbb{C}$-linear, continuous and $\mathrm{SO}(n)$-equivariant.*

*Proof.* The proof of linearity and continuity can be copied from the corresponding proof in Theorem 3.27 with the only change that the degree $n-1$ has to be replaced by the general degree $i$ now.

For $\mathrm{SO}(n)$-equivariance, recall that the representation $\mathrm{SO}(n) \to \mathrm{GL}(C(S^{n-1}))$ is given by

$$(gf)(u) = f\left(g^{-1}u\right)$$

for all $g \in \mathrm{SO}(n)$, $f \in C\left(S^{n-1}\right)$ and $u \in S^{n-1}$. Therefore, we obtain

$$\begin{aligned}\Phi_i^{gf}(K,\beta) &= \int_{\pi_1^{-1}(\beta)} \pi_2^*(gf)(x,u)\,\mathrm{d}\,\Theta_i\left(K,(x,u)\right) = \int_{\pi_1^{-1}(\beta)} f\left(g^{-1}u\right)\mathrm{d}\,\Theta_i\left(K,(x,u)\right)\\ &= \int_{\pi_1^{-1}(\beta)} \pi_2^* f\left(gx,g^{-1}u\right)\mathrm{d}\,\Theta_i\left(K,(x,u)\right) = \int_{\pi_1^{-1}\left(g^{-1}\beta\right)} \pi_2^* f(x,u)\,\mathrm{d}\,\Theta_i\left(g^{-1}K,(x,u)\right)\\ &= \Phi_i^f\left(g^{-1}K,g^{-1}\beta\right) = g\Phi_i^f(K,\beta)\end{aligned}$$

for all convex bodies $K \subseteq \mathbb{R}^n$ and Borel sets $\beta \subseteq \mathbb{R}^n$, where we have used that $\Theta_i$ is $\mathrm{SO}(n)$-equivariant in both components, see formulas (4.14) and (4.15) of [Sch14]. □

**Proposition 4.14.** $\mathrm{Curv}_{n-1}^{\infty} = \mathrm{Curv}_{n-1}^{\mathrm{sm}}$.

*Proof.* Let $\Phi = \Phi^f \in \mathrm{Curv}_{n-1}$ for some continuous function $f\colon S^{n-1} \to \mathbb{C}$, where we use the notation from Theorem 3.27. Moreover, assume that $\Phi$ is $\mathrm{GL}(n,\mathbb{R})$-smooth which particularly implies that $\Phi$ is $\mathrm{SO}(n)$-smooth. The previous lemma implies that the isomorphism in Theorem 3.27 is $\mathrm{SO}(n)$-equivariant. Hence, $\Phi^f$ is $\mathrm{SO}(n)$-smooth if and only if $f$ is $\mathrm{SO}(n)$-smooth. This is well-known to be equivalent to $f \in C^{\infty}\left(S^{n-1}\right)$, see for example Lemma 1.27 of [Wan11]. Therefore, the differential form $\pi_2^* f \wedge \omega_{n-1}$, see Section 3.6, is smooth which yields that

$$\Phi = \Phi^f = \left[\pi_2^* f \wedge \omega_{n-1}\right] \in \mathrm{Curv}_{n-1}^{\mathrm{sm}}$$

as required. □

As an immediate consequence we obtain that the smoothness conjecture is true in dimension $n = 1$.

**Corollary 4.15.** *If $n = 1$, then* $\mathrm{Curv}^{\infty} = \mathrm{Curv}^{\mathrm{sm}}$.

## 4.3 Smooth Curvature Measures on the Plane

Throughout this section, fix the dimension $n = 2$ which means that we study curvature measures on the plane $\mathbb{R}^2$. The goal is to confirm the smoothness conjecture in this case. By virtue of Proposition 4.12 and Proposition 4.14, we only have curvature measures of degree $i = 0$ left to treat.

Let us start with the odd 0-homogeneous case. In this case the idea of the proof is to use an embedding similar to the Klain embedding, see Theorem 2.20.

**Lemma 4.16.** *There exists a $\mathbb{C}$-linear, continuous and $\mathrm{SO}(2)$-equivariant embedding*

$$I\colon \mathrm{Curv}_0^- \hookrightarrow C^-\left(S^1\right).$$

*Proof.* Let $\Phi \in \mathrm{Curv}_0^-$ and $u \in S^1$. Moreover, let $l \subseteq \mathbb{R}^2$ denote the 1-dimensional linear subspace spanned by $u$. Then, the restriction $\Phi|_l$ of $\Phi$ to all convex bodies and Borel sets contained in $l$ is a curvature measure

$$\Phi|_l \in \mathrm{Curv}_0^-(l).$$

Therefore, there exists a constant $c = c(\Phi, u) \in \mathbb{C}$ such that $\Phi|_l\left([0,u]\right) = c\delta_0 - c\delta_u$. This construction induces the map

$$I\Phi\colon S^1 \to \mathbb{C},\ u \mapsto c(\Phi, u).$$

Note that for any curvature measure $\Phi \in \mathrm{Curv}_0^-$ and any $u \in S^1$, we have

$$I\Phi(u) = c(\Phi,u) = c(\Phi,u)\,\delta_0\left(\{0\}\right) - c(\Phi,u)\,\delta_u\left(\{0\}\right) = \Phi\left([0,u],\{0\}\right).$$

In particular, we obtain

$$I\Phi(-u) = \Phi\left([0,-u],\{0\}\right) = \Phi\left(-[0,u],-\{0\}\right) = -\Phi\left([0,u],\{0\}\right) = -I\Phi(u)$$

which shows that $I\Phi$ is an odd function on the circle. Hence, we have indeed constructed a map

$$I\colon \mathrm{Curv}_0^- \to C^-\left(S^1\right),\ \Phi \mapsto I\Phi.$$

Let $\lambda \in \mathbb{C}$ and $\Phi, \Psi \in \mathrm{Curv}_0^-$. For all $u \in S^1$, we compute

$$\begin{aligned} &I(\lambda\Phi + \Psi)(u) = (\lambda\Phi + \Psi)\left([0,u],\{0\}\right) = \lambda\Phi\left([0,u],\{0\}\right) + \Psi\left([0,u],\{0\}\right) \\ &= \lambda I\Phi(u) + I\Psi(u) \end{aligned}$$

which shows that $I$ is $\mathbb{C}$-linear.

Let $\Phi \in \mathrm{Curv}_0^-$ be arbitrary. Then, the computation

$$\begin{aligned} \|I\Phi\| &= \sup_{u\in S^1}\left|I\Phi(u)\right| = \sup_{u\in S^1}\left|\Phi\left([0,u],\{0\}\right)\right| \leq \sup_{u\in S^1}\left|\Phi\left([0,u]\right)\right|\left(\{0\}\right) \\ &\leq \sup_{u\in S^1}\left|\Phi\left([0,u]\right)\right|\left(\mathbb{R}^2\right) = \sup_{u\in S^1}\left\|\Phi\left([0,u]\right)\right\| \leq \sup_{K\subseteq B^2}\left\|\Phi(K)\right\| = \|\Phi\|_{B^2} \end{aligned}$$

shows that $I$ is continuous.

For $\mathrm{SO}(2)$-equivariance, assume that $\Phi \in \mathrm{Curv}_0^-$ and $g \in \mathrm{SO}(2)$. Then, we see that

$$\begin{aligned} &I(g\Phi)(u) = g\Phi\left([0,u],\{0\}\right) = \Phi\left(g^{-1}[0,u], g^{-1}\{0\}\right) \\ &= \Phi\left(\left[0, g^{-1}u\right],\{0\}\right) = I\Phi\left(g^{-1}u\right) = g(I\Phi)(u) \end{aligned}$$

for all $u \in S^1$ as required.

Lastly, we want to verify that $I$ is injective. To do so, assume that $\Phi \in \ker(I)$. Then, we have

$$\Phi|_l\left([0,u]\right) = I\Phi(u)\,\delta_0 - I\Phi(u)\,\delta_u = 0 \in \mathcal{M}(l)$$

for all $u \in S^1$. Since $\Phi\left([0,u],\beta\right) = \Phi\left([0,u],\beta\cap[0,u]\right)$ for all $\beta \in \mathcal{B}$, it follows that $\Phi\left([0,u]\right) = 0 \in \mathcal{M}$ for all $u \in S^1$. By 0-homogeneity, it follows that

$$\Phi\left([0,x],\beta\right) = \Phi\left(\left[0, \frac{x}{\|x\|}\right], \|x\|^{-1}\beta\right) = 0$$

holds for all $x \in \mathbb{R}^2 \setminus \{0\}$ and all $\beta \in \mathcal{B}$. Let $a \neq b$ be two distinct points in $\mathbb{R}^2$ now. Then, we have

$$\Phi\left([a,b],\beta\right) = \Phi\left([a,b]-a,\beta-a\right) = \Phi\left([0,b-a],\beta-a\right) = 0$$

for all $\beta \in \mathcal{B}$ due to translation-invariance of $\Phi$. Let $e_1 = (1,0) \in \mathbb{R}^2$. For any point $z \in \mathbb{R}^2$, we can write $\Phi\left(\{z\}\right)$ as the weak limit of $\Phi\left([z-\varepsilon e_1, z+\varepsilon e_1]\right)$ as $\varepsilon \to 0$. Since $\Phi\left([z-\varepsilon e_1, z+\varepsilon e_1]\right) = 0$ for all $\varepsilon > 0$, it follows that $\Phi\left(\{z\}\right) = 0$. Thus, we have proved that $\Phi$ vanishes on line segments and points, which means that $\Phi$ is simple. By the characterization of simple curvature measures, we obtain that

$$\Phi \in \mathrm{Curv}_0^- \cap \left(\ker\left(\mathrm{glob}_0^+\right) \oplus \mathrm{Curv}_1^- \oplus \mathrm{Curv}_2\right) = \{0\}\,.$$

Hence, $I$ is indeed injective. □

**Theorem 4.17.** *$I$ restricts to an isomorphism*

$$\mathrm{Curv}_0^{-,\infty} \cong C^{-,\infty}\left(S^1\right) \cong \mathrm{Curv}_0^{-,\mathrm{sm}}\,.$$

*In particular, the smoothness conjecture is true for 0-homogeneous odd curvature measures on $\mathbb{R}^2$, that is*

$$\mathrm{Curv}_0^{-,\infty} = \mathrm{Curv}_0^{-,\mathrm{sm}}\,.$$

*Proof.* Let $\Phi \in \mathrm{Curv}_0^-$ be $\mathrm{GL}(2,\mathbb{R})$-smooth. Then, $\Phi$ is in particular $\mathrm{SO}(2)$-smooth which implies that $I\Phi \in C^-\left(S^1\right)$ is $\mathrm{SO}(2)$-smooth by equivariance of $I$. This is well-known to be equivalent to

$$I\Phi \in C^{-,\infty}\left(S^1\right).$$

Combined with Theorem 4.10, we obtain that

$$\mathrm{Curv}_0^{-,\mathrm{sm}} \subseteq \mathrm{Curv}_0^{-,\infty} \hookrightarrow C^{-,\infty}\left(S^1\right),$$

where the injective map is given by $I$. Therefore, it is only left to show that the restriction of $I$ to smooth curvature measures maps surjectively into smooth functions, that is

$$I\colon\ \mathrm{Curv}_0^{-,\mathrm{sm}} \to C^{-,\infty}\left(S^1\right)$$

is surjective.

We start with some preliminary computations. For all smooth odd functions $g\colon S^1 \to \mathbb{C}$, we consider the curvature measure

$$\Phi_0^g\colon\ (K,\beta) \mapsto \int_{\pi^{-1}(\beta)} \pi_2^* g \,\mathrm{d}\,\Theta_0\,(K) = \int_{\mathrm{nc}(K)\cap\pi^{-1}(\beta)} \pi_2^* g \,\mathrm{d}\,\Theta_0\,(K)\,,$$

where the equality follows from the fact that the support of $\Theta_0(K, \cdot)$ is contained in $\mathrm{nc}(K)$. Since $g$ is smooth and odd, it follows that

$$\Phi_0^g \in \mathrm{Curv}_0^{-,\mathrm{sm}}.$$

Now note that for any $u \in S^1$, the normal cycle of the line segment $[0,u]$ satisfies

$$\begin{aligned}\mathrm{nc}\left([0,u]\right) = &\left((0,u) \times \left(u^\perp \cap S^1\right)\right) \cup \left(\{0\} \times \left\{v \in S^1 \mid \langle v,u\rangle \leq 0\right\}\right)\\ &\cup \left(\{u\} \times \left\{v \in S^1 \mid \langle v,u\rangle \geq 0\right\}\right),\end{aligned}$$

where this union is obviously disjoint. Of course, $u^\perp \cap S^1$ consists exactly of two antipodal points, say $u^\perp \cap S^1 = \{N, -N\}$. This yields that

$$\begin{aligned}&\int_{\left((0,u)\cap\beta\right)\times\left(u^\perp\cap S^1\right)} \pi_2^* g \,\mathrm{d}\,\Theta_0\left([0,u]\right)\\ &= \int_{\left((0,u)\cap\beta\right)\times\{N\}} \pi_2^* g \,\mathrm{d}\,\Theta_0\left([0,u]\right) + \int_{\left((0,u)\cap\beta\right)\times\{-N\}} \pi_2^* g \,\mathrm{d}\,\Theta_0\left([0,u]\right)\\ &= 0\end{aligned}$$

for all Borel sets $\beta \subseteq \mathbb{R}^n$, where the second equality follows from the assumption that $g$ is odd. Therefore, we obtain

$$\begin{aligned}&\Phi_0^g\left([0,u],\beta\right) = \int_{\mathrm{nc}\left([0,u]\right)\cap\pi_1^{-1}(\beta)} \pi_2^* g \,\mathrm{d}\,\Theta_0\left([0,u]\right)\\ &= \int_{\left(\{0\}\cap\beta\right)\times\left\{v\in S^1\mid\langle v,u\rangle\leq 0\right\}} \pi_2^* g \,\mathrm{d}\,\Theta_0\left([0,u]\right) + \int_{\left(\{u\}\cap\beta\right)\times\left\{v\in S^1\mid\langle v,u\rangle\geq 0\right\}} \pi_2^* g \,\mathrm{d}\,\Theta_0\left([0,u]\right)\\ &= \left(\int_{\left\{v\in S^1\mid\langle v,u\rangle\leq 0\right\}} g \,\mathrm{d}\,S_0\left([0,u]\right)\right)\delta_0(\beta) + \left(\int_{\left\{v\in S^1\mid\langle v,u\rangle\geq 0\right\}} g \,\mathrm{d}\,S_0\left([0,u]\right)\right)\delta_u(\beta)\\ &= \left(\int_{\left\{v\in S^1\mid\langle v,u\rangle\leq 0\right\}} g(v) \,\mathrm{d}\,v\right)\delta_0(\beta) + \left(\int_{\left\{v\in S^1\mid\langle v,u\rangle\geq 0\right\}} g(v) \,\mathrm{d}\,v\right)\delta_u(\beta),\end{aligned}$$

where the last equality follows from Theorem 4.2.3 of [Sch14].

Now fix an arbitrary function $f \in C^{-,\infty}\left(S^1\right)$. It is well-known that the hemispherical transform

$$\mathcal{H}\colon C^{-,\infty}\left(S^1\right) \to C^{-,\infty}\left(S^1\right),\ g \mapsto \left(\mathcal{H}g\colon u \mapsto \int_{\left\{v\in S^1\mid\langle v,u\rangle\geq 0\right\}} g(v) \,\mathrm{d}\,v\right),$$

restricted to smooth odd functions in domain and codomain, is a $\mathbb{C}$-linear isomorphism. Injectivity is a special case of Proposition 3.4.11 of [Gro96]; for surjectivity see p. 132

in the aforementioned reference. Hence, we find a smooth odd function $g_0 \colon S^1 \to \mathbb{C}$, depending on $f$, such that

$$\int_{\left\{v \in S^1 \mid \langle v,u \rangle \geq 0\right\}} g_0(v) \, \mathrm{d}\, v = -f(u)$$

for all $u \in S^1$. Since $g_0$ is odd, it follows that

$$\begin{aligned} &\int_{\left\{v \in S^1 \mid \langle v,u \rangle \leq 0\right\}} g_0(v) \, \mathrm{d}\, v = \int_{\left\{v \in S^1 \mid \langle v,u \rangle \geq 0\right\}} g_0(-v) \, \mathrm{d}\, v \\ &= -\int_{\left\{v \in S^1 \mid \langle v,u \rangle \geq 0\right\}} g_0(v) \, \mathrm{d}\, v = f(u) \end{aligned}$$

for all $u$. Plugging the last two equations into the computation above, we obtain

$$\Phi_0^{g_0}\left([0,u], \beta\right) = f(u) \, \delta_0(\beta) - f(u) \, \delta_u(\beta)$$

for all $u \in S^1$ and $\beta \in \mathcal{B}$. This shows that $I\left(\Phi_0^{g_0}\right) = f$ which yields the required surjectivity. □

We now turn to the even case which requires some more work than the odd one. The main part is to find an explicit description for the kernel of the globalization map

$$\mathrm{glob}_0^{+,\infty} \colon \mathrm{Curv}_0^{+,\infty} \to \mathrm{Val}_0^{+,\infty}.$$

Note that we naturally consider vector-valued valuations of codegree one via the Bernig embedding. Suppose you are given a valuation

$$\varphi \in \mathrm{Val}_{n-1}(\mathbb{R}^n) \otimes \mathbb{C}^m$$

for some $m, n \in \mathbb{N}$. A continuous map $f \colon S^{n-1} \to \mathbb{C}^m$ will be called a *McMullen map* of $\varphi$ if it satisfies

$$\varphi(K) = \int_{S^{n-1}} f \, \mathrm{d}\, S_{n-1}(K)$$

for all convex bodies $K \subseteq \mathbb{R}^n$. By Theorem 2.13, any valuation $\varphi$ admits a McMullen map.

The following lemma describes the class of McMullen maps explicitly in our special setting.

**Lemma 4.18.** *Let $\varphi \in \mathrm{Val}_1^- \otimes \mathbb{C}^2$ be a $\mathbb{C}^2$-valued valuation on $\mathbb{R}^2$, and let $f \colon S^1 \to \mathbb{C}^2$ be a continuous map. Then, $f$ is a McMullen map of $\varphi$ if and only if there exists a linear map $L \colon S^1 \to \mathbb{C}^2$ such that $f(u) = u_1 u_2 \varphi(\Delta_u) + L(u)$ for all unit vectors $u = (u_1, u_2) \in S^1 \setminus \{\pm e_1, \pm e_2\}$, where*

$$\Delta_u = \mathrm{conv}\left\{(0,0), \left(u_1^{-1}, 0\right), \left(0, u_2^{-1}\right)\right\}.$$

*Proof.* By McMullen's theorem, we have

$$\mathrm{Val}_1^- \otimes \mathbb{C}^2 \cong C^-\left(S^1, \mathbb{C}^2\right) \Big/ L\left(S^1, \mathbb{C}^2\right).$$

Therefore, it is satisfactory to show that for any valuation $\varphi \in \mathrm{Val}_1^- \otimes \mathbb{C}^2$ and any McMullen map $f\colon S^1 \to \mathbb{C}^2$ of $\varphi$, there exists a linear map $L\colon \mathbb{R}^2 \to \mathbb{C}^2$ such that

$$f(u) = u_1 u_2 \varphi(\Delta_u) + L(u)$$

for all unit vectors $u \in S^1 \setminus \{\pm e_1, \pm e_2\}$.

To do so, fix arbitrary $\varphi$ and $f$ as above. Moreover, assume that $u = (u_1, u_2) \in S^1$ satisfies $u_1, u_2 > 0$. This induces a simplex

$$\Delta_u = \mathrm{conv}\left\{(0,0),\ \left(u_1^{-1}, 0\right),\ \left(0, u_2^{-1}\right)\right\} \in \mathcal{K}.$$

An elementary geometric argument shows that

$$\Delta_u = \left\{x \in \mathbb{R}^2 \mid \langle u, x\rangle \leq 1,\ \langle e_1, x\rangle \geq 0,\ \langle e_2, x\rangle \geq 0\right\}.$$

Therefore, the normal vectors to the three facets of $\Delta_u$ are given by $-e_1$, $-e_2$, and $u$. Let $F_1$, $F_2$, and $F_u$ denote the corresponding facets, respectively. By virtue of the defining vertices, we have

$$\mathrm{vol}_1(F_1) = u_2^{-1},\ \mathrm{vol}_1(F_2) = u_1^{-1}.$$

The Pythagorean theorem therefore implies that

$$\mathrm{vol}_1(F_u) = \sqrt{\frac{1}{u_2^2} + \frac{1}{u_1^2}} = \sqrt{\frac{u_1^2 + u_2^2}{u_1^2 u_2^2}} = \frac{1}{u_1 u_2},$$

where we of course used that $u \in S^1$. Since $\varphi$ is odd by assumption, it follows that $f$ has to be odd. Hence, we obtain

$$\begin{aligned}\varphi(\Delta_u) &= \int_{S^1} f \,\mathrm{d}\, S_1(\Delta_u) = \mathrm{vol}_1(F_u) f(u) + \mathrm{vol}_1(F_1) f(-e_1) + \mathrm{vol}_1(F_2) f(-e_2)\\ &= \frac{1}{u_1 u_2} f(u) - \frac{1}{u_2} f(e_1) - \frac{1}{u_1} f(e_2).\end{aligned}$$

Let $L\colon S^1 \to \mathbb{C}^2$ be the unique linear map satisfying $L(e_1) = f(e_1)$ and $L(e_2) = f(e_2)$. Since

$$\int_{S^1} (f - L) \,\mathrm{d}\, S_1(K) = \int_{S^1} f \,\mathrm{d}\, S_1(K) - \int_{S^1} L \,\mathrm{d}\, S_1(K) = \int_{S^1} f \,\mathrm{d}\, S_1(K)$$

for all $K \in \mathcal{K}$, we see that $f$ is a McMullen map of $\varphi$ if and only if $f - L$ is. Hence, applying the computation above to $f - L$ yields

$$\begin{aligned}\varphi(\Delta_u) &= \frac{1}{u_1 u_2}(f-L)(u) - \frac{1}{u_2}(f-L)(e_1) - \frac{1}{u_1}(f-L)(e_2)\\ &= \frac{1}{u_1 u_2}\big(f(u) - L(u)\big)\end{aligned}$$

which is obviously equivalent to the required $f(u) = u_1 u_2 \varphi(\Delta_u) + L(u)$. The same argument works with the necessary but obvious sign changes if $u \in S^1 \setminus \{\pm e_1, \pm e_2\}$ lies in a different open quadrant. □

We will use the following notation. Given a convex body $K \subseteq \mathbb{R}^n$ and a boundary point $x \in \partial K$, we write $N(K, x)$ for the *normal cone* of $K$ in $x$; for a definition see p. 81 of [Sch14]. Since $N(K, x)$ is a subset of $\mathbb{R}^n$, we write

$$n(K, x) = N(K, x) \cap S^{n-1}$$

for the *set of unit normals* of $K$ in $x$.

The previous lemma combined with the Bernig embedding and McMullen's theorem gives rise to a useful embedding which is the first important tool for achieving our goal of an explicit characterization.

For the remainder of this section, let $A$ denote the vector space of odd continuous maps $f\colon S^1 \to \mathbb{C}^2$ which satisfy

$$\big\langle u, f(u)\big\rangle = 0$$

for all $u \in S^1$, where $\langle \cdot, \cdot \rangle$ denotes the standard inner product of $\mathbb{C}^2$ with the convention that the inner product is antilinear in the first argument and linear in the second.

**Proposition 4.19.** *There exists a $\mathbb{C}$-linear monomorphism*

$$\iota\colon \ker\left(\mathrm{glob}_0^+\right) \hookrightarrow A,\ \Phi \mapsto f_\Phi$$

*such that $f_\Phi$ is a McMullen map of $B\Phi$.*

*Proof.* Let $\Phi \in \mathrm{Curv}_0^+$ satisfy $\mathrm{glob}(\Phi) = 0$. Then, the Bernig embedding yields a valuation $B\Phi \in \mathrm{Val}_1^- \otimes \mathbb{C}^2$. Since $\Phi$ and $B\Phi$ are both simple, we assume that all convex bodies appearing in this proof are two-dimensional.

Suppose that $P \subseteq \mathbb{R}^2$ is a polytope with vertex set $\mathcal{F}_0(P) = \{v_0, \dots, v_m\}$. From equation (3.2) of [KW99], one obtains that there exist constants $c_0, \dots, c_m \in \mathbb{C}$ such that

$$\Phi(P, \beta) = \sum_{i=0}^{m} c_i \delta_{v_i}(\beta)$$

for all $\beta \in \mathcal{B}$, where every constant $c_i$ solely depends on the set of unit normals $n(P, v_i)$. Applying the equation above to the singletons $\beta = \{v_i\}$ yields that $c_i = \Phi\big(P, \{v_i\}\big)$ for all $i = 0, \dots, m$. Therefore, the Bernig embedding satisfies

$$B\Phi(P) = \sum_{i=0}^{m} c_i \int_{\mathbb{R}^2} x \,\mathrm{d}\, \delta_{v_i} = \sum_{i=0}^{m} \Phi\big(P, \{v_i\}\big)\, v_i.$$

Let $u \in S^1 \setminus \{\pm e_1, \pm e_2\}$ be arbitrary and consider the simplex $\Delta_u \in \mathcal{K}$ again which is the convex hull of

$$v_0 = (0,0)\,,\ v_1 = \left(u_1^{-1}, 0\right),\ v_2 = \left(0, u_2^{-1}\right).$$

The locality property of $\Phi$ yields that $\Phi\left(\Delta_u, \{v_0\}\right)$ is uniquely determined on any neighborhood of $v_0$. By definition of $\Delta_u$, it therefore follows that

$$c_\Phi = \Phi\left(\Delta_u, \{v_0\}\right)$$

does not depend on $u$.

We consider the map $f_\Phi \colon S^1 \to \mathbb{C}^2$ defined by

$$f_\Phi(u) = \begin{cases} u_1 u_2 B\Phi\left(\Delta_u\right) + L(u)\,, & \text{if } u \in S^1 \setminus \{\pm e_1, \pm e_2\}\,, \\ L(u)\,, & \text{if } u \in \{\pm e_1, \pm e_2\} \end{cases}$$

for all $u = (u_1, u_2) \in S^1$, where $L \colon S^1 \to \mathbb{C}$ is the linear map

$$L(u) = \frac{1}{2} c_\Phi \begin{pmatrix} u_2 \\ u_1 \end{pmatrix}.$$

Of course, $f_\Phi$ is continuous as $B\Phi$ is. Moreover, note that we have $\Delta_{-u} = -\Delta_u$ for all $u \in S^1 \setminus \{\pm e_1, \pm e_2\}$. Therefore, we obtain

$$(-u_1)(-u_2) B\Phi\left(\Delta_{-u}\right) = u_1 u_2 B\Phi\left(-\Delta_u\right) = -u_1 u_2 B\Phi\left(\Delta_u\right)$$

as $B\Phi$ is odd. Since $L$ is linear and therefore odd, it follows that $f_\Phi$ is odd.

Now, recall that the Bernig embedding is a $\mathbb{C}$-linear monomorphism and note that the previous lemma implies that $f_\Phi$ is a specific McMullen function of $B\Phi$. This shows that we have constructed a $\mathbb{C}$-linear monomorphism

$$\iota \colon \ker\left(\mathrm{glob}_0^+\right) \to C^-\left(S^1, \mathbb{C}^2\right),\ \Phi \mapsto f_\Phi.$$

It is left to show that the image of $\iota$ is contained in $A$. To do so, let $u = (u_1, u_2) \in S^1$ be arbitrary. Firstly, note that

$$\left\langle u, L(u) \right\rangle = \frac{1}{2} c_\Phi \left\langle \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}, \begin{pmatrix} u_2 \\ u_1 \end{pmatrix} \right\rangle = c_\Phi u_1 u_2.$$

In particular, for $u \in \{\pm e_1, \pm e_2\}$ we have

$$\left\langle u, f_\Phi(u) \right\rangle = \left\langle u, L(u) \right\rangle = c_\Phi u_1 u_2 = 0$$

as required, since $u_1 = 0$ or $u_2 = 0$. From now on, we assume that $u \neq \pm e_1, \pm e_2$. From the work of Kiderlen and Weil, more precisely from the last assertion in Theorem 3.1 of

[KW99], it follows that $\Phi(\Delta_u)$ is supported on the vertices of $\Delta_u$. Combining this with $\operatorname{glob}(\Phi) = 0$, we can conclude that

$$\Phi\left(\Delta_u, \{v_1\}\right) + \Phi\left(\Delta_u, \{v_2\}\right) = -\Phi\left(\Delta_u, \{v_0\}\right).$$

Moreover, the formula for $B\Phi(P)$ above applied to the polytope $P = \Delta_u$ yields that

$$B\Phi(\Delta_u) = \Phi\left(\Delta_u, \{v_1\}\right) v_1 + \Phi\left(\Delta_u, \{v_2\}\right) v_2$$

which implies

$$\begin{aligned}\iota(\Phi)(u) &= f_\Phi(u) = u_1 u_2 \left(\Phi\left(\Delta_u, \{v_1\}\right) v_1 + \Phi\left(\Delta_u, \{v_2\}\right) v_2\right) + L(u) \\ &= u_1 u_2 \begin{pmatrix} \Phi\left(\Delta_u, \{v_1\}\right) u_1^{-1} \\ \Phi\left(\Delta_u, \{v_2\}\right) u_2^{-1} \end{pmatrix} + L(u) = \begin{pmatrix} \Phi\left(\Delta_u, \{v_1\}\right) u_2 \\ \Phi\left(\Delta_u, \{v_2\}\right) u_1 \end{pmatrix} + L(u).\end{aligned}$$

Combining the equations above, we obtain

$$\begin{aligned}&\left\langle u, \iota(\Phi)(u)\right\rangle = \left\langle u, \begin{pmatrix} \Phi\left(\Delta_u, \{v_1\}\right) u_2 \\ \Phi\left(\Delta_u, \{v_2\}\right) u_1 \end{pmatrix}\right\rangle + \left\langle u, L(u)\right\rangle \\ &= u_1 u_2 \left(\Phi\left(\Delta_u, \{v_1\}\right) + \Phi\left(\Delta_u, \{v_2\}\right)\right) + c_\Phi u_1 u_2 = -u_1 u_2 \Phi\left(\Delta_u, \{v_0\}\right) + c_\Phi u_1 u_2 \\ &= 0\end{aligned}$$

as required, where the last equality follows from the definition of $c_\Phi$. □

The other important ingredient is the fact that we already know a plethora of examples for the curvature measures of interest.

**Proposition 4.20.** *The map*

$$J\colon \left\{g \in C^+\left(S^1\right) \,\middle|\, \int_{S^1} g(u)\,\mathrm{d}\,u = 0\right\} \to \ker\left(\operatorname{glob}_0^+\right),\ g \mapsto \Phi_0^g$$

*is a continuous* $\mathbb{C}$*-linear monomorphism which is* $\mathrm{SO}(2)$*-equivariant.*

*Proof.* Throughout this proof, we write

$$V = \left\{g \in C^+\left(S^1\right) \,\middle|\, \int_{S^1} g(u)\,\mathrm{d}\,u = 0\right\}$$

for the domain of $J$.

To show that $J$ is well-defined, let $g \in V$ be arbitrary. Since $g$ is even, we have $J(g) = \Phi_0^g \in \operatorname{Curv}_0^+$. Moreover, the assumption $\int g = 0$ yields

$$\operatorname{glob}\left(\Phi_0^g\right)(K) = \int_{S\,\mathbb{R}^2} \pi_2^* g\,\mathrm{d}\,\Theta_0(K) = \int_{S^1} g\,\mathrm{d}\,S_0(K) = \int_{S^1} g(u)\,\mathrm{d}\,u = 0$$

which shows that $J(g) \in \ker(\mathrm{glob})$. Hence, $J$ is well-defined.

Moreover, note that $\mathbb{C}$-linearity, continuity and $\mathrm{SO}(2)$-equivariance of $J$ follow immediately from Lemma 4.13.

Lastly, we want to prove injectivity. To do so, assume that $g \in V \setminus \{0\}$. By continuity, there exists an open set $U \subseteq S^1$ such that $g(u) \neq 0$ for all $u \in U$. Hence, there exists a compact and connected set $C \subseteq U$ with non-empty interior such that

$$\int_C g(u) \, \mathrm{d}\, u \neq 0.$$

We can now choose a simplex $\Delta = \operatorname{conv}\{0, v_1, v_2\}$ such that the set of unit normals of $\Delta$ in $v_1$ satisfies $n(\Delta, v_1) = C$. Then, we have

$$J(g)\big(\Delta, \{v_1\}\big) = \int_{\{v_1\} \times S^1} \pi_2^* g \, \mathrm{d}\, \Theta_0(\Delta) = \int_{n(\Delta, v_1)} g \, \mathrm{d}\, S_0(\Delta) = \int_C g(u) \, \mathrm{d}\, u \neq 0,$$

where the second equality follows from equation (4.3) in [Sch14]. Since $g$ is an arbitrary non-zero element of $V$, it follows that $\ker(J) = \{0\}$ as required. □

We need two more technical lemmas before we are able to prove the main result in the even case. The first one tells us that the space $A$ can in fact be described by $\mathbb{C}$-valued functions rather than $\mathbb{C}^2$-valued maps. The second one gives an integration by parts formula on the unit sphere.

**Lemma 4.21.** *A map $f\colon S^1 \to \mathbb{C}^2$ satisfies $f \in A$ if and only if there exists an even continuous function $h\colon S^1 \to \mathbb{C}$ such that*

$$f(u) = \big(h(u)\, u_2, -h(u)\, u_1\big)$$

*for all $u = (u_1, u_2) \in S^1$.*

*Proof.* Firstly, suppose that $h \in C^+\left(S^1\right)$ and define $f = f_h : S^1 \to \mathbb{C}^2$ by

$$f(u) = \begin{pmatrix} h(u)\, u_2 \\ -h(u)\, u_1 \end{pmatrix}$$

for all $u = (u_1, u_2) \in S^1$. Of course, $f$ is continuous as $h$ is continuous. Since $h$ is even, $f$ furthermore satisfies

$$f(-u) = \begin{pmatrix} h(-u)\,(-u_2) \\ -h(-u)\,(-u_1) \end{pmatrix} = -\begin{pmatrix} h(u)\, u_2 \\ -h(u)\, u_1 \end{pmatrix} = -f(u)$$

for all $u \in S^1$ which shows that $f$ is odd. Lastly, we need to check the vanishing inner product property of $A$. To this end, note that we have

$$\big\langle u, f(u) \big\rangle = u_1 f_1(u) + u_2 f_2(u) = u_1 h(u)\, u_2 + u_2\big(-h(u)\, u_1\big) = 0$$

for all $u \in S^1$. Thus, we have seen that $f \in A$.

Conversely, assume that $f = (f_1, f_2) \in A$. We define a map $h\colon S^1 \to \mathbb{C}$ by

$$h(u) = \begin{cases} -f_2(u)\, u_1^{-1}, & \text{if } u_1 \neq 0, \\ f_1(u)\, u_2^{-1}, & \text{if } u_2 \neq 0. \end{cases}$$

Firstly, we need to check that $h$ is well-defined. To do so, let $u \in S^1$ such that $u_1 \neq 0$ and $u_2 \neq 0$. Then, the condition $f \in A$ implies

$$0 = \langle u, f(u) \rangle = u_1 f_1(u) + u_2 f_2(u) \quad \Longrightarrow \quad \frac{f_1(u)}{u_2} = -\frac{f_2(u)}{u_1}.$$

This is also sufficient to prove that $g$ is continuous, since the first case ensures that $h$ is continuous on $S^1 \setminus \{\pm e_2\}$ and the second case ensures that $h$ is continuous on $S^1 \setminus \{\pm e_1\}$. Next, we want to show that $h$ is even. To do so, let $u = (u_1, u_2) \in S^1$ be arbitrary. If $u_1 \neq 0$, we obtain

$$h(-u) = -\frac{f_2(-u)}{-u_1} = \frac{f_2(-u)}{u_1} = -\frac{f_2(u)}{u_1} = h(u)$$

from the assumption that $f_2$ is odd. Similarly, if $u_2 \neq 0$, the assumption that $f_1$ is odd implies

$$h(-u) = \frac{f_1(-u)}{-u_2} = -\frac{f_1(-u)}{-u_2} = \frac{f_1(-u)}{u_2} = h(u).$$

Lastly, we need to check that $f$ indeed satisfies $f(u) = \big(h(u)\, u_2,\, -h(u)\, u_1\big)$ for all $u \in S^1$. To do so, note that $u_1 f_1(u) + u_2 f_2(u) = 0$ also implies

$$\begin{aligned} -f_2(u)\, u_1^{-1} u_2 &= f_1(u) \ \text{ if } u_1 \neq 0, \\ -f_1(u)\, u_2^{-1} u_1 &= f_2(u) \ \text{ if } u_2 \neq 0. \end{aligned}$$

If $u_1 \neq 0$, this implies that

$$\begin{pmatrix} h(u)\, u_2 \\ -h(u)\, u_1 \end{pmatrix} = \begin{pmatrix} -f_2(u)\, u_1^{-1} u_2 \\ f_2(u)\, u_1^{-1} u_1 \end{pmatrix} = \begin{pmatrix} f_1(u) \\ f_2(u) \end{pmatrix} = f(u).$$

Analogously, we have

$$\begin{pmatrix} h(u)\, u_2 \\ -h(u)\, u_1 \end{pmatrix} = \begin{pmatrix} f_1(u)\, u_2^{-1} u_2 \\ -f_1(u)\, u_2^{-1} u_1 \end{pmatrix} = \begin{pmatrix} f_1(u) \\ f_2(u) \end{pmatrix} = f(u)$$

for $u_2 \neq 0$ which finishes the proof. □

The following lemma is stated and proved in arbitrary dimension $n \in \mathbb{N}$ as there is no benefit to working in dimension 2 and it might be useful for different applications.

**Lemma 4.22.** *Let $g\colon S^{n-1} \to \mathbb{C}$ be a smooth function and let $h\colon \mathbb{R}^n \setminus \{0\} \to \mathbb{C}$ be a 1-homogeneous smooth function. Then,*

$$\int_{S^{n-1}} g(u)\, \nabla h(u)\, \mathrm{d}\, u = -\int_{S^{n-1}} h(u)\, \nabla g_{-n}(u)\, \mathrm{d}\, u,$$

*where the gradients are computed in $\mathbb{R}^n \setminus \{0\}$ and*

$$g_\alpha\colon \mathbb{R}^n \setminus \{0\} \to \mathbb{C},\ x \mapsto \|x\|^\alpha\, g\left(\frac{x}{\|x\|}\right)$$

*denotes the $\alpha$-homogeneous continuation of $g$.*

*Proof.* We start with an auxiliary result that tells us which homogeneous functions can be integrated on the unbounded domain $\Omega = \mathbb{R}^n \setminus B^n$. Assume that $\beta \in \mathbb{C}$ satisfies $\mathrm{Re}(\beta) < -n$ and let $f\colon \mathbb{R}^n \setminus \{0\} \to \mathbb{C}$ be a $\beta$-homogeneous smooth function. Then, we have

$$\begin{aligned}\int_\Omega \big|f(x)\big|\, \mathrm{d}\, x &= \int_1^\infty \int_{S^{n-1}} \big|f(ru)\big|\, r^{n-1}\, \mathrm{d}\, u\, \mathrm{d}\, r = \int_1^\infty \int_{S^{n-1}} r^{\mathrm{Re}(\beta)} \big|f(u)\big|\, r^{n-1}\, \mathrm{d}\, u\, \mathrm{d}\, r \\ &= \left(\int_1^\infty r^{\mathrm{Re}(\beta)+n-1}\, \mathrm{d}\, r\right)\left(\int_{S^{n-1}} \big|f(u)\big|\, \mathrm{d}\, u\right).\end{aligned}$$

The first integral is finite because of $\mathrm{Re}(\beta) + n - 1 < -1$ and the second integral is of course finite as $S^{n-1}$ is compact. Therefore, $f$ is integrable on $\Omega$.

Now, let $\alpha \in \mathbb{C}$ satisfy $\mathrm{Re}(\alpha) < -n-1$. Then, we can compute

$$\begin{aligned}\int_{S^{n-1}} g(u)\, \nabla h(u)\, \mathrm{d}\, u &= -(\alpha+n)\left(\int_1^\infty r^{\alpha+n-1}\, \mathrm{d}\, r\right)\int_{S^{n-1}} g(u)\, \nabla h(u)\, \mathrm{d}\, u \\ &= -(\alpha+n)\int_1^\infty \int_{S^{n-1}} r^\alpha g(u)\, \nabla h(u)\, r^{n-1}\, \mathrm{d}\, u\, \mathrm{d}\, r.\end{aligned}$$

Since $g_\alpha$ is homogeneous of degree $\alpha$ and $\nabla h$ is homogeneous of degree 0, this yields

$$\begin{aligned}\int_{S^{n-1}} g(u)\, \nabla h(u)\, \mathrm{d}\, u &= -(\alpha+n)\int_1^\infty \int_{S^{n-1}} g_\alpha(ru)\, \nabla h(ru)\, r^{n-1}\, \mathrm{d}\, u\, \mathrm{d}\, r \\ &= -(\alpha+n)\int_\Omega g_\alpha(x)\, \nabla h(x)\, \mathrm{d}\, x.\end{aligned}$$

Note that the considerations from the beginning of this proof yield

$$\int_\Omega \left(\big|g_\alpha(x)\, h(x)\big| + \big|\partial_i g_\alpha(x)\, h(x)\big| + \big|g_\alpha(x)\, \partial_i h(x)\big|\right) \mathrm{d}\, x < \infty$$

as all the summands are homogeneous of a degree smaller than $-n$. Hence, we can apply the integration by parts formula to this last integral. To do so, we let $N\colon \partial\Omega \to S^{n-1}$ denote the outer unit normal of $\Omega$ which in our case is just $N(u) = -u$. This yields

$$\begin{aligned}\int_{S^{n-1}} g(u)\, \nabla h(u)\, \mathrm{d}\, u &= -(\alpha+n)\left(-\int_\Omega h(x)\, \nabla g_\alpha(x)\, \mathrm{d}\, x + \int_{\partial\Omega} g(u)\, h(u)\, N(u)\, \mathrm{d}\, u\right) \\ &= (\alpha+n)\int_\Omega h(x)\, \nabla g_\alpha(x)\, \mathrm{d}\, x + (\alpha+n)\int_{S^{n-1}} g(u)\, h(u)\, u\, \mathrm{d}\, u.\end{aligned}$$

We now compute the first integral in the last term above. Note that $h$ is homogeneous of degree 1 and $\nabla g_\alpha$ is homogeneous of degree $\alpha - 1$. Therefore, we have

$$\begin{aligned}
&\int_\Omega h(x) \nabla g_\alpha(x) \,\mathrm{d}\, x = \int_1^\infty \int_{S^{n-1}} h(ru) \nabla g_\alpha(ru) r^{n-1} \,\mathrm{d}\, u \,\mathrm{d}\, r \\
&= \int_1^\infty \int_{S^{n-1}} r h(u) r^{\alpha-1} \nabla g_\alpha(u) r^{n-1} \,\mathrm{d}\, u \,\mathrm{d}\, r \\
&= \left( \int_1^\infty r^{\alpha+n-1} \,\mathrm{d}\, r \right) \left( \int_{S^{n-1}} h(u) \nabla g_\alpha(u) \,\mathrm{d}\, u \right) \\
&= -\frac{1}{\alpha+n} \int_{S^{n-1}} h(u) \nabla g_\alpha(u) \,\mathrm{d}\, u.
\end{aligned}$$

Combined with the computation above, we obtain

$$\int_{S^{n-1}} g(u) \nabla h(u) \,\mathrm{d}\, u = -\int_{S^{n-1}} h(u) \nabla g_\alpha(u) \,\mathrm{d}\, u + (\alpha+n) \int_{S^{n-1}} g(u) h(u) u \,\mathrm{d}\, u.$$

We will now prove that the right hand side depends analytically on $\alpha$. Since the second summand is clearly holomorphic in $\alpha$, this is the case if $\alpha \mapsto \nabla g_\alpha(u)$ is holomorphic for all $u \in S^{n-1}$. To prove this, note that

$$\begin{aligned}
&\partial_i g_\alpha(u) = \partial_i \left( g\left( \frac{u}{\|u\|} \right) \|u\|^\alpha \right) = \partial_i \left( g_0(u) \|u\|^\alpha \right) = \partial_i g_0(u) \|u\|^\alpha + g_0(u) \partial_i \left( \|u\|^\alpha \right) \\
&= \partial_i g_0(u) + g_0(u) \partial_i \left( \|u\|^\alpha \right)
\end{aligned}$$

for all $i = 1, \ldots, n$. Moreover, the chain rule implies that

$$\partial_i \left( \|u\|^\alpha \right) = \frac{\alpha}{2} \left( u_1^2 + \ldots + u_n^2 \right)^{\alpha/2-1} 2u_i = \alpha u_i$$

since $u \in S^{n-1}$. Combining the last two equations, we have proved that

$$\nabla g_\alpha(u) = \nabla g_0(u) + \alpha g_0(u) u$$

for all $u \in S^{n-1}$ which is clearly holomorphic in $\alpha$.

Recall that we have already proved the formula

$$\int_{S^{n-1}} g(u) \nabla h(u) \,\mathrm{d}\, u = -\int_{S^{n-1}} h(u) \nabla g_\alpha(u) \,\mathrm{d}\, u + (\alpha+n) \int_{S^{n-1}} g(u) h(u) u \,\mathrm{d}\, u$$

on the half plane $\mathrm{Re}(\alpha) < -n-1$. Since it depends analytically on $\alpha$, it follows that this formula indeed holds for all $\alpha \in \mathbb{C}$ by analytic continuation. In particular, we obtain

$$\int_{S^{n-1}} g(u) \nabla h(u) \,\mathrm{d}\, u = -\int_{S^{n-1}} h(u) \nabla g_{-n}(u) \,\mathrm{d}\, u$$

for $\alpha = -n$ as required. □

This finally puts us in position to verify the smoothness conjecture for even curvature measures of degree 0 in $\mathbb{R}^2$.

**Theorem 4.23.** *The map $J\colon g \mapsto \Phi_0^g$ induces an isomorphism*

$$\ker\left(\mathrm{glob}_0^{+,\infty}\right) \cong \left\{ g \in C^{+,\infty}\left(S^1\right) \,\middle|\, \int_{S^1} g(u)\,\mathrm{d}\,u = 0 \right\}$$

*as complex* $\mathrm{SO}(2)$*-representations. Moreover, the smoothness conjecture holds for even curvature measures of degree* 0 *on* $\mathbb{R}^2$*, that is*

$$\mathrm{Curv}_0^{+,\infty} = \mathrm{Curv}_0^{+,\mathrm{sm}}.$$

*Proof.* Throughout this proof, let

$$V^\infty = \left\{ g \in C^{+,\infty}\left(S^1\right) \,\middle|\, \int_{S^1} g(u)\,\mathrm{d}\,u = 0 \right\}$$

denote the space on the right hand side of the theorem. For all $g \in V^\infty$, the curvature measure $J(g)$ satisfies

$$J(g) = \Phi_0^g = \left[\pi_2^* g \wedge \omega_0\right],$$

where $\omega_0$ denotes the differential form on $S\mathbb{R}^2$ from Section 3.6. Since $g$ is smooth, it follows that

$$\pi_2^* g \wedge \omega_0 \in \Omega^{n-1}\left(S\mathbb{R}^2\right)^{\mathrm{tr}}$$

which implies $J(g) \in \mathrm{Curv}^{\mathrm{sm}} \subseteq \mathrm{Curv}^\infty$. Therefore, $J$ induces a map

$$J^\infty\colon V^\infty \to \ker\left(\mathrm{glob}_0^{+,\infty}\right),\ g \mapsto \Phi_0^g.$$

From Proposition 4.20, we obtain that $J^\infty$ is a $\mathbb{C}$-linear, $\mathrm{SO}(2)$-equivariant monomorphism. Hence, only surjectivity of $J^\infty$ is left to prove.

To do so, let $K \subseteq \mathbb{R}^2$ be a smooth and strictly convex body. Then, we have

$$B\Phi_0^g(K) = \int_{\mathbb{R}^2} x\,\mathrm{d}\,\Phi_0^g(K,x) = \int_{\mathbb{R}^2 \times S^1} g(u)\,x\,\mathrm{d}\,\Theta_0\left(K,(x,u)\right).$$

From the assumptions on $K$ it follows that for all $u \in S^1$ there exists a unique point $x = x(K,u) \in \partial K$ such that $u$ is a normal vector to $\partial K$ in $x$. Moreover, it is well-known that this point satisfies $x = \nabla h_K(u)$. Note that $h_K$ is smooth on $\mathbb{R}^2 \setminus \{0\}$ by the assumptions on $K$. Thus, we have

$$\begin{aligned} B\Phi_0^g(K) &= \int_{S^1} g(u)\,\nabla h_K(u)\,\mathrm{d}\,S_0(K,u) = \int_{S^1} g(u)\,\nabla h_K(u)\,\mathrm{d}\,u \\ &= -\int_{S^1} h_K(u)\,\nabla g_{-2}(u)\,\mathrm{d}\,u, \end{aligned}$$

where the first equality follows from Lemma 4.2.2 of [Sch14] and the last equality follows from the previous lemma. From Proposition 4.19, we obtain that for any curvature measure $\Phi \in \operatorname{Curv}_0^{+,\infty}$ which satisfies $\operatorname{glob}(\Phi) = 0$, there exists a smooth map $f \in A$ such that

$$B\Phi(K) = \int_{S^1} f \,\mathrm{d}\, S_1(K) = \int_{S^1} f(u) (\Box h_K)(u) \,\mathrm{d}\, u = \int_{S^1} h_K(u) (\Box f)(u) \,\mathrm{d}\, u,$$

where we use the self-adjoint box operator

$$\Box = \Delta + \mathrm{Id} \colon C^\infty\left(S^1, \mathbb{C}^2\right) \to C^\infty\left(S^1, \mathbb{C}^2\right)$$

where $\Delta$ denotes the spherical Laplacian on $S^1$. Note that the kernel of $\Box$ consists exactly of the restrictions of linear maps. Hence, $\Box f$ is uniquely determined as $f$ is uniquely determined up to addition of linear maps. Since $B$ is injective, all that is left to show is that for all smooth $f \in A$ there exists a map $g \in V^\infty$ such that

$$-\nabla g_{-2} = \Box f$$

holds on $S^1$. To do so, assume that $f \in A$ is smooth for the remainder of the proof.

We first verify that if this differential equation has a solution $g$, then this solution satisfies $g \in V^\infty$. Since $g_{-2} \colon \mathbb{R}^2 \setminus \{0\} \to \mathbb{C}$ is homogeneous of degree $-2$, Euler's theorem on homogeneous functions implies that

$$\left\langle x, \nabla g_{-2}(x) \right\rangle = -2 g_{-2}(x)$$

for all $x \in \mathbb{R}^2 \setminus \{0\}$. Now recall that $\left\langle u, f(u) \right\rangle = 0$ for all $u \in S^1$, since $f \in A$. If $g \colon S^1 \to \mathbb{C}$ solves the differential equation above, it satisfies $f = -(\nabla g_{-2} + \Delta f)$ on the circle $S^1$. We therefore obtain

$$\begin{aligned} 0 &= -\left\langle u, f(u) \right\rangle = -\left\langle u, -(\nabla g_{-2} + \Delta f)(u) \right\rangle = \left\langle u, \nabla g_{-2}(u) \right\rangle + \left\langle u, \Delta f(u) \right\rangle \\ &= -2 g_{-2}(u) + \left\langle u, \Delta f(u) \right\rangle \end{aligned}$$

which yields

$$g(u) = \frac{\left\langle u, \Delta f(u) \right\rangle}{2}.$$

Of course $g$ is smooth because $f$ is smooth. Moreover, recall that $f$ is odd which yields that $\Delta f$ is also odd. Hence, we obtain

$$g(-u) = \frac{\left\langle -u, \Delta f(-u) \right\rangle}{2} = \frac{-\left\langle u, -\Delta f(u) \right\rangle}{2} = \frac{\left\langle u, \Delta f(u) \right\rangle}{2} = g(u)$$

for all $u \in S^1$ which shows that $g$ is even. By self-adjointness of the Laplacian with respect to the $L^2$-inner product, we have

$$\int_{S^1} g(u) \,\mathrm{d}\, u = \frac{1}{2} \int_{S^1} \left\langle u, \Delta f(u) \right\rangle \mathrm{d}\, u = \frac{1}{2} \int_{S^1} \left\langle \Delta i(u), f(u) \right\rangle \mathrm{d}\, u,$$

where $i\colon S^1 \to \mathbb{C}^2$, $u \mapsto u$ denotes the natural inclusion. Obviously, $i$ is the restriction of a linear map $\mathbb{R}^2 \to \mathbb{C}^2$. Hence, it is an eigenvalue of the Laplacian which means that we find $\lambda \in \mathbb{C} \setminus \{0\}$ such that $\lambda i = \Delta i$ (in fact, one easily shows that $\lambda = -1$). Thus, we have

$$\int_{S^1} g(u)\,\mathrm{d}\,u = \frac{1}{2}\int_{S^1} \langle \lambda i(u),\, f(u)\rangle\,\mathrm{d}\,u = \frac{\lambda}{2}\int_{S^1} \langle u, f(u)\rangle\,\mathrm{d}\,u = 0$$

as $f \in A$. This proves that $g \in V^\infty$ as required.

Next, we have to prove that the differential equation $-\nabla g_{-2} = \Box f$ has a solution $g$. To do so, recall from Lemma 4.21 that there exists a smooth, even function $h\colon S^1 \to \mathbb{C}$ such that

$$f(u) = \begin{pmatrix} h(u)\,u_2 \\ -h(u)\,u_1 \end{pmatrix}$$

for all $u = (u_1, u_2) \in S^1$. In the following, we let $\gamma\colon \mathbb{R} \to \mathbb{R}^2$, $t \mapsto (\cos t, \sin t)$ denote the standard $2\pi$-periodic parametric representation of $S^1$. Then, $f$ can be identified with a $2\pi$-periodic smooth function $f\colon \mathbb{R} \to \mathbb{C}^2$ which satisfies $\langle \gamma(t), f(t)\rangle = 0$ and $f(t+\pi) = -f(t)$ for all $t \in \mathbb{R}$, where the last condition comes from the fact that $f$ is odd. Note that by abuse of notation we denote this map by $f$, too. Similarly, $h$ can be identified with a smooth $\pi$-periodic function $h\colon \mathbb{R} \to \mathbb{C}$, where the period $\pi$ comes from the fact that $h$ is even. Under these identifications, we have

$$f(t) = \begin{pmatrix} h(t)\sin(t) \\ -h(t)\cos(t) \end{pmatrix}$$

for all $t \in \mathbb{R}$. Applying the product rule twice to this identity, we obtain

$$\begin{aligned} f_1''(t) &= h''(t)\sin(t) + 2h'(t)\cos(t) - h(t)\sin(t), \\ f_2''(t) &= -h''(t)\cos(t) + 2h'(t)\sin(t) + h(t)\cos(t) \end{aligned}$$

for all $t \in \mathbb{R}$ which implies that

$$\Box f(t) = f''(t) + f(t) = \begin{pmatrix} h''(t)\sin(t) + 2h'(t)\cos(t) \\ -h''(t)\cos(t) + 2h'(t)\sin(t) \end{pmatrix}.$$

Applying this to a point $u = (\cos t, \sin t) \in S^1$ yields $\Box f(u) = 2h'(t)\,u - h''(t)\,Ru$, where $R \in \mathrm{SO}(2)$ denotes the rotation $R(x,y) = (-y, x)$. Note that $h'\colon \mathbb{R} \to \mathbb{C}$ is $\pi$-periodic and smooth itself as $h$ is. Therefore, $h'$ induces a smooth, even function $g\colon S^1 \to \mathbb{C}$. By expressing the gradient of $g_{-2}$ with respect to the orthonormal basis $\{u, Ru\}$ of $\mathbb{R}^2$, we can compute

$$\begin{aligned} -\nabla g_{-2}(u) &= -\left(\partial_u g_{-2}(u)\,u + \partial_{Ru} g_{-2}(u)\,Ru\right) = -\left(-2g(u)\,u + \frac{dg}{dt}(u)\,Ru\right) \\ &= 2h'(t)\,u - h''(t)\,Ru = \Box f(u), \end{aligned}$$

where we have again used Euler's formula for homogeneous functions. Since $u$ was an arbitrary point this shows that $g$ is indeed a solution to our differential equation.

Lastly, we want to prove the smoothness conjecture for even curvature measures of degree 0 on $\mathbb{R}^2$. Since $\mathrm{Val}_0^+ = \mathbb{C}$ is finite-dimensional, we obtain that $\mathrm{glob}_0^+$ is surjective onto $\mathrm{Val}_0^+$ as it has dense image. Therefore, the first isomorphism theorem yields

$$\mathrm{Curv}_0^+ \Big/ \ker\left(\mathrm{glob}_0^+\right) \cong \mathbb{C}$$

which implies that it is satisfactory to prove the smoothness conjecture for curvature measures $\Phi \in \mathrm{Curv}_0^+$ which satisfy $\mathrm{glob}(\Phi) = 0$. To do so, assume that $\Phi \in \mathrm{Curv}_0^{+,\infty}$ satisfies $\mathrm{glob}(\Phi) = 0$. Since $J^\infty$ is an isomorphism, there exists a function $g \in V^\infty$ such that

$$\Phi = \Phi_0^g = \left[\pi_2^* g \wedge \omega_0\right]$$

which shows that $\Phi \in \mathrm{Curv}_0^{+,\mathrm{sm}}$ as required. □

In total, we have now proved that the smoothness conjecture holds true in dimension $n = 2$.

**Theorem 4.24.** *If $n = 2$, then $\mathrm{Curv}^\infty = \mathrm{Curv}^{\mathrm{sm}}$.*

Let us mention that as a byproduct of the two proofs (even and odd) we have proved the following explicit description of 0-homogeneous smooth curvature measures in $\mathbb{R}^2$.

**Corollary 4.25.** *Let $n = 2$. Then, the map*

$$C^\infty\left(S^1\right) \to \mathrm{Curv}_0^\infty,\ g \mapsto \Phi_0^g$$

*is a $\mathbb{C}$-linear, $\mathrm{SO}(2)$-equivariant isomorphism.*

In particular, note that combined with Theorem 3.25 and Theorem 3.27 we obtain an explicit formula for any smooth curvature measure on $\mathbb{R}^2$.

## 4.4 Jordan–Hölder Series for Curvature Measures

Let $V$ be a continuous Fréchet space representation of a Lie group $G$. A finite increasing sequence

$$\{0\} = V_0 \subseteq V_1 \subseteq \ldots \subseteq V_{m-1} \subseteq V_m = V$$

of closed subrepresentations is called a *Jordan–Hölder series of length $m$* of $V$ if the representation $V_i/V_{i-1}$ is irreducible for all $i = 1, \ldots, m$.

**Definition 4.26.** Let $V$ be a continuous Fréchet space representation of a Lie group $G$ and let $m \in \mathbb{N}_0$. We say that $V$ has *length $m$* if $V$ admits a Jordan–Hölder series of length $m$.

For completeness, let us say that a representation $V$ has *infinite length* if it does not admit a Jordan–Hölder series. It follows from the Jordan–Hölder theorem, see for example Theorem 5.12 of [Rot95], that all the Jordan–Hölder series of a given representation $V$ have the same length. Therefore, any given representation $V$ indeed has a uniquely determined length $m = m(V) \in \mathbb{N}_0 \cup \{\infty\}$.

Note that $V$ has length $m = 0$ if and only if $V = \{0\}$. Moreover, $V$ has length $m = 1$ if and only if $V$ is irreducible.

With this in mind, Alesker's irreducibility theorem states that $\mathrm{Val}_i^\varepsilon$ has length at most one for all $i \in \{0, \dots, n\}$ and $\varepsilon \in \{+, -\}$.

We have already seen that the representation $\mathrm{Curv}_i^\varepsilon$ is not irreducible in general, see Corollary 4.8. Therefore, the length of $\mathrm{Curv}_i^\varepsilon$ will exceed one in general which means that the following conjecture is optimal.

**Conjecture 4.27.** *For all $i \in \{0, \dots, n\}$ and $\varepsilon \in \{+, -\}$, the representation $\mathrm{Curv}_i^\varepsilon$ has length at most two.*

If verified this conjecture could be seen as an analogue of Alesker's irreducibility theorem for the setting of curvature measures.

The conjecture is easily verified in the cases which we have fully characterized in Section 3, that is curvature measures of degree $i = n - 1$ and $i = n$.

**Theorem 4.28.**
- $\mathrm{Curv}_n^+$ *has length one.*
- $\mathrm{Curv}_n^-$ *has length zero.*
- $\mathrm{Curv}_{n-1}^+$ *has length one.*
- $\mathrm{Curv}_{n-1}^-$ *has length two.*

*Proof.* Recall from Theorem 3.25 that $\mathrm{Curv}_n^+ = \mathbb{C}\,\mathrm{VOL}_n$ which is irreducible as it has dimension one. Therefore, $\mathrm{Curv}_n^+$ indeed has length one.

Even more easily, we see that $\mathrm{Curv}_n^-$ has length zero due to $\mathrm{Curv}_n^- = \{0\}$ which was also proved in Theorem 3.25.

For the $(n-1)$-homogeneous cases, recall from Theorem 3.27 that the map

$$C\left(S^{n-1}\right) \to \mathrm{Curv}_{n-1},\ f \mapsto \Phi_{n-1}^f$$

is an isomorphism which preserves parity. Moreover, recall from McMullen's theorem (2.13) that we also have a linear map

$$C\left(S^{n-1}\right) \to \mathrm{Val}_{n-1},\ f \mapsto \varphi_f.$$

which is surjective and preserves parity. Lastly, note that

$$\mathrm{glob}\left(\Phi_{n-1}^f\right)(K) = \int_{S\,\mathbb{R}^n} \pi_2^* f \,\mathrm{d}\,\Theta_{n-1}(K) = \int_{S^{n-1}} f \,\mathrm{d}\, S_{n-1}(K) = \varphi_f(K)$$

for all $f \in C\left(S^{n-1}\right)$ and all $K \in \mathcal{K}$.

In the even case, this implies that the $\mathrm{GL}(n,\mathbb{R})$-equivariant map

$$\mathrm{glob}_{n-1}^{+}\colon \mathrm{Curv}_{n-1}^{+} \to \mathrm{Val}_{n-1}^{+}$$

is an isomorphism. From Alesker's irreducibility theorem, we therefore obtain that $\mathrm{Curv}_{n-1}^{+}$ is irreducible which implies that it has length one.

For the odd case, note that the discussion above implies that the $\mathrm{GL}(n,\mathbb{R})$-equivariant map

$$\mathrm{glob}_{n-1}^{-}\colon \mathrm{Curv}_{n-1}^{-} \to \mathrm{Val}_{n-1}^{-}$$

is surjective. Therefore, we have

$$\mathrm{Curv}_{n-1}^{-} \Big/ \ker\left(\mathrm{glob}_{n-1}^{-}\right) \cong \mathrm{Val}_{n-1}^{-}$$

which is irreducible by Alesker's irreducibility theorem. Moreover, recall from the proof of Theorem 3.27 that the Bernig embedding, which is $\mathrm{GL}(n,\mathbb{R})$-equivariant, induces an isomorphism

$$B_{n-1}^{-}\colon \ker\left(\mathrm{glob}_{n-1}^{-}\right) \to \mathrm{Val}_{n}^{+} \otimes \mathbb{C}^n = \mathbb{C}\,\mathrm{vol}_n \otimes \mathbb{C}^n\,.$$

Note that

$$(g\,\mathrm{vol}_n)(K) = \mathrm{vol}_n\left(g^{-1}K\right) = \left|\det\left(g^{-1}\right)\right| \mathrm{vol}_n(K)$$

for all $g \in \mathrm{GL}(n,\mathbb{R})$ and $K \in \mathcal{K}$. Therefore, $\mathbb{C}\,\mathrm{vol}_n$ is trivial when considered as an $\mathrm{SL}(n,\mathbb{R})$-representation which combined with the isomorphism induced by the Bernig embedding implies that $\ker(\mathrm{glob}_{n-1}^{-})$ is isomorphic to the standard representation on the level of $\mathrm{SL}(n,\mathbb{R})$-representations. The standard representation of $\mathrm{SL}(n,\mathbb{R})$ is of course well-known to be irreducible. In particular, $\ker(\mathrm{glob}_{n-1}^{-})$ is irreducible as a representation of $\mathrm{GL}(n,\mathbb{R})$. Thus, we have proved that

$$\{0\} \subseteq \ker\left(\mathrm{glob}_{n-1}^{-}\right) \subseteq \mathrm{Curv}_{n-1}^{-}$$

is a Jordan–Hölder series which shows that $\mathrm{Curv}_{n-1}^{-}$ has length two. □

In order to study other cases, we first need to study smooth vectors of continuous representations a bit more. Our main source for this topic is Section 4.4 of [War72]. Let $V$ be a continuous Banach space representation of a Lie group $G$ and write $V^{\infty} \subseteq V$ for the space of smooth vectors in $V$. The space $V^{\infty}$ is naturally a Fréchet space when equipped with the so called *Gårding topology* which can be introduced as follows. The injective map

$$V \to C(G,V)\,,\ v \mapsto (g \mapsto gv)$$

induces a map $V^{\infty} \to C^{\infty}(G,V)$. Therefore, we can identify $V^{\infty}$ with its image which is a closed subspace of the Fréchet space $C^{\infty}(G,V)$.

**Lemma 4.29.** *Let $V$ be a continuous Banach space representation of a Lie group $G$ and let $W \subseteq V$ be a closed subrepresentation. Then,*

$$V^\infty / W^\infty \cong \left(V/W\right)^\infty$$

*as Fréchet space representations of $G$.*

*Proof.* Let $\pi \colon V \to V/W$ be the natural projection. Since passing to smooth vectors is functorial, see Lemma 1.26 of [Wan11], this restricts to a map

$$\pi^\infty \colon V^\infty \to \left(V/W\right)^\infty, \, v \mapsto v + W$$

of $G$-representations. By Proposition 4.4.1.11 of [War72], this map is in fact surjective. From the definition of the Gårding topology one easily sees that $W^\infty$ is closed in $V^\infty$ as $W$ is closed in $V$. Therefore, $W^\infty \subseteq W$ implies that $\pi^\infty$ induces a $G$-equivariant map

$$\alpha \colon V^\infty / W^\infty \to \left(V/W\right)^\infty, \, v + W^\infty \mapsto v + W.$$

We want to prove that $\alpha$ is the required ismomorphism.

Let $v \in V^\infty$ such that $v + W^\infty \in \ker(\alpha)$. By definition of $\alpha$, this is equivalent to $v \in W$. Since $v$ is a smooth vector by assumption, it follows that $v \in W^\infty$ which is of course equivalent to $v + W^\infty = 0 \in V^\infty / W^\infty$. Hence, we have seen that $\alpha$ is injective.

For surjectivity, let $v + W \in \left(V/W\right)^\infty$ be arbitrary. Since $\pi^\infty$ is surjective, there exists $v^\infty \in V^\infty$ such that

$$v^\infty + W = \pi^\infty\left(v^\infty\right) = v + W.$$

This yields that $\alpha\left(v^\infty + W^\infty\right) = v + W$ as required.

Lastly, $\alpha$ is of course continuous as $\pi^\infty$ is. Therefore, it is also an isomorphism on the level of Fréchet spaces by the open mapping theorem. □

The following lemma is well-known in the theory of Banach space representations. A proof can be found on p. 254 of [War72] for example.

**Lemma 4.30.** *Let $V$ be a continuous Banach space representation of a Lie group $G$. Then, $V$ is irreducible if and only if $V^\infty$ is irreducible.*

The reason why these two lemmas are of importance to us is that they yield the following proposition.

**Proposition 4.31.** *Let $V$ be a continuous Banach space representation of a Lie group $G$. An increasing sequence*

$$\{0\} = V_0 \subseteq V_1 \subseteq \ldots \subseteq V_{m-1} \subseteq V_m = V$$

*of closed subrepresentations is a Jordan–Hölder series if and only if*

$$\{0\} = V_0^\infty \subseteq V_1^\infty \subseteq \ldots \subseteq V_{m-1}^\infty \subseteq V_m^\infty = V^\infty$$

*is a Jordan–Hölder series. In particular, $V$ and $V^\infty$ have the same length.*

*Proof.* Let $i \in \{1, \dots, m\}$ be arbitrary. By Lemma 4.30, $V_i/V_{i-1}$ is irreducible if and only if $(V_i/V_{i-1})^\infty$ is irreducible. By Lemma 4.29, we have

$$\left(V_i/V_{i-1}\right)^\infty \cong V_i^\infty/V_{i-1}^\infty.$$

Therefore, $V_i/V_{i-1}$ is irreducible if and only if $V_i^\infty/V_{i-1}^\infty$ is irreducible which yields the claim. □

The previous proposition yields that it is sufficient to compute the length of $\operatorname{Curv}_i^{\varepsilon,\infty}$ to verify Conjecture 4.27. This can be done for 0-homogeneous curvature measures on the plane $\mathbb{R}^2$ using the results from Section 4.3.

To do so, we need to recall some background on the so called principal series representations of $\mathrm{GL}(2,\mathbb{R})$. Our main source for this is [JL70], in particular Chapter I, §5. For a concise overview we also recommend the survey [Pra18]. We would also like to point out the great exposition in [Hum23]. The advantage of these texts is that they directly construct $\mathrm{GL}(2,\mathbb{R})$-representations. Another common way is to first construct principal series representations of $\mathrm{SL}(2,\mathbb{R})$ and then paste another character of $\mathbb{R}^\times$ to obtain the principal series representations of $\mathrm{GL}(2,\mathbb{R})$. This route is taken in Chapter II, §5 and Chapter VII of the book [Kna01] as well as the paper [Kna79].

It is well-known that any character $c\colon \mathbb{R}^\times \to \mathbb{C}^\times$ is of the form $x \mapsto \operatorname{sgn}(x)^p |x|^s$ for some $p \in \{0,1\}$ and some $s \in \mathbb{C}$, where $\operatorname{sgn}\colon \mathbb{R}^\times \to \{-1,1\}$ denotes the sign function. In the following, we write $c = [p,s]$ for the character associated with such $p$ and $s$.

To any pair of characters $c_1 = [p_1, s_1]$ and $c_2 = [p_2, s_2]$, there is an associated representation $\mathcal{P}(c_1, c_2)$ of $\mathrm{GL}(2,\mathbb{R})$. As a vector space $\mathcal{P}(c_1,c_2)$ consists of all smooth functions $f\colon \mathrm{GL}(2,\mathbb{R}) \to \mathbb{C}$ which satisfy

$$f\left(\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} h\right) = c_1(a)\, c_2(d) \sqrt{\left|ad^{-1}\right|} f(h)$$

for all $a, d \in \mathbb{R}^\times$, $b \in \mathbb{R}$ and $h \in \mathrm{GL}(2,\mathbb{R})$. The group $\mathrm{GL}(2,\mathbb{R})$ then acts on these functions by right multiplication, that is

$$(gf)(h) = f(hg)$$

for all $g, h \in \mathrm{GL}(2,\mathbb{R})$ and $f \in \mathcal{P}(c_1,c_2)$. It is standard in the theory of induced representations that $\mathcal{P}(c_1,c_2)$ is a representation of $\mathrm{GL}(2,\mathbb{R})$ with respect to this action.

**Definition 4.32.** The representation $\mathcal{P}(c_1,c_2)$ of $\mathrm{GL}(2,\mathbb{R})$ is called the *principal series representation* associated with $c_1$ and $c_2$.

Note that the space $\mathcal{P}(c_1,c_2)$ is naturally equipped with a topology as it is a subspace of $C^\infty\left(\mathrm{GL}(2,\mathbb{R})\right)$. It is also standard that the representation is continuous with respect to this topology. The following proposition is well-known, see for example Theorem 5.11 of [JL70].

**Proposition 4.33.** *Let $c_1 = [p_1, s_1]$ and $c_2 = [p_2, s_2]$ be two characters of $\mathbb{R}^\times$ and let $\mathcal{P}(c_1, c_2)$ be the associated principal series representation of $\mathrm{GL}(2, \mathbb{R})$. Then, $\mathcal{P}(c_1, c_2)$ is reducible if and only if there exists an integer $k \geq 2$ such that*

$$\begin{aligned} p_1 + p_2 &\equiv k \mod 2, \\ s_1 - s_2 &\in \{1-k,\, k-1\}. \end{aligned}$$

We now want to connect this to some representations which are closer to our studies of curvature measures. To do so, let $C^\infty(\mathbb{R}^2 \setminus \{0\})$ be the vector space of smooth functions $\mathbb{R}^2 \setminus \{0\} \to \mathbb{C}$. For given parameters $w \in \mathbb{C}$ and $\varepsilon \in \{-1, +1\}$, we let

$$C_w^{\varepsilon,\infty}\left(\mathbb{R}^2 \setminus \{0\}\right) \subseteq C^\infty\left(\mathbb{R}^2 \setminus \{0\}\right)$$

denote the subspace of smooth, positively $w$-homogeneous functions of parity $\varepsilon$.

Let us fix two characters $c_1 = [p_1, s_1]$ and $c_2 = [p_2, s_2]$ of $\mathbb{R}^\times$. Moreover, we let $w = s_2 - s_1 - 1 \in \mathbb{C}$ and $\varepsilon = (-1)^{p_1+p_2}$.

For a given function $f \in \mathcal{P}(c_1, c_2)$, we define a function $F \in C^\infty(\mathbb{R}^2 \setminus \{0\})$ by

$$F(x) = \frac{(-1)^\varepsilon}{\|x\|^{s_1-s_2+1}} f\begin{pmatrix} \frac{x_1}{\|x\|} & \frac{x_2}{\|x\|} \\ -\frac{x_2}{\|x\|} & \frac{x_1}{\|x\|} \end{pmatrix}$$

for all $x = (x_1, x_2) \in \mathbb{R}^2$. While this definition might seem clumsy, it can be easily expressed using the defining relation of $\mathcal{P}(c_1, c_2)$. This yields that for $x = (x_1, x_2)$ with $x_2 \neq 0$, we have

$$F(x) = f\left(\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} x_2 & -x_1 \\ 0 & \frac{1}{x_2} \end{pmatrix}\right) = f\begin{pmatrix} 0 & -\frac{1}{x_2} \\ x_2 & -x_1 \end{pmatrix}$$

while

$$F\begin{pmatrix} x_1 \\ 0 \end{pmatrix} = f\begin{pmatrix} \frac{1}{x_1} & 0 \\ 0 & x_1 \end{pmatrix}.$$

For all $\lambda > 0$ and $x \in \mathbb{R}^2 \setminus \{0\}$, the function $F$ satisfies

$$\begin{aligned} F(\lambda x) &= \lambda^{s_2-s_1-1} F(x) = \lambda^w F(x), \\ F(-x) &= (-1)^{p_1+p_2} F(x) = \varepsilon F(x) \end{aligned}$$

which shows that $F \in C_w^{\varepsilon,\infty}(\mathbb{R}^2 \setminus \{0\})$.

Conversely, let $F\colon \mathbb{R}^2 \setminus \{0\} \to \mathbb{C}$ be a smooth and $w$-homogeneous function of parity $\varepsilon$. Moreover, recall that by the QR decomposition any matrix $h \in \mathrm{GL}(2, \mathbb{R})$ can be uniquely written as

$$h = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix}\begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$$

for some $a > 0$, $d \in \mathbb{R}^\times$ and $b, \theta \in \mathbb{R}$. With respect to this decomposition, $F$ defines a function $f \in \mathcal{P}(c_1, c_2)$ by

$$f(h) = c_1(a)\, c_2(d) \sqrt{|ad^{-1}|} F(\cos\theta, \sin\theta).$$

The above construction yields an isomorphism

$$\mathcal{P}(c_1, c_2) \cong C_w^{\varepsilon,\infty}\left(\mathbb{R}^2 \setminus \{0\}\right)$$

on the level of vector spaces. Of course, one can now construct a $\mathrm{GL}(2,\mathbb{R})$-action on the vector space on the right hand side which ensures that this isomorphism becomes $\mathrm{GL}(2,\mathbb{R})$-equivariant. This action is given by

$$(gF)(x) = \operatorname{sgn}(\det g)^{p_2} |\det g|^{s_2 - \frac{1}{2}} F\left(g^{-1}x\right).$$

Therefore, we obtain a representation $\mathcal{PP}(c_1, c_2)$ of $\mathrm{GL}(2,\mathbb{R})$ with underlying vector space $C_w^{\varepsilon,\infty}(\mathbb{R}^2 \setminus \{0\})$ which is isomorphic to the principal series representation $\mathcal{P}(c_1, c_2)$. We will call $\mathcal{PP}(c_1, c_2)$ the *planar principal series representation* associated with $c_1$ and $c_2$.

On $C^\infty(\mathbb{R}^2 \setminus \{0\})$ consider the standard action given by

$$(gF)(x) = F\left(g^{-1}x\right)$$

for all $g \in \mathrm{GL}(2,\mathbb{R})$, $F \in C^\infty(\mathbb{R}^2 \setminus \{0\})$ and $x \in \mathbb{R}^2 \setminus \{0\}$. This turns the spaces $C_w^{\varepsilon,\infty}(\mathbb{R}^2 \setminus \{0\})$ into an a priori different representation of $\mathrm{GL}(2,\mathbb{R})$. These representations will be denoted by $\mathcal{V}_w^\varepsilon$.

Note that the representations $\mathcal{V}_w^\varepsilon$ only depend on $w$ and $\varepsilon$ and not on the characters $c_1$ and $c_2$. In fact, these representations are just a special case of the planar principal series representations, that means that for any given $w$ and $\varepsilon$ there exist suitable characters $c_1$ and $c_2$ such that

$$\mathcal{V}_w^\varepsilon = \mathcal{PP}(c_1, c_2) \cong \mathcal{P}(c_1, c_2).$$

As we do not need this here, we shall omit the proof. We only need two special cases which we will discuss separately.

For the odd case $\mathrm{Curv}_0^-$, we need the following lemma.

**Lemma 4.34.** *The* $\mathrm{GL}(2,\mathbb{R})$*-representation* $\mathcal{V}_0^-$ *is irreducible.*

*Proof.* We only need to specify what we have done above. To do so, we choose the character $c_1 = [p_1, s_1]$ as $p_1 = 1$, $s_1 = -1/2$ and the character $c_2 = [p_2, s_2]$ as $p_2 = 0$, $s_2 = 1/2$. Then, we have

$$w = s_2 - s_1 - 1 = 0, \quad \varepsilon = (-1)^{p_1 + p_2} = -1.$$

Moreover, for this choice of $c_2$ the action on the planar principal series representation $\mathcal{PP}(c_1, c_2)$ is given by

$$(gF)(x) = \operatorname{sgn}(\det g)^{p_2} |\det g|^{s_2 - \frac{1}{2}} F\left(g^{-1}x\right) = F\left(g^{-1}x\right)$$

for all $g \in \mathrm{GL}(2, \mathbb{R})$, $F \in C_0^{-,\infty}(\mathbb{R}^2 \setminus \{0\})$ and $x \in \mathbb{R}^2 \setminus \{0\}$. Since the latter action is just the action on $\mathcal{V}_0^\varepsilon$, we obtain

$$\mathcal{V}_0^\varepsilon = \mathcal{PP}(c_1, c_2) \cong \mathcal{P}(c_1, c_2).$$

The claim now follows from Proposition 4.33. □

**Theorem 4.35.** *Let $n = 2$. Then, $\mathrm{Curv}_0^-$ has length one.*

*Proof.* The task at hand is to rewrite the embedding $I\colon \mathrm{Curv}_0^- \to C^-(S^1)$ from Lemma 4.16 in a $\mathrm{GL}(2, \mathbb{R})$-equivariant fashion.

To do so, recall that

$$I\Phi(u) = \Phi\left([0, u], \{0\}\right)$$

for all $\Phi \in \mathrm{Curv}_0^-$ and $u \in S^1$. Of course, any point $x \in \mathbb{R}^2 \setminus \{0\}$ can be uniquely written as $x = \lambda u$ for some $\lambda > 0$ and $u \in S^1$. Since $\Phi$ is homogeneous of degree 0, we have

$$\Phi\left([0, x], \{0\}\right) = \Phi\left([0, \lambda u], \{0\}\right) = \Phi\left(\lambda\,[0, u], \lambda\,\{0\}\right) = \Phi\left([0, u], \{0\}\right)$$

for all $x = \lambda u \in \mathbb{R}^2 \setminus \{0\}$. This shows that $I\Phi\colon S^1 \to \mathbb{C}$ naturally extends to a function $I\Phi \in C_0^-(\mathbb{R}^2 \setminus \{0\})$.

From Theorem 4.17 we know that $I$ restricts to a $\mathbb{C}$-linear isomorphism

$$\mathrm{Curv}_0^{-,\infty} \cong C_0^{-,\infty}\left(\mathbb{R}^2 \setminus \{0\}\right).$$

Moreover, for given $g \in \mathrm{GL}(2, \mathbb{R})$ and $\Phi \in \mathrm{Curv}_0^-$ the map $I$ satisfies

$$\begin{aligned} I(g\Phi)(x) &= \Phi\left(g^{-1}[0, x], g^{-1}\{0\}\right) = \Phi\left(\left[0, g^{-1}x\right], \{0\}\right) \\ &= I\Phi\left(g^{-1}x\right) = g(I\Phi)(x) \end{aligned}$$

for all $x \in \mathbb{R}^2 \setminus \{0\}$ which shows that $I$ is $\mathrm{GL}(2, \mathbb{R})$-equivariant when considered as a map $I\colon \mathrm{Curv}_0^{-,\infty} \to \mathcal{V}_0^-$. Therefore, $\mathrm{Curv}_0^{-,\infty}$ is irreducible due to the previous lemma and thus has length one. The claim now follows from Proposition 4.31. □

Let us now get to the even case. More precisely, we want to verify Conjecture 4.27 for $\mathrm{Curv}_0^+$ under the assumption $n = 2$.

We start with two auxiliary lemmas. Their use will become evident later.

**Lemma 4.36.** *Let $M, N$ be two continuous representations of a Lie group $G$ such that $\dim(N) = 1$. Then, $M \otimes N$ is irreducible if and only if $M$ is irreducible.*

*Proof.* Assume that $M$ is reducible and let $U \subseteq M$ be a non-trivial, closed subrepresentation. Then, $U \otimes N$ is a non-trivial, closed subrepresentation of $M \otimes N$.

For the converse direction, note that the assumption $\dim(N) = 1$ implies that any linear subspace of $M \otimes N$ is of the form $U \otimes N$ for some linear subspace $U \subseteq M$. If such a subspace $U \otimes N$ is a non-trivial, closed subrepresentation of $M \otimes N$, then $U$ is a non-trivial, closed subrepresentation of $M$. □

Let us fix $V = \mathbb{R}^2$ for the moment. Recall that $\mathrm{GL}(2,\mathbb{R})$ acts on the dual space $V^*$ by $g\zeta(v) = \zeta\left(g^{-1}v\right)$ for all $g \in \mathrm{GL}(2,\mathbb{R})$, $\zeta \in V^*$ and $v \in V$. Obviously, this descends to an action of $\mathrm{GL}(2,\mathbb{R})$ on the projective space $\mathbb{P}(V^*)$.

We now let $\mathrm{GL}(2,\mathbb{R})$ act on $C^\infty(\mathbb{P}(V^*))$ by $gf(l) = f\left(g^{-1}l\right)$ for all $g \in \mathrm{GL}(2,\mathbb{R})$, $f \in C^\infty(\mathbb{P}(V^*))$ and $l \in \mathbb{P}(V^*)$. One easily sees that $C^\infty(\mathbb{P}(V^*))$ is a representation of $\mathrm{GL}(2,\mathbb{R})$ with respect to this action. Moreover, the trivial representation $\mathbb{C}$ is a closed subrepresentation of $C^\infty(\mathbb{P}(V^*))$ embedded as constant functions.

**Lemma 4.37.** *The representation* $C^\infty\left(\mathbb{P}(V^*)\right)/\mathbb{C}$ *of* $\mathrm{GL}(2,\mathbb{R})$ *is irreducible.*

*Proof.* For any $k \in \mathbb{Z}$, we write $\det^k$ for the one-dimensional real representation

$$\mathrm{GL}(2,\mathbb{R}) \to \mathrm{GL}(\mathbb{R}),\ g \mapsto \left(x \mapsto \det(g)^k x\right).$$

It is well-known that $V^* \cong \Lambda^1 V \otimes \det^{-1} = V \otimes \det^{-1}$ as $\mathrm{GL}(2,\mathbb{R})$-representations, see for example p. 40 of [KP96]. This implies that

$$C^\infty\left(\mathbb{P}(V^*)\right) \cong C^\infty\left(\mathbb{P}\left(V \otimes \det{}^{-1}\right)\right).$$

Note that any one-dimensional subspace of $V \otimes \mathbb{R}$ is of the form $l \otimes \mathbb{R}$ for some one-dimensional subspace $l \subseteq V$. Therefore, $\mathrm{GL}(2,\mathbb{R})$ acts on $\mathbb{P}(V \otimes \det^{-1})$ by

$$g(l \otimes \mathbb{R}) = gl \otimes \det{}^{-1}(g)\,\mathbb{R} = gl \otimes \mathbb{R}.$$

We consider the map

$$I\colon C^\infty\left(\mathbb{P}\left(V \otimes \det{}^{-1}\right)\right) \to C^\infty\left(\mathbb{P}(V)\right)$$

which is given by $If(l) = f(l \otimes \mathbb{R})$ for all $f \in C^\infty(\mathbb{P}(V \otimes \det^{-1}))$ and all $l \in \mathbb{P}(V)$.

The map $I$ is clearly bijective and $\mathbb{C}$-linear because it is the pullback of the diffeomorphism

$$\mathbb{P}(V) \to \mathbb{P}\left(V \otimes \det{}^{-1}\right),\ l \mapsto l \otimes \mathbb{R}.$$

Moreover, we have

$$I(gf)(l) = (gf)(l \otimes \mathbb{R}) = f\left(g^{-1}l \otimes \mathbb{R}\right) = If\left(g^{-1}l\right) = g(If)(l)$$

for all $g \in \mathrm{GL}(2,\mathbb{R})$, $f \in C^\infty(\mathbb{P}(V \otimes \det^{-1}))$ and $l \in \mathbb{P}(V)$ which shows that $I$ is $\mathrm{GL}(2,\mathbb{R})$-equivariant.

Of course $I$ maps the subspace of constant functions on $\mathbb{P}(V \otimes \det^{-1})$ to the space of constant functions on $\mathbb{P}(V)$. Therefore, we have proved that it is sufficient to show that

$$C^\infty\left(\mathbb{P}(V)\right) \Big/ \mathbb{C}$$

is irreducible. Let us consider a different model for this task. We consider the natural $\mathbb{C}$-linear isomorphism

$$I' \colon C^\infty\left(\mathbb{P}(V)\right) \to C_0^{+,\infty}\left(\mathbb{R}^2 \setminus \{0\}\right)$$

given by $I'f(x) = f\left(\mathrm{span}\{x\}\right)$ for all $f \in C^\infty(\mathbb{P}(V))$ and $x \in \mathbb{R}^2 \setminus \{0\}$. This clearly is an isomorphism $C^\infty\left(\mathbb{P}(V)\right) \cong \mathcal{V}_0^+$ of $\mathrm{GL}(2,\mathbb{R})$-representations. $I'$ also maps constant functions to constant functions. Therefore, it is sufficient to prove that the representation $\mathcal{V}_0^+ / \mathbb{C}$ is irreducible.

This can again be achieved by using principal series representations. To do so, we use the characters $c_1 = [p_1, s_1]$ and $c_2 = [p_2, s_2]$ given by

$$p_1 = 0,\ s_1 = -\frac{1}{2},\ p_2 = 0,\ s_2 = \frac{1}{2}.$$

Then, the underlying vector space of $\mathcal{PP}(c_1, c_2)$ is just $C_0^{\infty,+}\left(\mathbb{R}^2 \setminus \{0\}\right)$ and the group action on $\mathcal{PP}(c_1, c_2)$ is given by

$$gf(x) = f\left(g^{-1}x\right)$$

for all $g \in \mathrm{GL}(2,\mathbb{R})$, $f \in C^\infty(\mathbb{R}^2 \setminus \{0\})$ and $x \in \mathbb{R}^2 \setminus \{0\}$. This shows that we have

$$\mathcal{V}_0^+ \cong \mathcal{PP}(c_1, c_2) \cong \mathcal{P}(c_1, c_2).$$

For this principal series representation it is however well-known that the quotient $\mathcal{P}(c_1, c_2) / \mathbb{C}$ is the so called discrete series representation of weight two. This follows from Proposition 1 of [Pra18]. As the discrete series representation of weight 2 is known to be irreducible, the claim follows. □

Next, we need to recall that smooth valuations and curvature measures can be constructed in $\mathrm{GL}(n,\mathbb{R})$-equivariant fashion. To do so, we need to rewrite the constructions from Section 2.5 without making use of the inner product on $\mathbb{R}^n$. This construction can be found in more detail in Section 5 of [Ale06a].

Let us fix $V = \mathbb{R}^n$ and write $\mathbb{P}_V = V \times \mathbb{P}_+(V^*)$, where $\mathbb{P}_+(V^*)$ denotes the manifold of all oriented lines in $V^*$.

We let $V$ act on $\mathbb{P}_V$ by translations of the first component. This lifts to an action $V \curvearrowright \Omega^\bullet(\mathbb{P}_V)$ and we write $\Omega^\bullet(\mathbb{P}_V)^{\mathrm{tr}} \subseteq \Omega^\bullet(\mathbb{P}_V)$ for the subspace of differential forms which are invariant with respect to this action.

For any convex body $K \subseteq V$, the normal cycle $\mathrm{nc}(K)$ can be interpreted as a subset of $\mathbb{P}_V$. Using the same defining formulas as in Section 2.5, any differential form $\omega \in \Omega^{n-1}(\mathbb{P}_V)^{\mathrm{tr}}$ therefore defines a curvature measure $[\omega] \in \mathrm{Curv}^{\mathrm{sm}}$ and a valuation $[\![\omega]\!] \in \mathrm{Val}^{\mathrm{sm}}$.

Note that a choice of inner product on $V$ would induce an isomorphism $V \cong V^*$ and a diffeomorphism $\mathbb{P}_+(V^*) \cong S^{n-1}$. Under these identifications the construction above coincides with the original definition of $[\omega]$ and $[\![\omega]\!]$.

Let $\mathrm{or}(V)$ denote the complexification of the real vector space which consists of all maps $\rho\colon \Lambda^n V \to \mathbb{R}$ which satisfy $\rho(\lambda\tau) = \mathrm{sgn}(\lambda)\,\rho(\tau)$ for all $\lambda \in \mathbb{R}$ and $\tau \in \Lambda^n V$. This is naturally a one-dimensional complex representation of $\mathrm{GL}(n,\mathbb{R})$.

The crucial upshot of this construction of smooth curvature measures and valuations is the following lemma.

**Lemma 4.38.** *The $\mathbb{C}$-linear maps*

$$\Omega^{n-1}(\mathbb{P}_V)^{\mathrm{tr}} \otimes \mathrm{or}(V) \to \mathrm{Curv}^{\mathrm{sm}},\ \omega \mapsto [\omega],$$
$$\Omega^{n-1}(\mathbb{P}_V)^{\mathrm{tr}} \otimes \mathrm{or}(V) \to \mathrm{Val}^{\mathrm{sm}},\ \omega \mapsto [\![\omega]\!]$$

*are $\mathrm{GL}(n,\mathbb{R})$-equivariant.*

*Proof.* This follows immediately from the fact that with respect to this construction integration against the normal cycle is $\mathrm{GL}(n,\mathbb{R})$-equivariant, see Section 5.1 of Alesker's paper [Ale06a]. □

Note that $\Omega^{n-1}(\mathbb{P}_V)^{\mathrm{tr}}$ is naturally graded due to the product structure of the manifold $\mathbb{P}_V$, that is

$$\Omega^{n-1}(\mathbb{P}_V)^{\mathrm{tr}} = \bigoplus_{i=0}^{n-1} \Omega^{i,n-1-i}(\mathbb{P}_V)^{\mathrm{tr}}.$$

With respect to this grading we have $[\omega] \in \mathrm{Curv}_i$ and $[\![\omega]\!] \in \mathrm{Val}_i$ if $\omega \in \Omega^{i,n-1-i}(\mathbb{P}_V)^{\mathrm{tr}}$.

Moreover, the spaces $\Omega^{i,n-1-i}(\mathbb{P}_V)^{\mathrm{tr}}$ admit a natural decomposition with respect to parity, that is

$$\Omega^{i,n-1-i}(\mathbb{P}_V)^{\mathrm{tr}} = \Omega_+^{i,n-1-i}(\mathbb{P}_V)^{\mathrm{tr}} \oplus \Omega_-^{i,n-1-i}(\mathbb{P}_V)^{\mathrm{tr}}.$$

This decomposition is induced by the natural involution $\mathbb{P}_V \to \mathbb{P}_V$ which is given by changing the orientation of the oriented line $l \in \mathbb{P}_+(V)$. Moreover, this decomposition determines the parity of the induced curvature measure. For us only one case will be of importance which we discuss below.

We now return to the special setting $n = 2$. In particular, we fix $V = \mathbb{R}^2$. Since we are interested in 0-homogeneous curvature measures, we are interested in the $\mathrm{GL}(2,\mathbb{R})$-equivariant map

$$\Omega^{0,1}(\mathbb{P}_V)^{\mathrm{tr}} \otimes \mathrm{or}(V) \to \mathrm{Curv}_0^{\mathrm{sm}},\ \omega \mapsto [\omega].$$

In this case the curvature measure $[\omega]$ is even if and only if the differential form $\omega$ is even, see Theorem 1.2.11 of [AF14]. This means that we get an induced $\mathrm{GL}(2,\mathbb{R})$-equivariant map

$$\Omega_+^{0,1}(\mathbb{P}_V)^{\mathrm{tr}} \otimes \mathrm{or}(V) \to \mathrm{Curv}_0^{+,\mathrm{sm}}.$$

**Proposition 4.39.** *The representation*

$$\ker\left(\mathrm{glob}_0^+\right)$$

*of* $\mathrm{GL}(2,\mathbb{R})$ *is irreducible.*

*Proof.* By Lemma 4.30 this is equivalent to showing that the representation

$$\ker\left(\mathrm{glob}_0^{+,\infty}\right) = \ker\left(\mathrm{glob}_0^{+,\mathrm{sm}}\right)$$

is irreducible, where we use Theorem 4.23 for the equality.

To do so, we want to analyze the $\mathrm{GL}(2,\mathbb{R})$-equivariant map

$$A\colon \Omega_+^{0,1}(\mathbb{P}_V)^{\mathrm{tr}} \otimes \mathrm{or}(V) \to \mathrm{Curv}_0^{+,\mathrm{sm}}$$

which we have constructed above.

The kernel of the map $\Omega^{n-1}(\mathbb{P}_V)^{\mathrm{tr}} \to \mathrm{Curv}^{\mathrm{sm}}$ can be explicitly characterized in terms of the natural contact structure of the manifold $\mathbb{P}_V$, see for example Theorem 2.2 of [BFS14]. In our case, we obtain $\ker(A) = \{0\}$. Moreover, the map is surjective by construction. This shows that $A$ is an isomorphism.

Let $\omega \in \Omega_+^{0,1}(\mathbb{P}_V)^{\mathrm{tr}}$. A special case of the kernel theorem due to Bernig and Bröcker, see [BB07], states that

$$\mathrm{glob}\left([\omega]\right) = [\![\omega]\!] = 0$$

if and only if $\omega$ is exact. Therefore, $A$ restricts to a $\mathrm{GL}(2,\mathbb{R})$-equivariant isomorphism

$$\bar{A}\colon \left\{\omega \in \Omega_+^{0,1}(\mathbb{P}_V)^{\mathrm{tr}} \middle| \omega \text{ is exact } \right\} \otimes \mathrm{or}(V) \to \ker\left(\mathrm{glob}_0^{+,\mathrm{sm}}\right).$$

Since $\mathrm{or}(V)$ is one-dimensional, it is sufficient to prove that

$$\left\{\omega \in \Omega_+^{0,1}(\mathbb{P}_V)^{\mathrm{tr}} \middle| \omega \text{ is exact } \right\}$$

is irreducible by virtue of Lemma 4.36. To this end, we start with the following observation. Since the only translation-invariant functions $f\colon V \to \mathbb{C}$ are constant functions, we have

$$\Omega_+^{0,1}(\mathbb{P}_V)^{\mathrm{tr}} \cong \Omega^0(V)^{\mathrm{tr}} \otimes \Omega_+^1\left(\mathbb{P}_+(V^*)\right) \cong \mathbb{C} \otimes \Omega_+^1\left(\mathbb{P}_+(V^*)\right) = \Omega_+^1\left(\mathbb{P}_+(V^*)\right).$$

Moreover, note that any exact form $\omega \in \Omega_+^1\left(\mathbb{P}_+(V^*)\right)$ can be written as $\omega = \mathrm{d}f$ for some $f \in C^{+,\infty}(\mathbb{P}_+(V^*))$ and we have $\omega = 0$ if and only if $f$ is constant. Therefore, we obtain

$$\left\{\omega \in \Omega_+^{0,1}(\mathbb{P}_V)^{\mathrm{tr}} \middle| \omega \text{ is exact } \right\} \cong C^{+,\infty}\left(\mathbb{P}_+(V^*)\right) \Big/ \mathbb{C} \cong C^{\infty}\left(\mathbb{P}(V^*)\right) \Big/ \mathbb{C}.$$

This implies the claim by virtue of Lemma 4.37. □

We are now able to prove the final result of this thesis.

**Theorem 4.40.** *Let $n = 2$. The representation $\mathrm{Curv}_0^+$ of $\mathrm{GL}(2, \mathbb{R})$ has length two.*

*Proof.* We have seen in the previous proposition that

$$\ker\left(\mathrm{glob}_0^+\right)$$

is irreducible. Moreover, recall that $\mathrm{Val}_0^+ \cong \mathbb{C}$ is finite dimensional. Therefore, the globalization map is surjective which yields that

$$\mathrm{Curv}_0^+ \Big/ \ker\left(\mathrm{glob}_0^+\right) \cong \mathrm{Val}_0^+ \cong \mathbb{C}\,.$$

Since the trivial representation is of course irreducible, it follows that the sequence $\{0\} \subseteq \ker(\mathrm{glob}_0^+) \subseteq \mathrm{Curv}_0^+$ is a Jordan–Hölder series of length two which finishes proof. □

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